

Title: Direct propylene epoxidation via water activation over Pd-Pt electrocatalysts

Authors: Minju Chung¹, Joseph H. Maalouf¹, Jason S. Adams², Chenyu Jiang², Yuriy Román-Leshkov¹, Karthish Manthiram^{2*}

Affiliations:

¹Department of Chemical Engineering, Massachusetts Institute of Technology; Cambridge, MA 02139, United States

²Division of Chemistry and Chemical Engineering, California Institute of Technology; Pasadena, CA 91125, United States

*Corresponding author. Email: karthish@caltech.edu

Abstract: Direct electrochemical propylene epoxidation using water oxidation intermediates presents a sustainable alternative to existing routes that involve hazardous chlorine or peroxide reagents. Here we report an oxidized palladium-platinum alloy catalyst (PdPtO_x/C), which reaches a Faradaic efficiency of 66±5% toward propylene epoxidation at 50 mA/cm² at ambient temperature and pressure. Embedding platinum into the palladium oxide crystal structure stabilized oxidized platinum species, resulting in improved catalyst performance. The reaction kinetics suggest that epoxidation on PdPtO_x/C proceeds through electrophilic attack by metal-bound peroxo (OO*) intermediates. This work demonstrates an effective strategy for selective electrochemical oxygen-atom transfer from water, without mediators, for diverse oxygenation reactions.

One-Sentence Summary: Our work contributes toward advancing the sustainable synthesis of epoxides using water as the oxygen atom source.

Main Text:

Propylene oxide (PO) is an important chemical building block used in the production of many commodity chemicals such as polyurethanes and polyesters (1). Currently, PO production relies on highly reactive, corrosive, or explosive chemical oxidants such as molecular chlorine (Cl_2) or peroxides (2). While direct oxidation of propylene using molecular oxygen has been explored as an alternative solution, achieving high selectivity for PO is challenging due to the allylic hydrogen stripping in propylene and consequent further oxidation (2). In this regard, a selective direct propylene epoxidation pathway using water as a cleaner and easily accessible oxygen source would present a compelling alternative to the current epoxidation methods.

Electrochemistry offers a promising approach for using water as a sustainable source of oxygen atoms at ambient temperature and pressure. In water electrolysis, typically oxygen and hydrogen are generated, but alternatively, commodity chemicals can be synthesized via oxidative processes at the anode of an electrochemical cell combined with the hydrogen evolution at the cathode. For example, chloride oxidation can be conducted at the anode to generate Cl_2 , which acts as a redox mediator in alkene epoxidation (Fig. 1A, bottom) (3–5). This electrified version of the chlorohydrin process was demonstrated to be effective at industrially relevant current densities and selectivity (4). However, this method produces byproducts such as hypochlorite, chlorinated organic compounds, and brine streams that pose serious environmental challenges unless 100% selectivity and recycling of the chloride electrolytes can be achieved. Rather than relying on Cl_2 as a redox mediator, it should be possible to directly transfer reactive intermediates typically seen in water oxidation to the alkene before the intermediates are fully oxidized into molecular oxygen (O_2) (Fig. 1A, top) (6–8). This direct electrochemical alkene epoxidation approach does not involve hazardous oxidizing agents, such as Cl_2 , making it an attractive avenue for high-demand propylene epoxidation.

Catalytic performance of PdPtO_x/C and design principle

Previously studied catalysts for electrochemical propylene epoxidation can be clustered into palladium- (Pd) (9–12), platinum- (Pt) (13–15), and silver- (Ag) (16, 17) derived groups (Fig. 1B and table S1), and the best catalytic performance for electrochemical epoxidation has been observed with oxidized Pd and Pt catalysts. Studies on Pd catalysts have shown that epoxidation occurs on the surface oxide layer at anodic potentials above 1.1 V vs. RHE, while allylic oxidation producing acrolein and acrylic acid occurs on the metallic surface at lower potentials (9, 12). The performance enhancement of oxidized Pt catalysts was correlated with an increase in surface presence of Pt(II) and Pt(IV) species, while pure PtO_2 was less active (14). Moreover, a theoretical study has identified weak oxygen-binding metal oxides as promising catalyst candidates for propylene epoxidation, and cautioned that the metallic phase of strong binding metals, such as Pt and Pd, may facilitate the production of unwanted side products through dehydrogenation (18). These findings suggest that the catalytic performance may be further optimized by adjusting the oxidation states of the metal. However, preparing Pt oxide catalysts with high concentrations of oxidized Pt species is challenging due to their instability. On the other hand, stable Pd oxide catalysts can be obtained through high-temperature annealing, but they generally exhibit lower rates of epoxidation.

Alloying Pt with Pd can be a strategy that leverages complementing properties of the two elements, enabling the attainment of optimal oxidation states and structures for an epoxidation catalyst (Fig. 1C). Under aerobic conditions, Pt nanoparticles tend to sinter into a metallic form when annealed above 400 °C (19) because volatile PtO_2 species mediate the sintering process by facilitating

Ostwald ripening (20, 21). In contrast, stable PdO nanoparticles can be produced through annealing of Pd nanoparticles without significant sintering (19, 22). Importantly, PdO can trap mobile PtO₂ species, which results in less sintering of Pd-Pt alloy nanoparticles compared to Pt nanoparticles (20, 23). In addition, Pd and Pt share very similar crystal structures and atomic sizes, allowing homogeneous distributions of Pd and Pt components in their alloys.

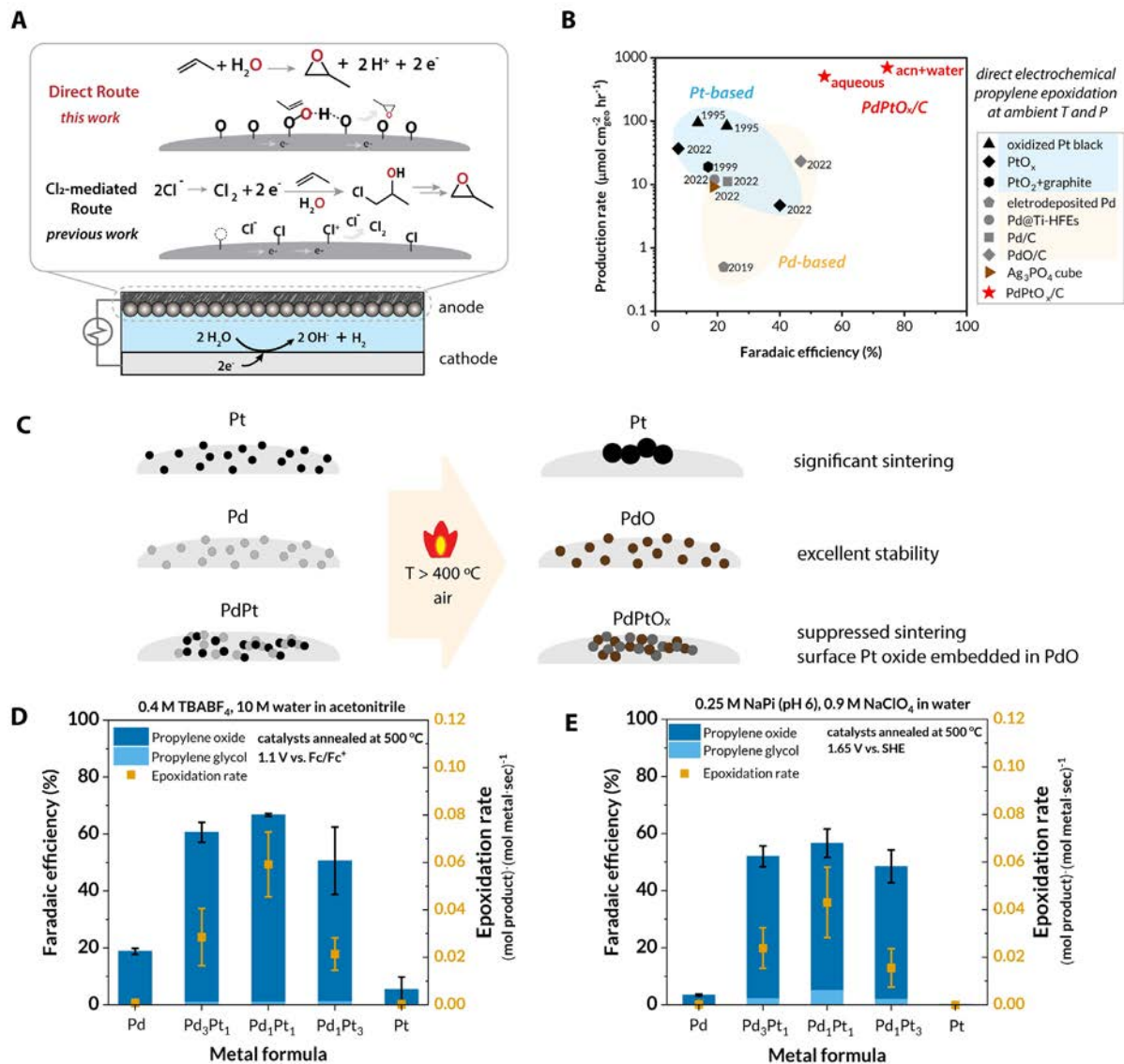


Fig. 1. Background and demonstration of direct electrochemical propylene epoxidation. (A) Comparison of chlorine-mediated and direct electrochemical propylene epoxidation. (B) Compilation of previously reported direct electrochemical propylene epoxidation and our work. (C) Effect of annealing on Pt, Pd, and Pd-Pt nanoparticles. (D and E) Direct electrochemical propylene epoxidation performance with different compositions of Pd-Pt-on-carbon catalysts in (D) water-acetonitrile electrolyte and (E) aqueous electrolyte. Potentials in this figure were 100% iR-compensated (i = current, R = resistance). The vertical error bars represent standard deviations from the mean of multiple replicates ($n=3$) of the same experiment.

In this study, we designed a Pd-Pt alloy catalyst containing both Pd and Pt oxide by embedding and stabilizing Pt oxide species in Pd oxide. A series of alloys with varying ratios of Pd and Pt were synthesized using a co-reduction method (24) on an amorphous carbon substrate and subsequently annealed under static air before being tested for their activity in propylene

epoxidation (Supplementary Materials Methods, fig. S1-3). Our catalyst is compatible with a blended electrolyte comprised of acetonitrile and water (Fig. 1D) that can accommodate a broad spectrum of alkene substrates, as well as aqueous electrolytes (Fig. 1E) that can minimize overall cell voltage for gaseous alkene epoxidation (fig. S4). The alloy catalyst with an equimolar composition of Pd and Pt annealed at 500 °C (PdPtO_x/C) was most active toward propylene epoxidation and outperformed previously reported catalysts for direct anodic epoxidation under ambient conditions (Fig. 1B and table S1). Furthermore, alloys with metal compositions of Pd₃Pt₁ and Pd₁Pt₃ exhibited epoxidation activities on par with PdPtO_x/C; when Pd and Pt are integrated at the nanometer scale (fig S2 and S3), Pd-Pt alloy catalysts can achieve enhanced epoxidation rates and Faradaic efficiencies relative to either Pd or Pt alone (Fig. 1D-E, fig S5).

In an electrolytic cell equipped with circulating liquid electrolytes (fig. S4), we demonstrated continuous propylene oxide production with high Faradaic efficiency averaging 66±5% (propylene glycol; 1.3±0.3%) for 3 hours at a constant current density of 50 mA/cm². Additionally, PdPtO_x/C catalyzed cyclooctene epoxidation with a Faradaic efficiency of ~90%, a notable improvement over the results observed in our previous studies using Ir-MnO_x (~50%) and MnO_x (~30%) catalysts (table S2) (7, 8).

Catalyst characterizations

We examined the bulk and surface properties of our annealed PdPt/C, Pd/C, and Pt/C catalysts (PdPtO_x/C, PdO_x/C, and PtO_x/C, respectively) to shed light on the enhanced performance of PdPtO_x/C compared to PdO_x/C and PtO_x/C.

X-ray diffraction (XRD) patterns showed that although PdO_x/C and PtO_x/C only exhibited oxide and metallic phases, respectively, the PdPtO_x/C contained both phases (Fig. 2A and fig. S6). The oxide phase of PdPtO_x/C resembles that of PdO_x/C, as the (101) peak in PdPtO_x/C at 33.7° occurs at a similar angle as the PdO_x/C peak at 33.9°. Similarly, the metallic phase of PdPtO_x/C parallels that of PtO_x/C, with the (111) peak shifted from 39.8° in PtO_x/C to 40.0° in PdPtO_x/C. Furthermore, the broader peaks in the PdPtO_x/C XRD pattern indicate that the PdPtO_x/C has smaller crystallites than PtO_x/C (table S3), which is consistent with our hypothesis that the sintering of PdPt/C is less severe than that of Pt/C upon annealing.

Extended X-ray absorption fine structure (EXAFS) measurements revealed that, within fitting errors, Pd and Pt coordinate with the same number of oxygens in PdPtO_x/C; the Pd-O and Pt-O coordination numbers were estimated to be 3.7±0.4 and 2.8±0.6, respectively (Fig. 2B-C and table S4-6). The local coordination environments of Pd in PdPtO_x/C and PdO_x/C were similar to that of PdO, while PtO_x/C resembled the Pt foil standard and did not show any contribution from the Pt-O scattering path. In contrast, the local environment of Pt in PdPtO_x/C was different from that in Pt foil or PtO₂ standard, and the first-shell scattering path of PdPtO_x/C was fit with a combination of Pt-O and Pt-Pd/Pt scattering paths. These EXAFS data suggest that blending Pt in the PdO structure induced a local coordination environment of Pt resembling that of PtO, which is unstable alone.

The mixing of Pd and Pt on the surface, matching the bulk composition, was confirmed through cyclic voltammetry (CV) and X-ray photoelectron spectroscopy (XPS) analyses. A single peak of surface oxide reduction (at ~ 0 V vs. Fc/Fc⁺) was observed in the CV of the PdPtO_x/C electrode, suggesting that Pd and Pt coexist on the catalyst surface (Fig. 2D and fig. S7). Furthermore, near-surface compositions derived from XPS survey spectra also indicate that Pd and Pt are present in

an equimolar ratio near the surface (table S7), matching the bulk composition determined using inductively coupled plasma-optical emission spectrometry (table S8).

XPS analysis showed that alloying Pt with Pd stabilizes oxidized Pt on the catalyst surface. Pt(0) species were dominant in the PtO_x/C , while a higher content of Pt(II) species was observed in PdPtO_x/C (Fig. 2F). The ratio of Pt(II):Pt(0) increased from 1.0 to 6.4 as the catalyst treatment temperature was increased from 400 °C to 500 °C (table S9). A small amount of Pd(0) was present in PdPtO_x/C annealed at 400 °C, while Pd(II) was exclusively observed in the catalyst annealed at 500 °C (Fig. 2E). The epoxidation rate and selectivity also improved with the treatment temperature for PdPt/C (fig. S8-9), indicating that the higher ratio of oxidized Pd and Pt species is contributing to the enhancement in the catalyst performance.

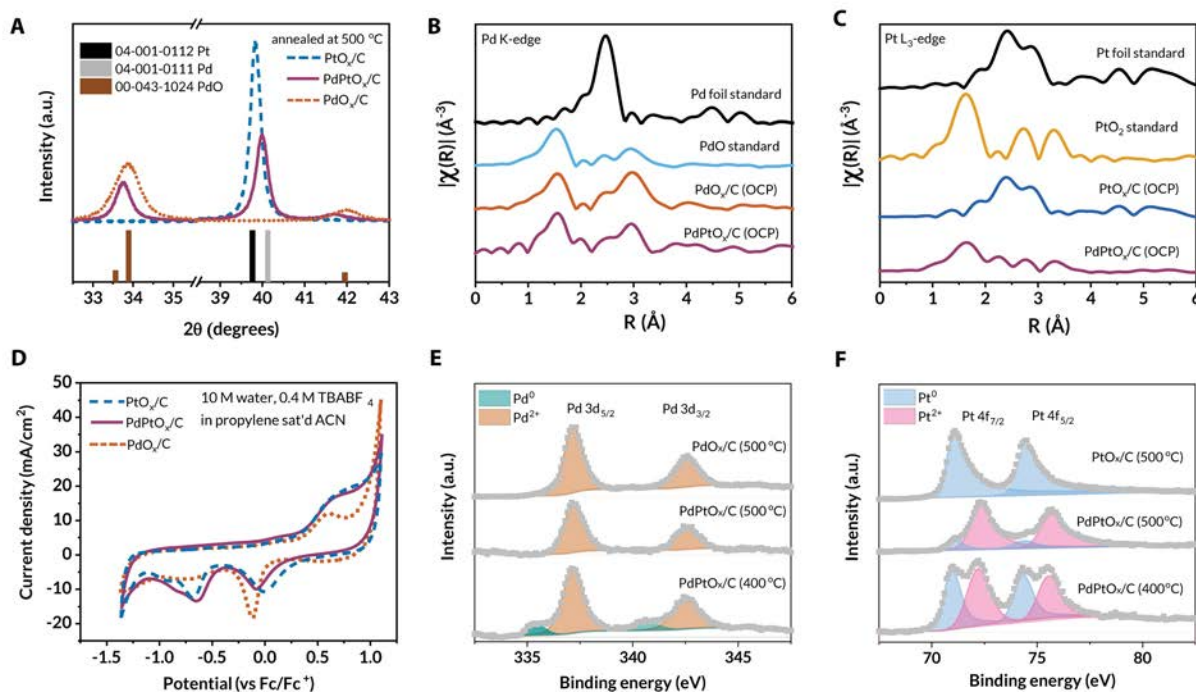


Fig. 2. Ex-situ characterizations of annealed PdPt/C, Pd/C, and Pt/C catalysts. (A) Magnified XRD patterns around the peak associated with the oxide and metallic phases of annealed catalysts at 500 °C. (B and C) EXAFS spectra of standards and catalysts at (B) Pd K-edge and (C) Pt L₃-edge. (D) Cyclic voltammetry traces of the catalysts. The scans were recorded at 50 mV/sec scan rate with 85% automatic iR-compensation (i = current, R= resistance). (E) High-resolution Pd 3d XPS spectra of annealed Pd/C and PdPt/C catalysts. (F) High-resolution Pt 4f XPS spectra of annealed Pt/C and PdPt/C catalysts.

Operando x-ray absorption spectroscopy (XAS) experiments probing the Pd K-edge and Pt L₃-edges during epoxidation highlighted again that achieving more oxidized Pt may have a critical role in improving catalytic activity toward propylene epoxidation. Progressively more oxidative potentials were applied from 0.45 V to 1.15 V vs. Fc/Fc⁺, and then a reductive potential of -1.35 V followed by oxidative potentials back to 1.15 V were applied during the measurements. The average metal valency at each condition was estimated from linear combination fitting of x-ray absorption near edge structure (XANES) spectra using reference spectra from standard materials. The results indicate that Pd oxide stabilizes oxidized Pt species in Pd-Pt alloys, leading to a higher average oxidation state of Pt in PdPtO_x/C (+2), compared to that in PtO_x/C (0). This difference is maintained under anodic potentials (0.45-1.15 V vs. Fc/Fc⁺, Fig. 3A). Unlike the stark contrast in Pt valency between PdPtO_x/C and PtO_x/C , the Pd valency stayed the same (+2) in both PdPtO_x/C

and PdO_x/C (Fig. 3B). These observations suggest that the more oxidized Pt species contribute to the enhancement in the production rate and selectivity of PO.

Furthermore, EXAFS fitting of catalysts under operating conditions showed that coordination numbers and scattering path lengths of Pd and Pt did not change during electrolysis (table S5-6). Applying a cathodic potential followed by an anodic potential demonstrated that the edge energies and the coordination environment of Pd and Pt respond to applied potentials in a partially, albeit not fully, reversible fashion (Fig. 3, fig. S10-11).

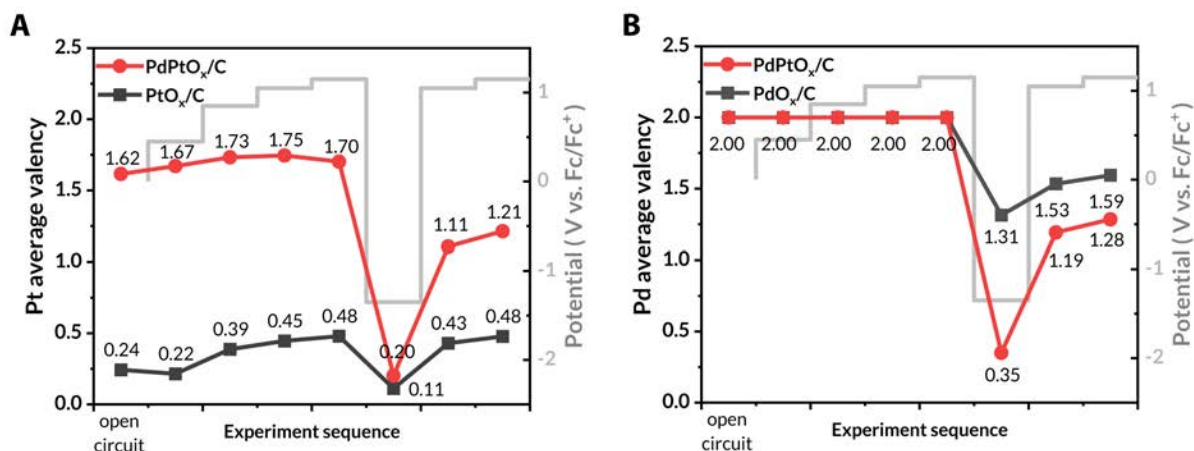


Fig. 3. Operando X-ray absorption spectroscopy. (A) Changes of Pt average valency in PdPtO_x/C and PtO_x/C in responses to applied potentials. (B) Changes of Pd average valency in PdPtO_x/C and PdO_x/C in responses to applied potentials. Potentials in this figure were 100% iR-compensated (i = current, R = resistance).

In summary, the results from XRD, CV, and the elemental analysis collectively suggest that Pd and Pt are mixed in both the bulk and surface of the PdPtO_x/C catalyst. XANES and XPS results revealed that a larger quantity of oxidized Pt species is stabilized in PdPtO_x/C compared to PtO_x/C. XAS data suggest that Pd stabilizes Pt at a similar oxidation state (around +2) and crystal structure as PdO, which together improve catalytic performance.

Electrochemical kinetic study and proposed reaction mechanism

Understanding the mechanism of direct propylene epoxidation on PdPtO_x/C can help us develop more efficient catalysts and process conditions for producing PO. The reaction mechanism was investigated using a multi-faceted approach including kinetic rate measurements, deuterium kinetic isotope effect measurements, probe substrate analysis, and substrate-based descriptor assessment.

Electrochemical kinetic rate measurements were conducted to examine electrolyte species dependences for direct anodic epoxidation on PdPtO_x/C (Fig. 4, fig. S12-13). The reaction rates, represented by the partial current density toward epoxidation, were measured using chronoamperometry in conjunction with product quantification by ¹H-NMR (Supplementary Materials Methods).

We hypothesized that the direct alkene epoxidation pathway at the anode may involve multiple coupled proton-electron transfer steps from water to the electrode, followed by oxygen-atom transfer from intermediates typically seen in water oxidation to propylene. O₂ was observed to be

the major byproduct with the PdPtO_x/C catalyst, while a small amount of CO₂ was also observed (fig. S14). It is worth noting that the selectivity of PO over other propylene-derived products (propylene glycol, acetone, and CO₂) reaches 92%. Therefore, in our proposed mechanism, only the competition between epoxidation and oxygen evolution was considered.

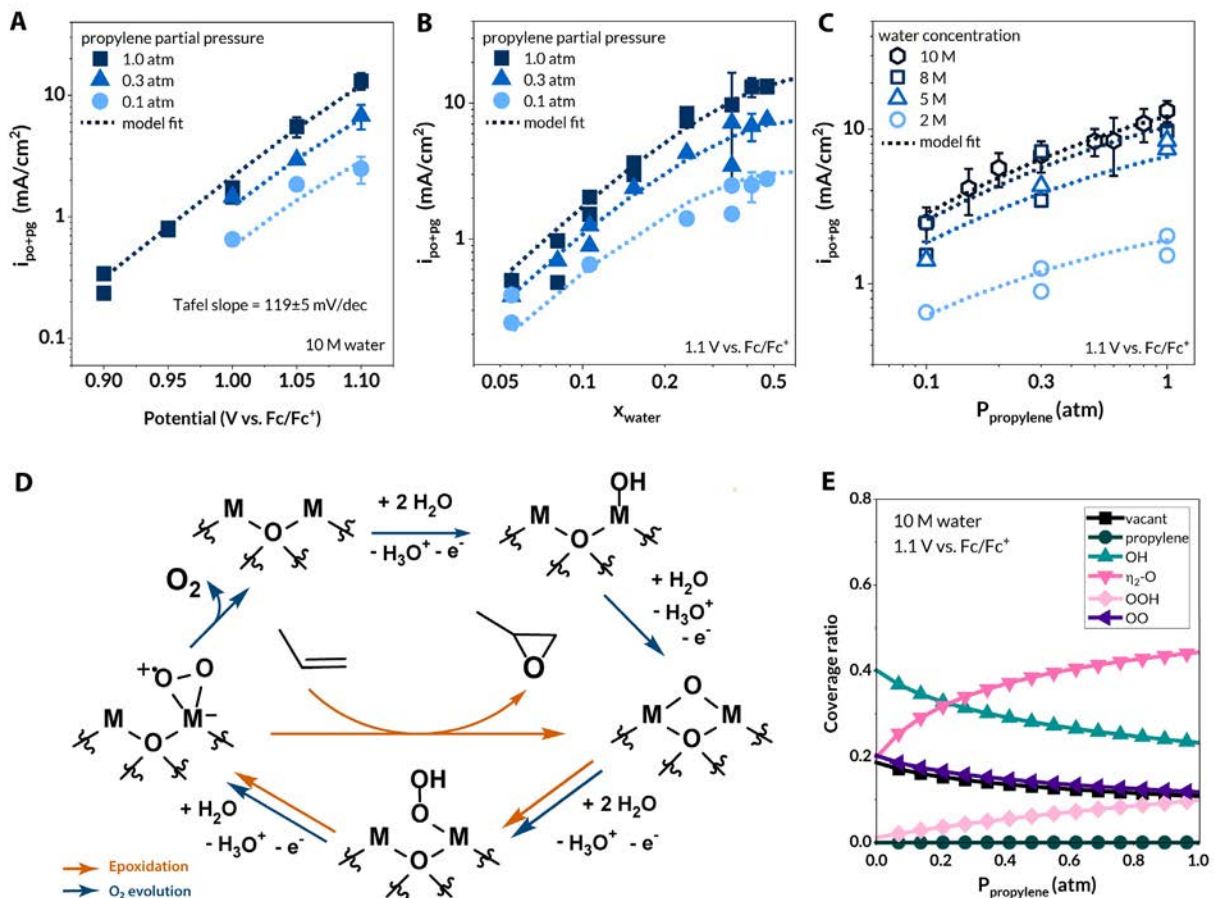


Fig. 4. Kinetic data on direct anodic epoxidation with PdPtO_x/C catalysts. (A) Anodic potential dependence of current density at 1 atm propylene in a 10 M water, 0.4 M tetrabutylammonium tetrafluoroborate (TBABF₄) acetonitrile solution (B) Water molar concentration dependences of propylene epoxidation current density at different propylene partial pressures. (C) Propylene partial pressure dependences of propylene epoxidation current density at different water concentrations. (D) A putative reaction mechanism including the direct anodic epoxidation and oxygen evolution pathways. (E) Simulated propylene partial pressure dependences of intermediate species coverage at 10 M water, 1.1 V vs. Fc/Fc⁺ condition. Potentials in this figure were 100% iR-compensated (i = current, R = resistance). Dotted lines in A–C represent best-fit from kinetic model of proposed mechanism in D (more details are shown in Supplementary Text S1, Case 2d). The vertical error bars represent standard deviations from the mean of multiple replicates (n=3) of the same experiment.

In cases where the surface coverage is primarily determined by a single reactive intermediate and a rate-limiting step is evident, the measured Tafel slope of 119 ± 5 mV/dec implies that the rate-limiting step involves a single electron transfer, and that no pre-equilibrated electron transfer steps occur between the abundant reactive intermediate and the rate-limiting step (Fig. 4A). Nevertheless, non-integer and non-linear reaction orders were observed in the water and propylene dependencies, signaling the presence of more intricate kinetic control.

At 1 atm propylene condition, the observed dependence of rates on water molar content exhibited $\sim 2^{\text{nd}}$ order behavior at lower water ratio, which decreased to $\sim 1^{\text{st}}$ order at higher water ratio (Fig. 4B and fig. S12). At lower propylene pressures, the water dependence showed more linear $\sim 1^{\text{st}}$ order dependence in the whole water concentration range. For propylene, the rate showed 1^{st} order dependence on propylene activity at lower propylene pressures, but the dependence decreased at elevated propylene pressures (Fig. 4C and fig. S13).

To address the complexity arising from non-integer reaction orders and variations in orders with reactant concentrations, we implemented a multiparameter kinetic model and fitted it to the kinetic data collected over a wide range of experimental conditions. We initially hypothesized that surface-adsorbed hydroperoxo (OOH^*), peroxo (OO^*), or oxo (O^*) species might serve as the reactive oxygen species for epoxidation. We then proposed various elementary steps and corresponding rates for each hypothesis (Supplementary Text S1). Given that acetonitrile is a less effective proton acceptor than water, we suggested elementary steps where two distinct water molecules play explicit roles in the reaction, one as a reactant and the other as a proton acceptor. Then, we employed kinetic modeling to deduce which intermediate would most likely react with propylene. This approach was viable as the reaction of different species (OOH^* , OO^* , and O^*) with propylene generates distinct surface intermediates following epoxidation (OH^* , O^* , and $*$, respectively). In contrast, the evolution of O_2 invariably leaves vacant ($*$) surface sites. The goodness-of-fit of model was evaluated by the root-mean-square-error (RMSE) between the experimental data and the simulated data from the model, with the lowest value indicating the best-fit model.

Our experimental data best matched the mechanism suggesting OO^* as the key intermediate for epoxidation, with an RMSE of 0.48 mA/cm^2 . Diverse derivatives of this mechanism involving the OO^* epoxidation intermediate are plausible, dependent on whether a dual or single site is presumed for OO^* and O^* intermediates. We selected the model with the lowest RMSE as our representative mechanism (Fig. 4D), and the model provides a quantitative explanation of the data illustrated in Fig. 4A-C. We also simulated surface coverage using the model (Fig. 4E, Supplementary Text S1). At higher propylene pressures where O^* and OH^* are abundant, the overall epoxidation rate would be largely determined by the rate of water oxidation to form OOH^* . However, the comparable prevalence of different surface oxygen intermediates underscores the necessity for a comprehensive kinetic model that does not assume a single dominant reactive intermediate and a rate-limiting step.

Kinetic isotope labeling and stilbene control experiments further support our hypothesis asserting OO^* as the key intermediate in epoxidation. The use of deuterated water lowered the epoxidation rate ($k_{\text{H}} / k_{\text{D}} = 1.8 \pm 0.5$; fig. S15 and table S10) while the Faradaic efficiency remained similar. The difference observed in the epoxidation rate is consistent with the kinetically-relevant water oxidation steps involving O-H cleavages in our proposed mechanism. Moreover, our mechanism identifies OO^* as the branching intermediate for both epoxide and O_2 evolution (Fig. 4D), providing an explanation for the marginal selectivity difference, since neither of these pathways from OO^* involves O-H cleavage. In addition, cis-stilbene was used as a probe substrate to help determine whether epoxidation proceeds through a concerted mechanism with OOH^* or a stepwise pathway with OO^* (25, 26). Cis-stilbene (96%) epoxidation with PdPtO_x/C produced 23% cis-stilbene oxide and 77% trans-stilbene oxide, which suggests that alkene epoxidation over PdPtO_x/C occurs through a stepwise mechanism (fig. S16) with OO^* reactive intermediates.

Correlation between alkene electrophilicity and epoxidation rate

Additional insights into direct anodic epoxidation on PdPtO_x/C were obtained from evaluating the correlation between the electrophilicity of alkene substrates and the epoxidation rate. Previous studies on thermochemical epoxidation suggest that either electrophilic (27–29) or nucleophilic (30, 31) epoxidation can be a reasonable reaction pathway for electrochemical propylene epoxidation on late-transition metal catalysts (Supplementary Text S2). By understanding the electronic nature of the oxygen transfer during the reaction, we can enhance our understanding of the mechanism and guide future catalyst development.

The global electrophilicity index (ω) can be calculated by taking the square of alkene chemical potential divided by chemical hardness, and is expressed in units of energy (32, 33). A higher value of ω indicates that the molecule is more electrophilic. Furthermore, the Fukui functions can be used to determine the local reactivity of each atom in a molecule by predicting the redistribution of electron density during electrophilic, nucleophilic, and radical attacks. The local electrophilicity (ω^-) at each atom center can be obtained by multiplying the global electrophilicity by the value of the Fukui function for the removal of an electron.

The average ω^- of the vinyl carbons as a substrate-based descriptor was assessed against the epoxidation rates of a wide range of alkene substrates. A total of 11 distinct aliphatic alkenes with various structures and functional groups were selected, with ω^- values ranging from 12–44 kJ/mol. In order to ensure consistent reactant chemical potential across different substrates, the gaseous substrates were fed until saturation, and the liquid substrates were added to the electrolyte near the phase-separation point, corresponding to approximately 0.2 M substrate for a 0.11 M tetrabutylammonium tetrafluoroborate (TBABF₄), 10 M water in acetonitrile (table S11). The substrates were tested at the same potential referenced to the ferrocene redox couple (1.05 V vs. Fc/Fc⁺).

A negative correlation between the logarithm of rate and the alkene electrophilicity was observed, with more nucleophilic alkenes displaying higher reactivity towards epoxidation (Fig. 5). The results suggest that the epoxidation may proceed through an electrophilic attack by oxygen species, and that the energy barriers to this attack are lower for more nucleophilic alkenes. Furthermore, the correlation suggests that a frontier-orbital interaction between the LUMO of the oxygen species and the HOMO of the alkene modulates the reactivity.

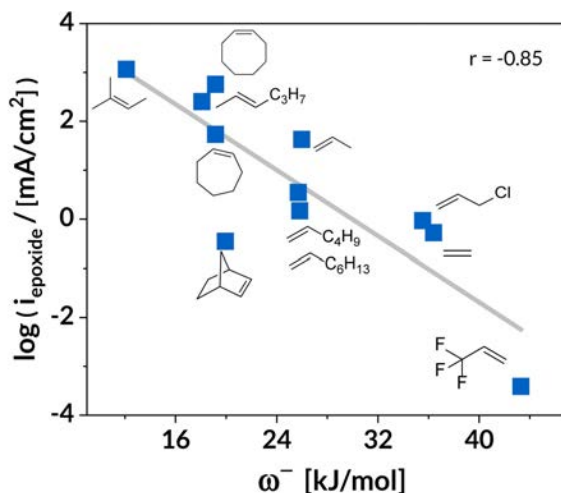


Fig. 5. Correlation between the alkene electrophilicity index and the direct anodic epoxidation rate. Pearson's r value = -0.85.

Electrolyte choice and the effect of solvent medium on the reaction kinetics

The successful operation of direct epoxidation via water activation hinges on whether the electrolyte system is compatible with a particular combination of catalyst and alkene substrate. Water-acetonitrile blended electrolytes were employed to investigate the epoxidation of various alkene substrates, since the acetonitrile co-solvent enhances the solubility of hydrophobic alkene substrates in the water-rich phase. However, the use of an organic solvent can lead to an increase in the ohmic resistance throughout the electrolyte (fig. S4). The use of water alone as the solvent for the electrolyte, without an organic co-solvent, would be ideal as water is required as a reactant and able to effectively solvate electrolyte ions, minimizing the ohmic resistance of the overall electrolyte. For gaseous substrates that are sparingly soluble in water, like propylene, gas-diffusion electrodes (GDEs) could be utilized to reduce the distance that reactant gases must diffuse through the electrolyte to reach the catalyst, and therefore aqueous electrolytes would be sufficient for epoxide production. Indeed, the PdPtO_x/C catalyst showed ~60% FE toward propylene epoxidation at 1.65 V vs. SHE in a sodium phosphate aqueous electrolyte when incorporated into a carbon-paper GDE with Nafion binder (Fig. 1E and fig. S4). Adding the ionomer onto the electrode was critical to achieve epoxidation performance commensurate with water-acetonitrile electrolytes, as the hydrophobic backbone of the ionomer generates partially wetted channels that are effective for transport of the reactant gases to the catalyst.

The microenvironment of electrolytes also affects the conversion of epoxides into glycols. A noticeable amount of propylene glycol (PG) was observed in the aqueous propylene epoxidation (~10% FE), higher than observed in the blended water-acetonitrile electrolyte. Furthermore, using 0.1 M HClO₄ or any acidic electrolyte results in the exclusive formation of PG, with no detected PO (table S1). As epoxides are known to undergo acid-catalyzed hydrolysis to afford glycols, it is imperative to find a pH regime where hydrolysis is minimized to ensure high selectivity toward the product of interest in aqueous electrolytes. The reaction rates of PO to PG in aqueous electrolytes with pH 4 to 7 were measured using ¹H-NMR; the hydrolysis of PO to generate PG was found to be significantly faster in the pH 4 buffer solution with almost 40% consumption of PO after 7 hours whereas the hydrolysis in less acidic solutions, at pH 5-7, was largely suppressed with less than around 10% PO to PG conversion (fig. S18). Thus, maintaining a near-neutral pH in the aqueous electrolyte is essential to optimize PO production.

The practical industrial operation of this system will likely require the use of a membrane-electrode assembly (MEA), resembling a water electrolyzer with humidified propylene gas fed to the anode. The MEA is advantageous for minimizing ohmic loss and separating PO from liquid electrolytes. However, our analysis on the influence of pH on the PO hydrolysis rate suggests that minimizing the interfacial pH gradient at the anode would be the key challenge for maximizing PO production over PG in MEAs operated at high current densities.

As long as the issues of mass-transport limitation and hydrolysis of PO are addressed, mechanistic insights on propylene epoxidation obtained in water-acetonitrile electrolyte may be extended to aqueous systems. The investigation on water activity in water-acetonitrile electrolyte suggests that aqueous electrolyte may be viewed as an extreme case of the water-acetonitrile electrolyte with the water mole ratio approaching unity. In the water-acetonitrile electrolyte system, positive deviation from Raoult's law (fig. S17) indicates that the water in this electrolyte prefers self-

interactions over mixing, meaning that the activity coefficient of water in this case is greater than unity ($\gamma_{\text{water}} > 1$), and that it approaches unity with increasing water concentration. The comparison between the rate order with respect to water activity and water concentration provides information about the activity coefficient of the transition state ($\gamma_{\ddagger}$) as well (Supplementary Text S4). At lower water concentrations (1-5 M), the propylene epoxidation rate showed $\sim 4^{\text{th}}$ order dependence on water activity while $\sim 2^{\text{nd}}$ order dependence on water mole ratio was observed. The 2^{nd} order dependence is consistent with our hypothesis that the predominant rate determining step leading to the formation of OOH^* involves a first water molecule that is a reactant and a second water molecule that is a proton acceptor. The abnormally high apparent 4^{th} order dependence on water activity suggests that the stabilization of the transition state may be achieved by augmenting the water content. However, $\sim 2^{\text{nd}}$ order dependences in both water activity and concentration were observed at higher water concentrations (5-12 M). This can be rationalized if $\gamma_{\ddagger}$ scales with γ_{water} at low water concentrations but remains relatively unchanged at higher water concentrations. This hypothesis aligns with our chemical intuition: once a sufficient quantity of water is present around the transition state, adding more water does not significantly alter the near-transition-state chemical environment. Therefore, once the water concentration exceeds 5 M in acetonitrile, the chemical environment at the active site should be akin to that of an aqueous electrolyte. In this case, the transition from water-acetonitrile electrolyte to aqueous electrolyte may be a simple extrapolation of water mole ratio to one, without a discontinuous change in the solvent properties. This interpretation is supported by the measured reaction rates in aqueous electrolytes that is consistent with the predictions from the mechanistic model derived from kinetic studies in water-acetonitrile electrolytes (fig. S12).

Collectively, the results from multiple orthogonal analyses provide a comprehensive understanding of the mechanism of electrochemical epoxidation on PdPtO_x/C . Our findings are consistent with a mechanistic hypothesis that OO^* is the key species involved in the electrophilic epoxidation of propylene. The PdO crystal structure embedding Pt(II) species was found to be effective for catalyzing direct anodic epoxidation, suggesting that other materials with similar structures and stronger inductive effects on Pt may also produce effective catalysts by promoting more electrophilic epoxidation pathways. Additionally, we discovered through fundamental kinetic analysis that the identity of reactive oxygen species may vary between catalysts. For instance, the proposed reactive oxygen species for epoxidation on PdPtO_x/C and MnO_x -based catalysts are OO^* and O^* , respectively. Developing species that stabilize OO^* may enhance epoxidation on noble metals while avoiding oxygen evolution. Our work contributes toward advancing the sustainable synthesis of epoxides, which currently have significant energy and environmental footprints. The molecular-level understanding of direct anodic epoxidation obtained from this study may have implications to other electrocatalytic oxygen-atom transfer reactions driven by reactive water-activation intermediates.

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Author contributions:

Conceptualization: M.C., K.M.

Methodology: M.C.

Investigation: M.C., J.H.M., J.S.A.

Validation: M.C., J.S.A.

Visualization: M.C.

Funding acquisition: K.M.

Project administration: M.C., K.M.

Supervision: K.M., Y.R.-L.

Writing – original draft: M.C.

Writing – review & editing: M.C., J.H.M., J.S.A., C.J., Y.R.-L., K.M.

Competing interests: M.C. and K.M. have filed a provisional patent application (no. 63/544,664) regarding the design of catalysts and electrodes.

Data and materials availability: There are no restrictions on materials. All data needed to evaluate the conclusions in the paper are present in the paper or the supplementary materials. Tabular data underlying the figures are deposited in the Dryad data repository (34).

Supplementary Materials

Materials and Methods

Supplementary Text

Figs. S1 to S25

5 Tables S1 to S12

References (35–39)

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Supplementary Materials for

Direct propylene epoxidation via water activation over Pd-Pt electrocatalysts

Minju Chung, Joseph H. Maalouf, Jason S. Adams, Chenyu Jiang, Yuriy Román-Leshkov,
Karthish Manthiram*

Corresponding author: karthish@caltech.edu

The PDF file includes:

Materials and Methods
Supplementary Text
Figs. S1 to S25
Tables S1 to S12

Materials and Methods

Materials

Catalyst preparation

Palladium(II) chloride (PdCl_2 , Sigma Aldrich), Chloroplatinic acid hexahydrate ($\text{H}_2\text{PtCl}_6 \cdot 6\text{H}_2\text{O}$, Sigma Aldrich), sodium citrate dihydrate (Sigma Aldrich), Vulcan XC-72R (Fuel cell store, SKU:590106-1), Sodium borohydride (NaBH_4 , Sigma Aldrich), Hydrochloric acid (HCl , ACS reagent, 37%, Sigma Aldrich), ethanol (Anhydrous, KOPTEC USP), 5% Nafion 117 solution (Sigma Aldrich) were purchased and used without additional treatment.

Electrochemical study

Electrolyte / Calibration / Reference material/quantification

Tetrabutylammonium tetrafluoroborate (TBABF_4 , 98+%, TCI America), Acetonitrile (ACN, anhydrous, 99.8+%, Alfa Aesar), cis-cyclooctene oxide (99%, Sigma Aldrich), cyclooctanone (98%, Sigma Aldrich), 1,3,5-Trimethoxybenzene (TMB, 99 %, Sigma Aldrich), Deuterated chloroform (CDCl_3 , 99.8 atom % D, Sigma Aldrich), Deuterated acetonitrile (CD_3CN , 99 atom % D, Oakwood Chemical), Deuterium oxide (D_2O , 99.9 atom % D, Oakwood Chemical), Ferrocene(Fc, 99%, Alfa Aesar), Nitric acid (HNO_3 , 70% trace metal grade, Fisher Chemical), Hydrochloric acid (HCl , 36.5-38%, VWR BDH Chemicals) were purchased and used without additional treatment. Argon UHP grade 5.0 gas (Airgas, AR UHP300) was used to dilute the propylene stream for the rate-order dependence study.

Olefin substrates

cis-cyclooctene (cyclooctene, 95%, Alfa Aesar), cycloheptene (95%, Acros organics), 1-hexene (99+%, Sigma Aldrich), cis-2-hexene (95%, Sigma Aldrich), 3,3,3-trifluoropropene (99%, Oakwood Chemical), 2-cyclohepten-1-one (90%, Thermo Scientific), 2-cyclohexen-1-one (95+%, Sigma Aldrich), Propylene (Industrial grade, Airgas), Ethylene (UHP grade 3.0, Airgas)

X-ray absorption spectroscopy reference materials

Platinum foil (Pt, 0.025 mm thick, 99.99% trace metals basis, Beantown Chemical), Platinum(IV) oxide (PtO_2 , ≥ 80.0 wt% Pt, Sigma Aldrich), Palladium foil (Pd, 0.1mm thick, 99.9% trace metals basis, Beantown Chemical), and Palladium(II) oxide (PdO , 99.9 % Pd, Strem Chemicals) were purchased and mixed with poly(ethylene glycol) (average mol wt 10000, Sigma Aldrich) and pressed into pellets for transmission-mode measurements.

Methods

Catalyst preparation

PdPt/C synthesis

Pd precursor solution was prepared by adding 0.25 mmol PdCl_2 in 5 mL of 0.1 M HCl aqueous solution, and the solution was sitted overnight to ensure PdCl_2 is completely dissolved. Pt precursor solution was prepared by dissolving 0.25 mmol $\text{H}_2\text{PtCl}_6 \cdot 6\text{H}_2\text{O}$ in 5 mL Milli-Q water. When precursor solutions were ready, 4 mmol sodium citrate dihydrate was dissolved into 200 mL water in a 500 mL round-bottom flask. Pd and Pt precursor solutions were added to the sodium citrate solution while it was vigorously stirred (>1000 rpm) using a magnetic stir bar (egg-shaped, 38.1mm length x 15.9 mm diameter, PTFE). Then, 113.1 mg of Vulcan XC-72R carbon black powder was added to the solution to achieve 40 wt% metal on carbon. The amount of carbon black

was adjusted for making catalysts with different metal-to-carbon ratios. The flask was plugged with a rubber septum and then sonicated for 30 min. The chunks of carbon black were still observed after sonication. Then, the solution mixture was vigorously stirred at 1500 rpm, and freshly prepared 50 mL of 0.1 M NaBH₄ solution was added dropwise (1 mL/min) into the mixture using a syringe pump at room temperature (fig. S22). A needle was placed on the rubber septum to avoid pressure build-up inside the flask due to the evolution of H₂ from NaBH₄. After the suspension was stirred for 8 h including the NaBH₄ injection time, the black precipitate was vacuum-filtered with a copious amount of MilliQ water (>1 L). The filter paper with the washed precipitate cake was placed on a glass petri dish, covered with a piece of aluminum foil with holes, and dried overnight in a vacuum oven at 40 °C. The dried PdPt/C powder was collected from the filter paper, and stored in a borosilicate vial until being used for making catalyst ink. The recipe for alloys with other ratios of Pd and Pt are summarized in Table S12.

carbon paper electrode preparation

Hydrophobic carbon paper electrodes (Toray Carbon Paper 120, Wet-proofed, Fuel Cell Store) were punched in 14 mm-diameter disks. When the electrode was assembled into the electrochemical cell, only 1 cm² of the electrode was exposed to the electrolyte. The punched hydrophobic electrodes were annealed in a muffle furnace with a temperature ramp rate of 5 °C/min from room temperature to 600 °C, and dwelled for 1 hour to increase the wettability of the electrodes by the catalyst ink. The electrodes were naturally cooled to room temperature after the annealing. The resulting hydrophilic carbon paper electrodes were used as a substrate for catalyst drop-casting.

catalyst ink and catalyst-fixed electrode preparation

A known amount of metal-on-carbon catalyst powder was dispersed in a water-ethanol solution (water:ethanol=3:1 volumetric ratio) with the help of sonication. The amount of the catalyst powder was adjusted for the desired concentration of the ink. 30 µL, 30 µL, and 40 µL of the prepared ink were drop-casted on one side of a 14 mm-diameter hydrophilic carbon paper electrode in a drying oven at 80°C. The electrode was dried in a closed oven between each drop-casting. The total volume of drop-casted ink was always fixed to 100 µL, so the ink concentration was varied to achieve desired catalyst loading. A typical concentration of the catalyst ink was 10 mg powder/mL (4 mg metal//mL).

The electrode was then transferred to a muffle furnace. Ramp rates for a furnace were set to be 10 °C/min from room temperature to 400 °C or 500 °C and dwelled for 3 hours under stagnant air (kinetic studies and X-ray absorption spectroscopy: 400 °C, Figure 1 data: 500 °C). During the annealing process, a significant amount of the carbon support was combusted, resulting in a final carbon weight percent of 10.63% and 7.54% for catalysts annealed at 400 °C and 500 °C, respectively (table S8). The oxidized metal-on-carbon electrode was naturally cooled to room temperature in the furnace and used for electrochemical experiments. The prepared electrodes were used within one day after annealing.

For electrodes used in aqueous electrolytes, 50 µL of 2 wt% Nafion solution was drop-casted onto an electrode after annealing. Subsequently, the electrode was dried in an oven at 80°C for 3 hours. The 2 wt% Nafion solution was formulated by diluting a commercially purchased 5 wt% Nafion solution with water.

Catalyst characterization

ICP-OES analysis

Pd and Pt loadings in the PdPtO_x/C catalysts were determined using an inductively coupled plasma-optical emission spectrometer (ICP-OES; Agilent 5100 VDV) coupled with ICP Expert software. A few milligrams of the powder catalyst were mixed with 4 mL of 70% HNO₃ added to 1 mL of 37% HCl (inverted acid ratio of aqua regia formula) and digested overnight. The sample was then diluted into a 2% HNO₃ solution and filtered with a 0.22 μm polypropylene syringe filter (ThermoFisher Scientific, CH2213-PP). Pd (1000 ppm, 5% v/v HNO₃(aq), RICCA Chemical Company) and Pt (1000 ppm, 5% v/v HNO₃(aq), Inorganic Ventures) standards were also diluted into 2% HNO₃ solution to make the final concentrations of 0-100 ppm. Pd and Pt concentrations in the samples were quantified using the calibration curves generated from the standards.

X-ray photoelectron spectroscopy (XPS)

The electrodes were cut into small pieces and affixed to an XPS sample holder using conductive Cu tape. For the post-electrolysis samples, the electrodes were rinsed with acetonitrile several times and dried under ambient air overnight before attaching them to the tape. Then, the holder with samples was transferred to the XPS sample introduction chamber. XPS spectra were collected using a Physical Electronics Versaprobe II X-ray Photoelectron Spectrometer and the data were analyzed with Casa XPS software.

Powder X-ray diffraction (XRD) analysis

The X-ray diffraction patterns of the synthesized catalysts were obtained using an X-ray powder diffractometer (PANalytical X'Pert PRO) with Cu Kα radiation ($\lambda = 1.54056 \text{ \AA}$). For the powder XRD analysis, the synthesized catalysts were annealed in powder form in a ceramic bowl using the same annealing method employed for electrodes. The crystallite sizes (L) were determined using Scherrer equation:

$$L = \frac{K\lambda}{B\cos\theta}.$$

Scherrer constant (K) of 0.94, full-width-at-half-maximum (B) determined by fitting the diffraction pattern peaks to Gaussian function and subtracting instrumental broadening, $\lambda = 1.54056 \text{ \AA}$ for Cu Kα radiation, and Bragg angle (θ) were used for the analysis. L was determined for each peak and the average and standard deviation of multiple peaks were reported (table S3).

X-ray absorption spectroscopy (XAS)

Operando XAS spectra at Pd K-edge and Pt L₃-edge were collected in X-ray fluorescence mode at beamline 7-BM and 8-ID of the National Synchrotron Light Source II. The same set of electrochemical cells used for the kinetic studies was used, except for a backplate with a hole for the X-ray entrance. A Kapton polyimide film was placed between the window and electrode to prevent electrolyte leakage. Electrode samples were prepared on-site and used within one day after annealing. All XAS data were processed using the Athena software for background removal and normalization. The EXAFS data were further modeled and analyzed using the Artemis software. The amplitude reduction factor (S_0^2) for each metal edge was determined by fitting the reference material with known coordination numbers (table S4). The S_0^2 values were used in other simulations to estimate the coordination numbers of the samples.

TEM analysis

High-resolution elemental mapping of Pd-Pt alloy catalysts was obtained using energy-dispersive X-ray spectroscopy (fig. S2 and S3) with an aberration-corrected scanning transmission electron microscope (Titan Themis Z G3 Cs-corrected S/TEM, Thermo Fisher Scientific). To prepare the samples, catalyst particles on carbon paper electrodes were scraped off using a plastic spatula and then dispersed in ethanol via sonication. The resulting solution was sufficiently diluted such that there was no noticeable color in the ethanol solution. For analysis, 13 μ L of this diluted solution was drop-casted onto a carbon film-supported copper grid. The sample was then dried in an oven set at 80 $^{\circ}$ C overnight.

Electrochemical methods

Electrochemical cell setup and electrolysis methods

Electrochemical experiments were conducted with a sandwich-type electrochemical cell (fig. S23). Platinum foil (Pt, 0.025 mm thick, 99.99% trace metals basis, BeanTown Chemical) and metal-on-carbon loaded carbon paper were used as the counter and working electrode, respectively. An Ag/AgCl electrode (LF-2, 3.4 M KCl leak-free 2.0 mm diameter, Innovative Instruments) was used as the pseudo-reference electrode, and aluminum foil or plate was used as the current collector. Hydrophobic carbon paper was placed behind the nanoparticle-loaded carbon paper to prevent leakage of the solvent. For kinetic rate measurements and *operando* XAS, acetonitrile with 0.4 M TBABF₄ was used as the solvent with varying pressures/concentrations of propylene and water. The partial pressure of propylene was varied between 0 and 1 atm by diluting the propylene stream with argon to make the total gas flow is 10 standard cubic centimeters per minute (sccm). The electrolyte solution was pre-saturated with the gas before 3.6 mL of the solution was added to the cell for each experiment. For kinetic measurements, propylene gas has flowed with 10 sccm during the run as well. Reaction conditions for the substrate scope study is described in Table S11.

The electrochemical measurements were conducted with VMP3 Multi-channel potentiostat from BioLogic. The resistance value at open circuit potential (OCP) was measured by electrochemical impedance spectroscopy techniques before the start of any electrolysis, and the potentials in the following experiments were 100% iR-compensated. In most of the experiments for kinetic study, 10 coulombs were passed unless noted otherwise.

All cell parts including the counter electrode were washed with acetone and MilliQ water after every experiment and sonicated for at least 20 minutes in 10-20% nitric acid aqueous solution to remove any metal contaminants. The cell parts were washed with MilliQ water again after the sonication and dried in 80 $^{\circ}$ C oven for at least 20 minutes before use.

Reference electrode calibration

All of the anodic potentials reported for the water-acetonitrile electrolyte are referenced to the redox potential of ferrocene (Fc/Fc⁺). There is no physical device that can act as a ferrocene redox reference electrode, so a leak-free Ag/AgCl pseudo-reference was used in actual experiments, and this pseudo-reference was calibrated against the redox potential of ferrocene. The redox potential was found by averaging the potential at the cathodic and anodic peaks shown in cyclic voltammetry (CV) scans. Calibration solutions were roughly 5 mM in ferrocene and reflected electrolyte conditions other than that. For the reference calibration, glassy carbon was used as the anode, and the calibrations were performed in a dedicated set of PEEK cell parts to avoid ferrocene contamination in other electrochemical experiments. The CV started from open circuit potential and swept to 0 V vs. Ag/AgCl pseudo-reference, followed by three consecutive

scans from 0 – 0.8 V vs. Ag/AgCl pseudo-reference, and the CV ended after sweeping back to the open circuit potential. The second scan from each CV was used for determining the redox peaks. The example of ferrocene redox calibration different electrolytes with different water concentrations were shown as below (fig. S24). It should be noted that the pseudo-Ag/AgCl electrode is meant to be stable at Ag/AgCl potential in saturated KCl, but its actual potential showed some discrepancy, usually within 50mV error, when its potential was measured against rigorously maintained “master” Ag/AgCl electrode in saturated KCl.

Double-layer capacitance measurement

We used double-layer capacitance (DLC) measurement to estimate the electrochemically active surface area (ECSA) across our electrodes:

$$\text{Electrochemically active surface area (ECSA [cm}^2\text{/cm}^2_{\text{geo}}\text{])} = \frac{\text{double-layer capacitance (Cd [mF/cm}^2_{\text{geo}}\text{])}}{\text{specific capacitance (Cs [mF/cm}^2\text{])}}.$$

CV scans spanning ± 50 mV of the OCP for each DLC measurement were recorded at scan rates of 20, 40, 80, and 120 mV/s. The resulting capacitive CV traces (fig. S25) were used to calculate the capacitive current as the difference of anodic and cathodic currents averaged across ± 10 mV of the OCP. These capacitive currents at each scan rate were plotted versus the scan rate, and the slope of the linear regression line was divided by 2 to obtain the DLC of the electrode. Since the entire surface of the carbon paper exposed to the electrolyte was covered with the catalysts, we assumed that the bare carbon paper did not contribute to the DLC. It is a common practice to assume a constant specific capacitance across various metals, which makes DLC directly correlated to ECSA. However, it is important to note that the constant specific capacitance used for converting DLC to ECSA varies across different reports, ranging from 0.01 to 0.1 mF/cm² (35, 36). Therefore, we opted to present DLC values along with ECSA-normalized current densities, assuming specific capacitance value of 0.04 mF/cm² (fig. S5).

Product analysis

Reaction workup and quantification of liquid products

After the electrolysis, post-electrolysis electrolytes were collected into a 5 mL volumetric flask using a glass pipette. The collected electrolyte was often less than 3.6 mL, so ~ 1.8 mL of additional acetonitrile was added to the electrolyte chamber, rinsed the chamber, and then added to the post-electrolysis electrolyte in the volumetric flask until the total volume of collected electrolyte becomes 5 mL. For propylene oxide or ethylene oxide quantification, securing a product peak at ¹H-NMR without overlapping other electrolyte peaks was possible, so an aliquot of 5 mL electrolyte solution was directly used for the NMR study. NMR samples were prepared by combining 500 μ L of the electrolyte solution with a 200 μ L of 5 mM internal standard 1,3,5-trimethoxybenzene solution in CD₃CN. The ¹H probe was automatically tuned, the lock was achieved on the CD₃CN in the sample, and gradient shimming and auto-gain were used. Solvent suppression was done for the largest three peaks in the spectra to get a better signal-to-noise ratio for the peaks of our interest. 32 scans were collected per sample, and the relaxation delay was set to 6 seconds. Products were quantified by comparing the areas of the peaks of interest to those of the internal standard, 1,3,5-trimethoxybenzene. Auto phase correction and baseline correction with Whittaker Smoother method from the MestReNova software was performed on the spectra before the integration of each peak. If the baseline was not achieved by the automatic correction, manual correction was used to get a reliable baseline for the peak integration.

For other epoxides including cyclooctene oxide, extraction with hexane was required to exclude TBABF₄ that can overlap with the products on ¹H-NMR spectra. We sampled 3 mL out of the collected 5 mL electrolyte into a 50 mL polypropylene centrifuge tube. Then, a known amount of internal standard 1,3,5-trimethoxybenzene were added to the electrolyte. MilliQ water was filled to ~ 20 mL and ~ 5 mL of hexane was added to the electrolyte. Olefin substrate, its oxidation products, and the internal standard moved to the hexane layer after extraction with the help of centrifugation at 3000 rpm for 30 seconds. Hexane extraction was performed three more times (~ 20 mL of hexane was used in total), and the combined hexane solution was quantified using ¹H-NMR. The ¹H-NMR spectra were measured on a Bruker Avance Neo spectrometer operating at 500.34 MHz with solvent suppression of the three largest peaks from hexane. Samples were prepared by combining 50 µL of the extracted hexane layer with a 600 µL CDCl₃. The ¹H probe was automatically tuned, the lock was achieved on the CDCl₃ in the sample, and gradient shimming and auto-gain were used. Solvent suppression was done for the largest three peaks in the spectra to get a better signal-to-noise ratio for the peaks of our interest. 32 scans were collected per sample, and relaxation delay was set to 6 seconds. Products were quantified by comparing the areas of the peaks of interest to those of the internal standard, 1,3,5-trimethoxybenzene. Auto phase correction and baseline correction with Whittaker Smoother method from the MestReNova software was performed on the spectra before the integration of each peak.

Quantification of gas products

An on-line gas chromatograph (GC; SRI Instruments, MG #5, Model 8610C) with auto-sampling capabilities was used to quantify the gaseous anodic side products CO₂, O₂, and the cathodic reaction product H₂ evolved during the electrolysis. Only several representative experiments were done with on-line gas product quantification to verify closure in Faradaic efficiency. After filling the electrolyte, the electrolytic cell was closed and the gas outlet was connected to the GC. Propylene was flowed into the cell through the anode for roughly 10 minutes before the measurement to allow air to flush out of the gas lines. The GC sample was taken every 13 minutes (12.5 minutes of sequence + 0.5 minutes of break). A blank sample was taken in the beginning, and then the electrolysis started after 4 minutes passed. GC sample was taken every 13 minutes (12.5 minutes of sequence + 0.5 minutes of break) thereafter. The gas products were quantified by averaging data points except for the blank sample.

After a sample was collected, argon gas was used as the carrier gas and sent onto a 6' HayeSepD pre-column. After light gases pass the pre-column, we started backward flowing in the pre-column and the eluted forward flow was sent to a 6' MS-5A column. After hydrogen and oxygen are eluted from the MS-5A column, another sample was collected and injected into a 6' HayeSep D column. The eluted gas after the main columns (MS-5A, 6' HayeSep D) was analyzed with both a thermal conductivity detector (TCD) and a flame ionization detector (FID). The column temperatures were held at 90 °C throughout the measurements. The gas-phase ppm determined from GC was converted into the partial current for a given product and Faradaic efficiencies were calculated from the partial current divided by the average current.

$$\text{Partial current [mA]} = (\text{ppm}) \times \frac{\frac{\text{moles product}}{\text{total moles}}}{10^6 \text{ ppm}} \times \frac{1 \text{ total moles}}{24.5 \text{ L at RTP}} \times \frac{1 \text{ L}}{10^3 \text{ cm}^3} \times \frac{(\text{flow rate in sccm}) \text{ cm}^3}{\text{min}} \times \frac{1 \text{ min}}{60 \text{ sec}} \times \frac{n \text{ moles } e^-}{1 \text{ mole product}} \times \frac{96485 \text{ C}}{1 \text{ mole } e^-} \times \frac{1000 \text{ mA}}{\frac{1 \text{ C}}{\text{sec}}}$$

Calculation of rate, faradaic efficiency, and selectivity

Faradaic efficiencies were calculated as following:

$$\text{Faradic efficiency [\%]} = \frac{\text{moles of product} \times nF}{Q}$$

Q is the amount of passed charge, n is the number of electrons required for the production of a molecule of product, and F is Faraday's constant. n=2 for propylene oxide, propylene glycol, and acetone. n=4 for oxygen, and n=6 for CO₂.

The epoxidation rates in current density were calculated by multiplying the total average current density by the epoxidation Faradaic efficiency:

$$\text{partial current density} = \text{total average current density} \times \text{Faradaic efficiency}.$$

The selectivity for PO was determined by dividing the molar amount of PO by the total molar amount of products derived from propylene (PO, propylene glycol, acetone, CO₂).

Experimental measurements of water activity

The activity of water was experimentally measured (fig. S17) using previously reported headspace gas sampling protocol (37). In short, 4 mL of electrolyte samples were prepared in 20 mL headspace sampling vials. The vials were filled with propylene gas streams (0.1 to 1 v/v diluted with Ar) and the electrolyte solution was pre-saturated with the gas before the vials were sealed. The electrolyte samples in sealed vials were equilibrated at room temperature for at least 12 hours. 4 mL of pure MilliQ water was used as the reference having activity of unity. Headspace GC-TCD was then used to quantify the vapor pressure of water in the various samples. The activity of water for any given sample was simply calculated as:

$$\text{water activity} = \frac{\text{water peak area of sample}}{\text{water peak area of pure water reference}}.$$

Rate measurements of propylene oxide hydrolysis in aqueous electrolytes

The reaction rates of PO to PG in aqueous electrolytes with pH 4 to 7 were measured using ¹H-NMR. The 1 M buffer solutions were prepared by dissolving the theoretically determined amounts of Na₂HPO₄ and NaH₂PO₄ in water and adjusted to the desired pH value with 1 M NaOH and 1 M H₃PO₄ solutions. To prepare the NMR samples for the kinetic measurement, 400 μL of the buffer solution, 50 μL of 0.1 M PO in water were combined, resulting in 10 mM of PO in each sample. An automation program was set up using IconNMR to collect NMR spectra of the samples at the specific time points – 30-min increment for the first 3 hours and 1-hour increment for the rest 4 hours. Water suppression was done by excitation sculpting to generate NMR spectra with minimal residual water signal and flat baseline, ensuring the accuracy of the quantification. The consumption of PO was normalized by taking the ratio between the peak area of the protons corresponding to the methyl group in PO and the sum of peak areas of that in PO and PG.

Model fitting

Seven potential mechanistic pathways were hypothesized (Supplementary Text S1; Case 1, 2a-e, and 3), each represented by a distinct rate equation. Inputs to the rate equation model were propylene partial pressure normalized by 1 atm, calculated water molar ratio, and applied voltage (V vs. Fc/Fc⁺). In the model, 0 V for the applied voltage was arbitrarily set to 0.85 V vs Fc/Fc⁺. Although any reference for 0 V could have been adopted, it is judicious to select a value relevant to the experimental voltage range (0.9-1.1 V vs. Fc/Fc⁺) to ensure optimal numerical precision. Transfer coefficients were set to 0.5 for all of the elementary electrochemical reactions. Depending on the model, 6-8 parameters were estimated by model fitting over 74 sample data.

Potential-independent rate constants and equilibrium constants of each elementary reaction were parameterized, and the parameters were estimated using a least-squares curve fitting function (lsqcurvefit) in MATLAB R2023a. The model rate equations were fit to the experimental epoxidation current density data in mA/cm² unit, which is proportional to the epoxidation rate.

Since other model inputs are unitless, the estimated rate constants for each elementary step also have units of mA/cm². Initial guesses for the parameters were chosen to ensure that the resulting final rate constants across a series of elementary reactions are within the same order of magnitudes when integrated with potential-dependent terms. Lower bounds for all parameters were set to zero.

Root-mean-square error (RMSE) was obtained from the difference between experimental and simulated data. Lower RMSE means less deviation of the simulated results from the experimental results. When different initial guesses resulted in different fitting results, the best fit corresponding to the lowest RMSE was selected and reported.

Alkene electrophilicity calculation

For each alkene, three different types of calculations were performed with different numbers of electrons, which are required to calculate the ionization potential and electron affinities. First, the N=0 (number of electrons for zero-charge and multiplicity of 1) case was considered. In this case, for each alkene, the initial starting geometry was obtained by sampling 100 conformers using the RDKit library, followed by further optimization using RDKit's implementation of the MMFF94s force field. Then using Gaussian 16, a geometry optimization, frequency calculation, and population analysis (charge =0, multiplicity=1) were performed at the M062X/def2-SVPD level of theory with acetonitrile as the implicit solvent implemented by the SMD algorithm. Calculations were distributed across Intel Xeon Platinum 8260 nodes provided by the MIT Super Cloud.

Next, the N+1 case was considered. Here, we use the geometry-optimized structure from the N case and perform a single-point energy calculation where the system has one more electron than the N case. In other words, charge =-1 and multiplicity =2. Similarly, for the N-1 case, the geometry optimized structure from the N case was used to perform a single point energy calculation where the system has 1 less electron than the N case (charge =1 and multiplicity =2). For each alkene, the electron affinity and ionization potential were calculated using the free energies by:

$$IP = E_{N-1} - E_N$$

$$EA = E_{N+1} - E_N$$

These quantities were used to calculate the chemical potential and chemical hardness as follows:

$$\mu = IP - EA$$

$$\eta = - \frac{(IP + EA)}{2}$$

From these two values, we were able to calculate the global electrophilicity index ω using the relation:

$$\omega = \frac{\mu^2}{2\eta}$$

Formally, the Fukui functions can be defined as the spatially dependent change in electron density as a function of the number of electrons N at a constant potential $v(r)$:

$$f = \left(\frac{\partial \rho(r)}{\partial N} \right)_{v(r)}$$

Based on this, the Fukui function for a discrete change in the number of electrons can be defined:

$$f^- = \rho_N(r) - \rho_{N-1}(r)$$

This definition is then typically calculated using a population analysis with the following:

$$f_k^- = q_k(N) - q_k(N-1)$$

Where q_k is the gross population on atom k. Gross populations were calculated using the cclib library using the Lowdin population analysis. Mulliken charge analysis was also performed, but we found Lowdin population analysis performed better. Finally, the local electrophilicity index was calculated by $\omega^- = \omega f_k^-$.

Supplementary Text

S1. Discussion on the propylene epoxidation mechanism

We enumerated a series of mechanisms invoking OOH* (Case 1), OO*(Case 2a-e), or O*(Case 3) as the key epoxidation intermediates:

index	Key features	Fitted number of parameters	RMSE
Case 1	OOH* catalyzes epoxidation, single site mechanism	6	5.8864e-01
Case 2a	OO* catalyzes epoxidation, single site mechanism	7	4.9425e-01
Case 2b	OO* catalyzes epoxidation OOH*+O* $\rightarrow$ OO*+OH* rearrangement included	8	5.1518e-01
Case 2c	OO* catalyzes epoxidation, dual site mechanism η_2 -O* and η_2 -OO* intermediates	7	4.9870e-01
Case 2d	OO* catalyzes epoxidation η_2 -O* and η_1 -OO intermediates	7	4.8398e-01
Case 2e	OO* catalyzes epoxidation η_1 -O* and η_2 -OO* intermediates	7	5.2258e-01
Case 3	O* catalyzes epoxidation single site mechanism	7	6.0747e-01

RMSE is the root mean square error obtained from the difference between experimental and simulated results. Lower RMSE means less deviation of the simulated results from the experimental results. Based on the RMSE, mechanisms in which OO* catalyzes epoxidation best explain the experimental data, compared to the ones invoking OOH* or O* as the key epoxidation intermediate. However, considering the errors within the experimental replicates are mostly in 0.1

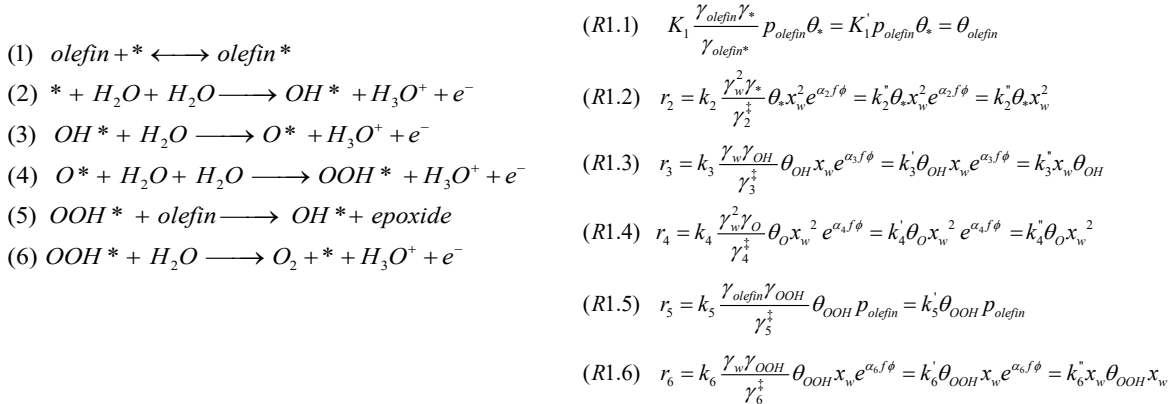
mA/cm² range, the difference in RMSE within Case 2a-e may not be significant enough to determine which case is most accurate. In our main text, we chose Case 2d as a representative mechanism, while Cases 2a-e could also be plausible mechanisms.

We formulated rate expressions based on reactant concentrations, incorporating the activity coefficients of reactants and transition states for each elementary step into the rate constants. While we successfully measured water activities under various electrolyte conditions using the headspace-GC technique (fig. S17), the activity coefficients for other surface reactants, solvated propylene, and transition states remain undetermined. Consequently, formulating rate expressions based on water activity implies an assumption that other activity coefficients are either unity or offset each other, which is a speculative presumption without adequate backing. Instead of selectively neglecting some of the activity coefficients, we lumped all of the activity coefficient terms as a Gibbs excess energy contribution to the rate constant and performed parameter estimation using concentrations as variables.

A series of elementary steps for epoxidation and oxygen evolution were proposed based on Eley-Rideal kinetics for each case, and the overall rate equation for epoxidation was formulated by invoking a pseudo-steady-state hypothesis (PSSH) on intermediate species. Molar ratio of water is expressed as x_w , unitless propylene pressure normalized by 1 atm is expressed as p_{olefin} in the rate expressions. Irreversible epoxide and O₂ desorption steps were assumed, given that the epoxidation rate showed no dependency on the concentrations of either the epoxide or O₂ (fig. S20).

Case 1: OOH* catalyzes epoxidation, single site mechanism

A series of elementary steps and rate equations for each step are proposed as below:



Potential-dependent exponential terms and activity coefficients are imbedded into the rate constant for simplicity of derivation. Quasi-equilibrium between the vacant site and the propylene-adsorbed site was invoked, and PSSH was applied for each water oxidation species.

We apply the PSSH to the various water oxidation intermediates:

$$\frac{d\Gamma_{OH}}{dt} = r_2 - r_3 + r_5 = 0, \quad \frac{d\Gamma_O}{dt} = r_3 - r_4 = 0, \quad \frac{d\Gamma_{OOH}}{dt} = r_4 - r_5 - r_6 = 0$$

Further rearrangement after substituting in equations R1.2 - R1.6 yields:

$$\theta_{OOH} = \frac{k_2'' x_w}{k_6''} \theta_*$$

$$\theta_{OH} = \frac{k_2'' x_w + (k_2'' k_5' / k_6'') p_{olefin}}{k_3''} \theta_*$$

$$\theta_O = \frac{k_3''}{k_4'' x_w} \theta_{OH} = \frac{k_2'' x_w + (k_2'' k_5' / k_6'') p_{olefin}}{k_4'' x_w} \theta_*$$

Assuming quasi-equilibrium between the vacant site and the propylene-adsorbed site (R1.1), now all of the sites can be expressed in terms of the density of vacant sites:

$$1 = \theta_* + \theta_{olefin} + \theta_{OH} + \theta_O + \theta_{OOH}$$

$$1 = \theta_* \left(1 + K_1' p_{olefin} + \frac{k_2'' x_w + (k_2'' k_5' / k_6'') p_{olefin}}{k_3''} + \frac{k_2'' x_w + (k_2'' k_5' / k_6'') p_{olefin}}{k_4'' x_w} + \frac{k_2'' x_w}{k_6''} \right)$$

5

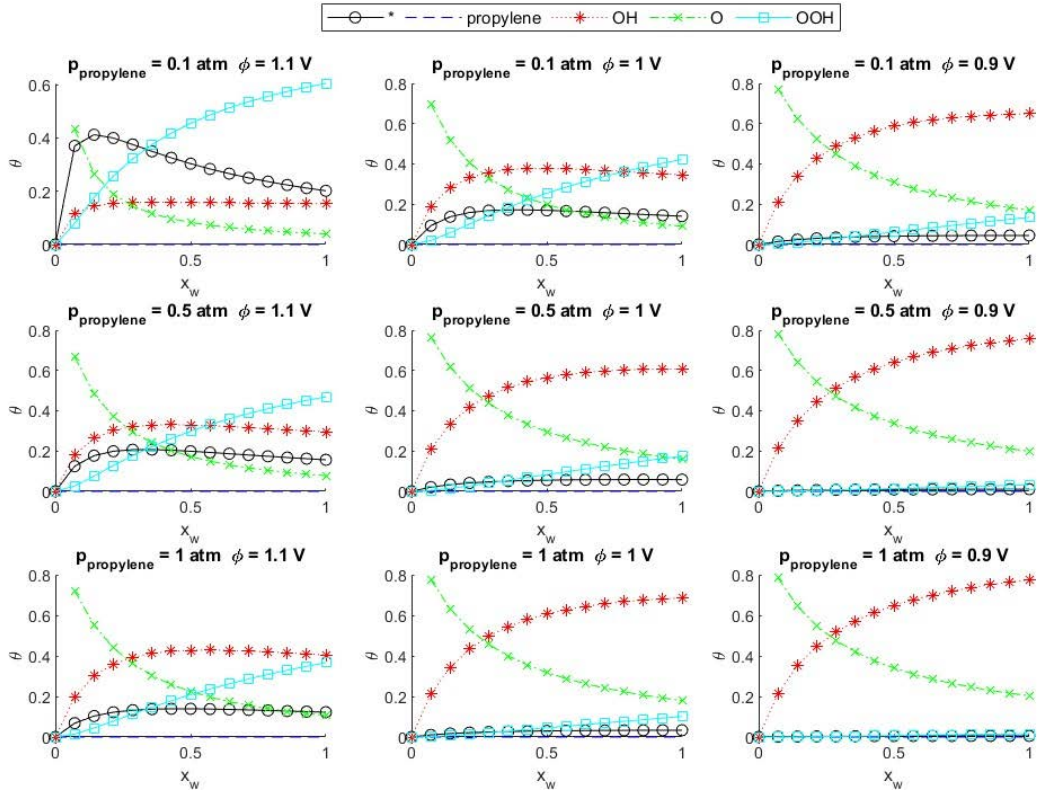
Then, the epoxidation rate can be expressed as below:

$$r_{epoxidation} = r_5 = k_5' \theta_{OOH} p_{olefin} = \frac{\frac{k_2'' k_5'}{k_6''} x_w p_{olefin}}{\left(1 + K_1' p_{olefin} + \frac{k_2'' x_w + (k_2'' k_5' / k_6'') p_{olefin}}{k_3''} + \frac{k_2'' x_w + (k_2'' k_5' / k_6'') p_{olefin}}{k_4'' x_w} + \frac{k_2'' x_w}{k_6''} \right)}$$

10

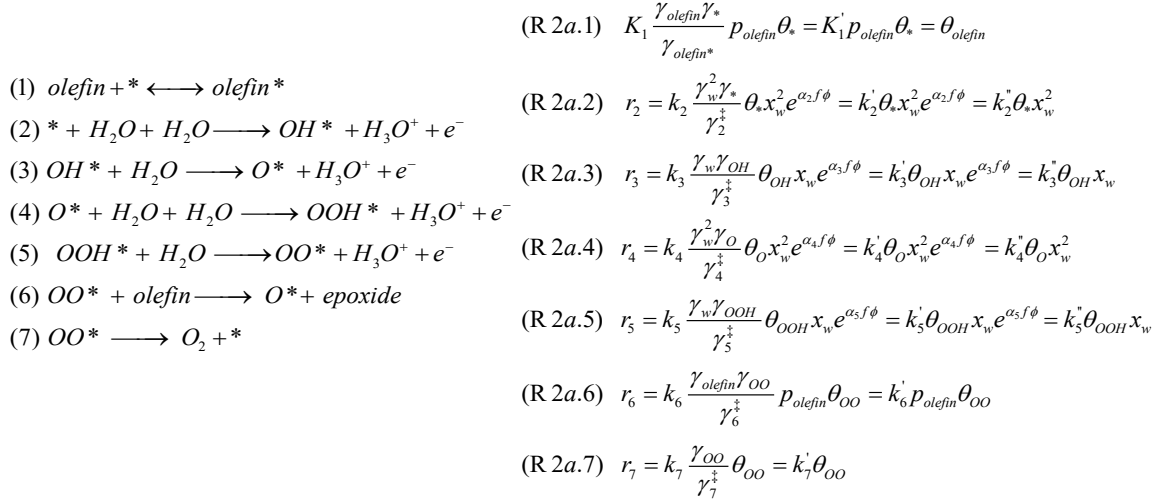
Rate constants were estimated by fitting the experimental data to the rate equation, and the coverages of various surface sites at different water concentrations, propylene pressures, and applied potentials (vs. Fc/Fc⁺) are simulated as follows:

case 1: OOH catalyzes epoxidation, single site mechanism



Case 2a: OO^* catalyzes epoxidation, single site mechanism

A series of elementary steps and rate equations for each step are proposed as below:



5

We apply the PSSH to the water oxidation intermediates:

$$\frac{d\Gamma_{OH}}{dt} = r_2 - r_3 = 0, \quad \frac{d\Gamma_O}{dt} = r_3 - r_4 + r_6 = 0, \quad \frac{d\Gamma_{OOH}}{dt} = r_4 - r_5 = 0, \quad \frac{d\Gamma_{OO}}{dt} = r_5 - r_6 - r_7 = 0$$

Further rearrangement after substituting in equations R2a.2-7 yields:

$$\begin{aligned}
 k_2 \frac{\gamma_w^2 \gamma^*}{\gamma_2^*} e^{\alpha_2 f \phi} \theta_* x_w^2 &= k_3 \frac{\gamma_w \gamma_{OH}}{\gamma_3^*} e^{\alpha_3 f \phi} \theta_{OH} x_w \\
 \theta_{OH} &= \frac{k_2}{k_3} \frac{\gamma_w \gamma_3^* \gamma^*}{\gamma_2^* \gamma_{OH}} e^{(\alpha_2 - \alpha_3) f \phi} x_w \theta_* \\
 k_4 \frac{\gamma_w^2 \gamma_O}{\gamma_4^*} e^{\alpha_4 f \phi} \theta_O x_w^2 &= k_5 \frac{\gamma_w \gamma_{OOH}}{\gamma_5^*} e^{\alpha_5 f \phi} \theta_{OOH} x_w \\
 \theta_{OOH} &= \frac{k_4}{k_5} \frac{\gamma_w \gamma_O}{\gamma_4^*} x_w e^{(\alpha_4 - \alpha_5) f \phi} \theta_O \\
 k_5 \frac{\gamma_w \gamma_{OOH}}{\gamma_5^*} e^{\alpha_5 f \phi} \theta_{OOH} x_w &= k_6 \frac{\gamma_{olefin} \gamma_{OO}}{\gamma_6^*} p_{olefin} \theta_{OO} + k_7 \frac{\gamma_{OO}}{\gamma_7^*} \theta_{OO} \\
 \theta_{OO} &= \frac{k_5 \frac{\gamma_w \gamma_{OOH}}{\gamma_5^*} e^{\alpha_5 f \phi} x_w}{k_6 \frac{\gamma_{olefin} \gamma_{OO}}{\gamma_6^*} p_{olefin} + k_7 \frac{\gamma_{OO}}{\gamma_7^*}} \theta_{OOH} = \frac{k_4 \frac{\gamma_w^2 \gamma_O}{\gamma_4^*} e^{\alpha_4 f \phi} x_w^2}{k_6 \frac{\gamma_{olefin} \gamma_{OO}}{\gamma_6^*} p_{olefin} + k_7 \frac{\gamma_{OO}}{\gamma_7^*}} \theta_O \\
 k_3 \frac{\gamma_w \gamma_{OH}}{\gamma_3^*} \theta_{OH} x_w e^{\alpha_3 f \phi} &= k_7 \frac{\gamma_{OO}}{\gamma_7^*} \theta_{OO} = \frac{k_7 \frac{\gamma_{OO}}{\gamma_7^*} k_4 \frac{\gamma_w^2 \gamma_O}{\gamma_4^*} e^{\alpha_4 f \phi} x_w^2}{k_6 \frac{\gamma_{olefin} \gamma_{OO}}{\gamma_6^*} p_{olefin} + k_7 \frac{\gamma_{OO}}{\gamma_7^*}} \theta_O \\
 \theta_O &= k_3 \frac{\gamma_w \gamma_{OH}}{\gamma_3^*} e^{\alpha_3 f \phi} \frac{k_6 \frac{\gamma_{olefin} \gamma_{OO}}{\gamma_6^*} p_{olefin} + k_7 \frac{\gamma_{OO}}{\gamma_7^*}}{k_7 \frac{\gamma_{OO}}{\gamma_7^*} k_4 \frac{\gamma_w^2 \gamma_O}{\gamma_4^*} e^{\alpha_4 f \phi} x_w} \theta_{OH} = \frac{k_6 \frac{\gamma_{olefin} \gamma_{OO}}{\gamma_6^*} p_{olefin} + k_7 \frac{\gamma_{OO}}{\gamma_7^*}}{k_7 \frac{\gamma_{OO}}{\gamma_7^*} k_4 \frac{\gamma_O}{\gamma_4^*} \frac{\gamma_2^*}{k_2 \gamma^*} e^{(\alpha_4 - \alpha_2) f \phi}} \theta_*
 \end{aligned}$$

$$\theta_{OOH} = \frac{k_4 \frac{\gamma_w \gamma_O}{\gamma_4^\ddagger} x_w}{k_5 \frac{\gamma_{OOH}}{\gamma_5^\ddagger}} e^{(\alpha_4 - \alpha_5) f \phi} \theta_O = \frac{k_6 \frac{\gamma_{olefin} \gamma_{OO}}{\gamma_6^\ddagger} p_{olefin} + k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger}}{k_5 \frac{\gamma_{OOH}}{\gamma_5^\ddagger} k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger} \frac{\gamma_2^\ddagger}{k_2 \gamma_*} e^{(\alpha_5 - \alpha_2) f \phi}} \gamma_w x_w \theta_*$$

$$\theta_{OO} = \frac{k_4 \frac{\gamma_w^2 \gamma_O}{\gamma_4^\ddagger} e^{\alpha_4 f \phi} x_w^2}{k_6 \frac{\gamma_{olefin} \gamma_{OO}}{\gamma_6^\ddagger} p_{olefin} + k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger}} \theta_O = \frac{k_4 \frac{\gamma_w^2 \gamma_O}{\gamma_4^\ddagger} e^{\alpha_4 f \phi} x_w^2}{k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger} k_4 \frac{\gamma_O}{\gamma_4^\ddagger} \frac{\gamma_2^\ddagger}{k_2 \gamma_*} e^{(\alpha_4 - \alpha_2) f \phi}} \theta_* = \frac{k_2 \frac{\gamma_w^2 \gamma_*}{\gamma_2^\ddagger} e^{\alpha_2 f \phi} x_w^2}{k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger}} \theta_*$$

With the assertion of quasi-equilibrium on step 1, all of the sites can be expressed in terms of the density of vacant sites. Also, we assume transfer coefficients $\alpha_2 = \alpha_3 = \alpha_4 = \alpha_5 = \alpha$. Then, the rate equation can be formulated as below:

$$1 = \theta_* + \theta_{olefin} + \theta_{OH} + \theta_O + \theta_{OOH} + \theta_{OO}$$

$$1 = \theta_* \left(1 + K_1 \frac{\gamma_{olefin} \gamma_*}{\gamma_{olefin}^*} p_{olefin} + \frac{k_2}{k_3} \frac{\gamma_w \gamma_3^\ddagger \gamma_*}{\gamma_2^\ddagger \gamma_{OH}} x_w + \frac{k_6 \frac{\gamma_{olefin} \gamma_{OO}}{\gamma_6^\ddagger} p_{olefin} + k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger}}{k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger} k_4 \frac{\gamma_O}{\gamma_4^\ddagger} \frac{\gamma_2^\ddagger}{k_2 \gamma_*}} + \frac{k_6 \frac{\gamma_{olefin} \gamma_{OO}}{\gamma_6^\ddagger} p_{olefin} + k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger}}{k_5 \frac{\gamma_{OOH}}{\gamma_5^\ddagger} k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger} \frac{\gamma_2^\ddagger}{k_2 \gamma_*}} \gamma_w x_w + \frac{k_2 \frac{\gamma_w^2 \gamma_*}{\gamma_2^\ddagger} e^{\alpha f \phi} x_w^2}{k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger}} \right)$$

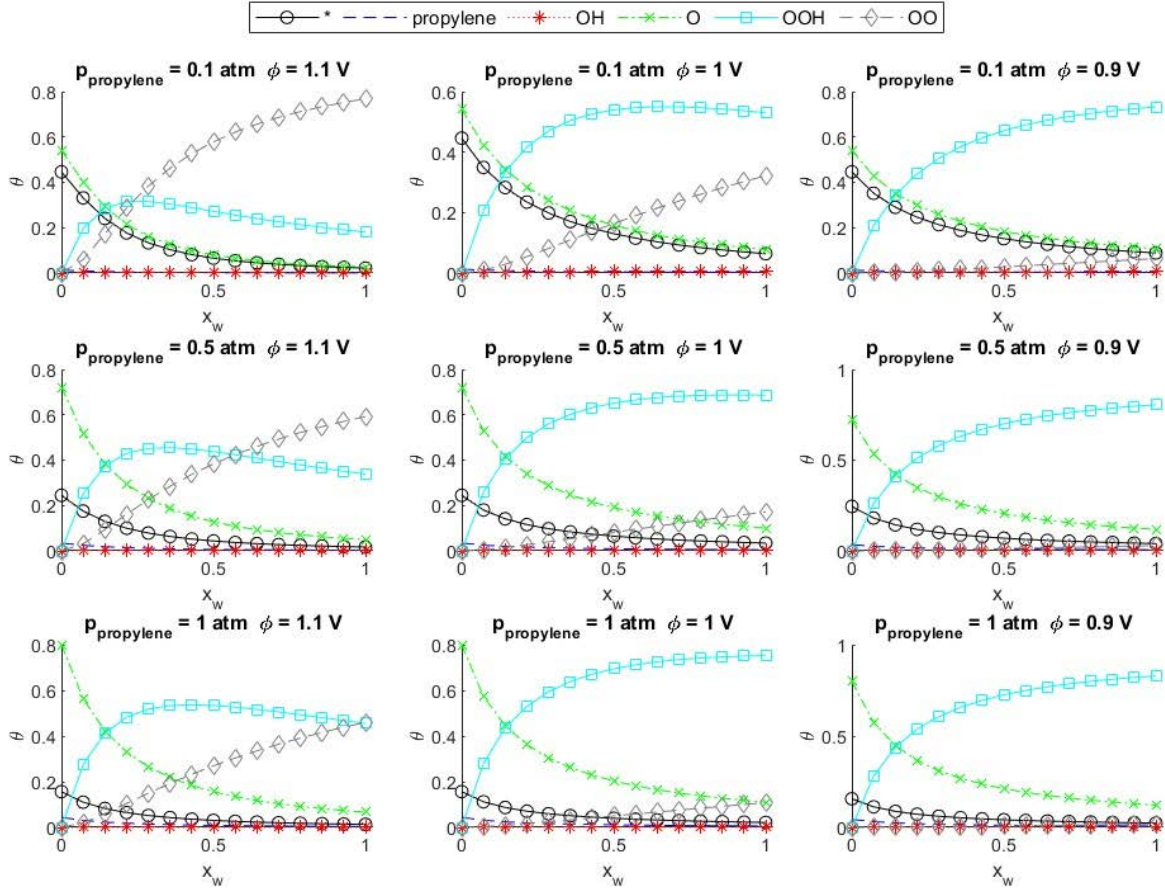
$$r = r_6 = k_6 \frac{\gamma_{olefin} \gamma_{OO}}{\gamma_6^\ddagger} p_{olefin} \theta_{OO}$$

$$= \frac{k_6 \frac{\gamma_{olefin} \gamma_{OO}}{\gamma_6^\ddagger} \frac{k_2 \frac{\gamma_w^2 \gamma_*}{\gamma_2^\ddagger} e^{\alpha f \phi}}{k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger}} p_{olefin} x_w^2}{1 + K_1 \frac{\gamma_{olefin} \gamma_*}{\gamma_{olefin}^*} p_{olefin} + \frac{k_2}{k_3} \frac{\gamma_w \gamma_3^\ddagger \gamma_*}{\gamma_2^\ddagger \gamma_{OH}} x_w + \frac{k_6 \frac{\gamma_{olefin} \gamma_{OO}}{\gamma_6^\ddagger} p_{olefin} + k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger}}{k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger} k_4 \frac{\gamma_O}{\gamma_4^\ddagger} \frac{\gamma_2^\ddagger}{k_2 \gamma_*}} + \frac{k_6 \frac{\gamma_{olefin} \gamma_{OO}}{\gamma_6^\ddagger} p_{olefin} + k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger}}{k_5 \frac{\gamma_{OOH}}{\gamma_5^\ddagger} k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger} \frac{\gamma_2^\ddagger}{k_2 \gamma_*}} \gamma_w x_w + \frac{k_2 \frac{\gamma_w^2 \gamma_*}{\gamma_2^\ddagger} e^{\alpha f \phi} x_w^2}{k_7 \frac{\gamma_{OO}}{\gamma_7^\ddagger}}}$$

$$= \frac{k_6 \frac{k_2}{k_7} p_{olefin} x_w^2}{1 + K_1' p_{olefin} + \frac{k_2''}{k_3} x_w + \frac{k_6' p_{olefin} + k_7'}{k_7' k_4' / k_2''} + \frac{k_6' p_{olefin} + k_7'}{k_5' k_7' / k_2''} x_w + \frac{k_2'' x_w^2}{k_7'}}$$

In this mechanism, to illustrate the complete rate equation without lumped parameters, all activity coefficients terms were displayed up to the final derivation step. Coverages of various surface sites at different water concentrations, propylene pressures, and applied potentials (vs. Fc/Fc⁺) are simulated as follows:

case 2a: OO catalyzes epoxidation, single site mechanism



Case 2b: OO* catalyzes epoxidation, OOH*+O*→OO*+OH* rearrangement included

A series of elementary steps and rate equations for each step are proposed as below:

- (1) $olefin + * \rightleftharpoons olefin *$
- (2) $* + * + H_2O + H_2O \longrightarrow OH * + * + H_3O^+ + e^-$
- (3) $OH * + * + H_2O \longrightarrow O * + * + H_3O^+ + e^-$
- (4) $O * + * + H_2O + H_2O \longrightarrow OOH * + * + H_3O^+ + e^-$
- (5) $OOH * + O * \longrightarrow OO * + OH *$
- (6) $OO * + * + olefin \longrightarrow O * + * + epoxide$
- (7) $OOH * + * + H_2O \longrightarrow OO * + * + H_3O^+ + e^-$
- (8) $OO * + * \longrightarrow O_2 + * + *$

$$\begin{aligned}
(R2b.1) \quad & K_1 \frac{\gamma_{olefin} \gamma_*}{\gamma_{olefin}^*} p_{olefin} \theta_* = K_1' p_{olefin} \theta_* = \theta_{olefin} \\
(R2b.2) \quad & r_2 = k_2 \frac{\gamma_w^2 \gamma_*^2}{\gamma_2^*} \theta_*^2 x_w^2 e^{\alpha_2 f \phi} = k_2' x_w^2 \theta_*^2 e^{\alpha_2 f \phi} = k_2'' x_w^2 \theta_*^2 \\
(R2b.3) \quad & r_3 = k_3 \frac{\gamma_w \gamma_{OH} \gamma_*}{\gamma_3^*} \theta_{OH} \theta_* x_w e^{\alpha_3 f \phi} = k_3' x_w \theta_{OH} \theta_* e^{\alpha_3 f \phi} = k_3'' x_w \theta_{OH} \theta_* \\
(R2b.4) \quad & r_4 = k_4 \frac{\gamma_w \gamma_O \gamma_*}{\gamma_4^*} \theta_O \theta_* x_w^2 e^{\alpha_4 f \phi} = k_4' x_w^2 \theta_O \theta_* e^{\alpha_4 f \phi} = k_4'' x_w^2 \theta_O \theta_* \\
(R2b.5) \quad & r_5 = k_5 \frac{\gamma_O \gamma_{OOH}}{\gamma_5^*} \theta_{OOH} \theta_O = k_5' \theta_{OOH} \theta_O \\
(R2b.6) \quad & r_6 = k_6 \frac{\gamma_{olefin} \gamma_{OO} \gamma_*}{\gamma_6^*} p_{olefin} \theta_{OO} \theta_* = k_6' p_{olefin} \theta_{OO} \theta_* \\
(R2b.7) \quad & r_7 = k_7 \frac{\gamma_w \gamma_{OOH} \gamma_*}{\gamma_7^*} \theta_{OOH} \theta_* x_w e^{\alpha_7 f \phi} = k_7' x_w \theta_{OOH} \theta_* e^{\alpha_7 f \phi} = k_7'' x_w \theta_{OOH} \theta_* \\
(R2b.8) \quad & r_8 = k_8 \frac{\gamma_{OO} \gamma_*}{\gamma_8^*} \theta_{OO} \theta_* = k_8' \theta_{OO} \theta_*
\end{aligned}$$

Quasi-equilibrium between the vacant site and the propylene-adsorbed site was invoked, and PSSH was applied for each water oxidation species.

5 We apply the PSSH to the various water oxidation intermediates:

$$\frac{d\Gamma_{OH}}{dt} = r_2 - r_3 + r_5 = 0, \quad \frac{d\Gamma_O}{dt} = r_3 - r_4 - r_5 + r_6 = 0, \quad \frac{d\Gamma_{OOH}}{dt} = r_4 - r_5 - r_7 = 0, \quad \frac{d\Gamma_{OO}}{dt} = r_5 - r_6 + r_7 - r_8 = 0$$

Further rearrangement after substituting in equations R2b.2– R2b.8 yields:

$$\begin{aligned}
\theta_{olefin} &= K_1' p_{olefin} \theta_* & \theta_{OH} &= \left(\frac{k_2''}{k_3''} + \frac{\left(k_2'' + \frac{k_2'' k_6'}{k_8'} \right)^2}{k_3'' \left(k_2'' + \frac{k_2'' k_6'}{k_8'} + \frac{k_4'' k_7''}{k_5'} x_w \right)} \right) x_w \theta_* \\
\theta_O &= \left(\frac{k_2''}{k_4''} + \frac{k_2'' k_6'}{k_4'' k_8'} \right) \theta_* & \theta_{OOH} &= \frac{\left(k_2'' + \frac{k_2'' k_6'}{k_8'} \right) x_w^2}{\left(\frac{k_2'' k_5'}{k_4''} + \frac{k_2'' k_5' k_6'}{k_4'' k_8'} + k_7'' x_w \right)} \theta_* & \theta_{OO} &= \frac{k_2''}{k_8'} x_w^2 \theta_*
\end{aligned}$$

10 With the assertion of quasi-equilibrium on step 1, all of the sites can be expressed in terms of the density of vacant sites. Then, the rate equation can be formulated as below:

$$1 = \theta_* + \theta_{olefin} + \theta_{OH} + \theta_O + \theta_{OOH} + \theta_{OO}$$

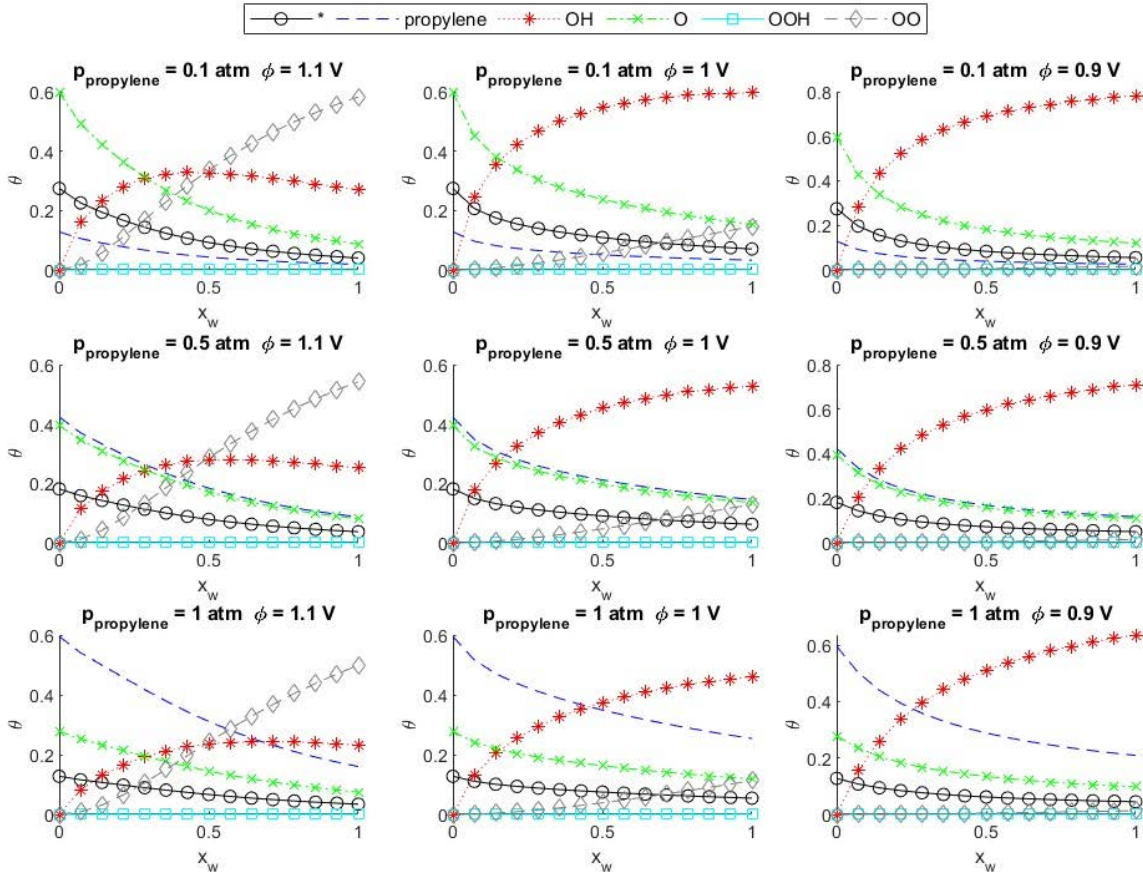
$$1 = \theta_* \left(1 + K_1' p_{olefin} + \frac{\frac{k_2''}{k_3''} + \frac{\left(k_2'' + \frac{k_2'' k_6'}{k_8'}\right)^2}{k_3'' \left(k_2'' + \frac{k_2'' k_6'}{k_8'} + \frac{k_4'' k_7''}{k_5''} x_w\right)}}{x_w} + \left(\frac{k_2''}{k_4''} + \frac{k_2'' k_6'}{k_4'' k_8'}\right) + \frac{\left(k_2'' + \frac{k_2'' k_6'}{k_8'}\right) x_w^2}{\left(\frac{k_2'' k_5''}{k_4''} + \frac{k_2'' k_5'' k_6'}{k_4'' k_8'} + k_7'' x_w\right)} + \frac{k_2''}{k_8'} x_w^2 \right)$$

$$r_{epoxidation} = r_6 = k_6' p_{olefin} \theta_{OO} \theta_* = \frac{k_2'' k_6'}{k_8'} p_{olefin} x_w^2 \theta_*^2$$

$$= \frac{\frac{k_2'' k_6'}{k_8'} p_{olefin} x_w^2}{\left(1 + K_1' p_{olefin} + \frac{\frac{k_2''}{k_3''} + \frac{\left(k_2'' + \frac{k_2'' k_6'}{k_8'}\right)^2}{k_3'' \left(k_2'' + \frac{k_2'' k_6'}{k_8'} + \frac{k_4'' k_7''}{k_5''} x_w\right)}}{x_w} + \left(\frac{k_2''}{k_4''} + \frac{k_2'' k_6'}{k_4'' k_8'}\right) + \frac{\left(k_2'' + \frac{k_2'' k_6'}{k_8'}\right) x_w^2}{\left(\frac{k_2'' k_5''}{k_4''} + \frac{k_2'' k_5'' k_6'}{k_4'' k_8'} + k_7'' x_w\right)} + \frac{k_2''}{k_8'} x_w^2 \right)^2}$$

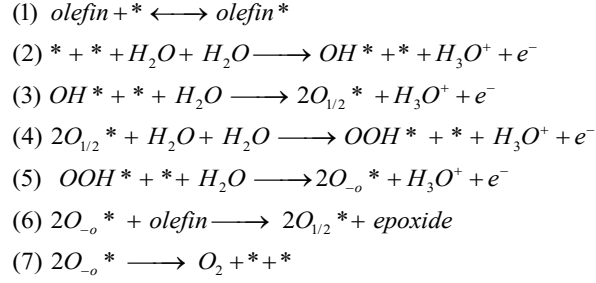
Coverages of various surface sites at different water concentrations, propylene pressures, and applied potentials (vs. Fc/Fc⁺) are simulated as follows:

case 2b: OO catalyzes epoxidation, OOH+O -> OO+OH assumed



Case 2c: assuming OO^* as the reactive oxygen species for epoxidation, bridged O^* and OO^*

A series of elementary steps and rate equations for each step are proposed as below:



$$(R2c.1) \quad K_1 \frac{\gamma_{olefin} \gamma^*}{\gamma_{olefin}^*} p_{olefin} \theta_* = K_1' p_{olefin} \theta_* = \theta_{olefin}$$

$$(R2c.2) \quad r_2 = k_2 \frac{\gamma_w^2 \gamma^*}{\gamma_2^*} \theta_*^2 x_w^2 e^{\alpha_2 f \phi} = k_2' x_w^2 \theta_*^2 e^{\alpha_2 f \phi} = k_2'' x_w^2 \theta_*^2$$

$$(R2c.3) \quad r_3 = k_3 \frac{\gamma_w \gamma_{OH} \gamma^*}{\gamma_3^*} \theta_{OH} \theta_* x_w e^{\alpha_3 f \phi} = k_3' x_w \theta_{OH} \theta_* e^{\alpha_3 f \phi} = k_3'' x_w \theta_{OH} \theta_*$$

$$(R2c.4) \quad r_4 = k_4 \frac{\gamma_w^2 \gamma_{O_{1/2}}^2}{\gamma_4^*} \theta_{O_{1/2}}^2 x_w^2 e^{\alpha_4 f \phi} = k_4' \theta_{O_{1/2}}^2 x_w^2 e^{\alpha_4 f \phi} = k_4'' \theta_{O_{1/2}}^2 x_w^2$$

$$(R2c.5) \quad r_5 = k_5 \frac{\gamma_w \gamma_{OOH} \gamma^*}{\gamma_5^*} \theta_{OOH} \theta_* x_w e^{\alpha_5 f \phi} = k_5' \theta_{OOH} \theta_* x_w e^{\alpha_5 f \phi} = k_5'' \theta_{OOH} \theta_* x_w$$

$$(R2c.6) \quad r_6 = k_6 \frac{\gamma_{olefin} \gamma_{O_{-o}}^2}{\gamma_6^*} p_{olefin} \theta_{O_{-o}}^2 = k_6' p_{olefin} \theta_{O_{-o}}^2$$

$$(R2c.7) \quad r_7 = k_7 \frac{\gamma_{O_{-o}}^2}{\gamma_7^*} \theta_{O_{-o}}^2 = k_7' \theta_{O_{-o}}^2$$

η_2-O^* and η_2-OO^* are shown as $O_{1/2}^*$ and O_{-o}^* , respectively. Quasi-equilibrium between the vacant site and the propylene-adsorbed site was invoked, and PSSH was applied for each water oxidation species.

We apply the PSSH to the various water oxidation intermediates:

$$\frac{d\Gamma_{OH}}{dt} = r_2 - r_3 = 0, \quad \frac{d\Gamma_{\eta_2-O}}{dt} = 2r_3 + 2r_6 - 2r_4 = 0, \quad \frac{d\Gamma_{OOH}}{dt} = r_4 - r_5 = 0, \quad \frac{d\Gamma_{\eta_2-OO}}{dt} = 2r_5 - 2r_6 - 2r_7 = 0$$

Further rearrangement after substituting in equations R2c.2– R2c.7 yields surface coverages and rate equation as below:

$$\begin{aligned}
 \theta_{olefin} &= K_1' p_{olefin} \theta_* & \theta_{OH} &= \frac{k_2''}{k_3''} x_w \theta_* & \theta_{O_{1/2}} &= \sqrt{\frac{k_2'' (k_6' p_{olefin} + k_7')}{k_4'' k_7'}} \theta_* \\
 \theta_{OOH} &= \frac{k_2'' (k_6' p_{olefin} + k_7')}{k_5'' k_7'} x_w \theta_* & \theta_{O_{-o}} &= \sqrt{\frac{k_2''}{k_7'}} x_w \theta_*
 \end{aligned}$$

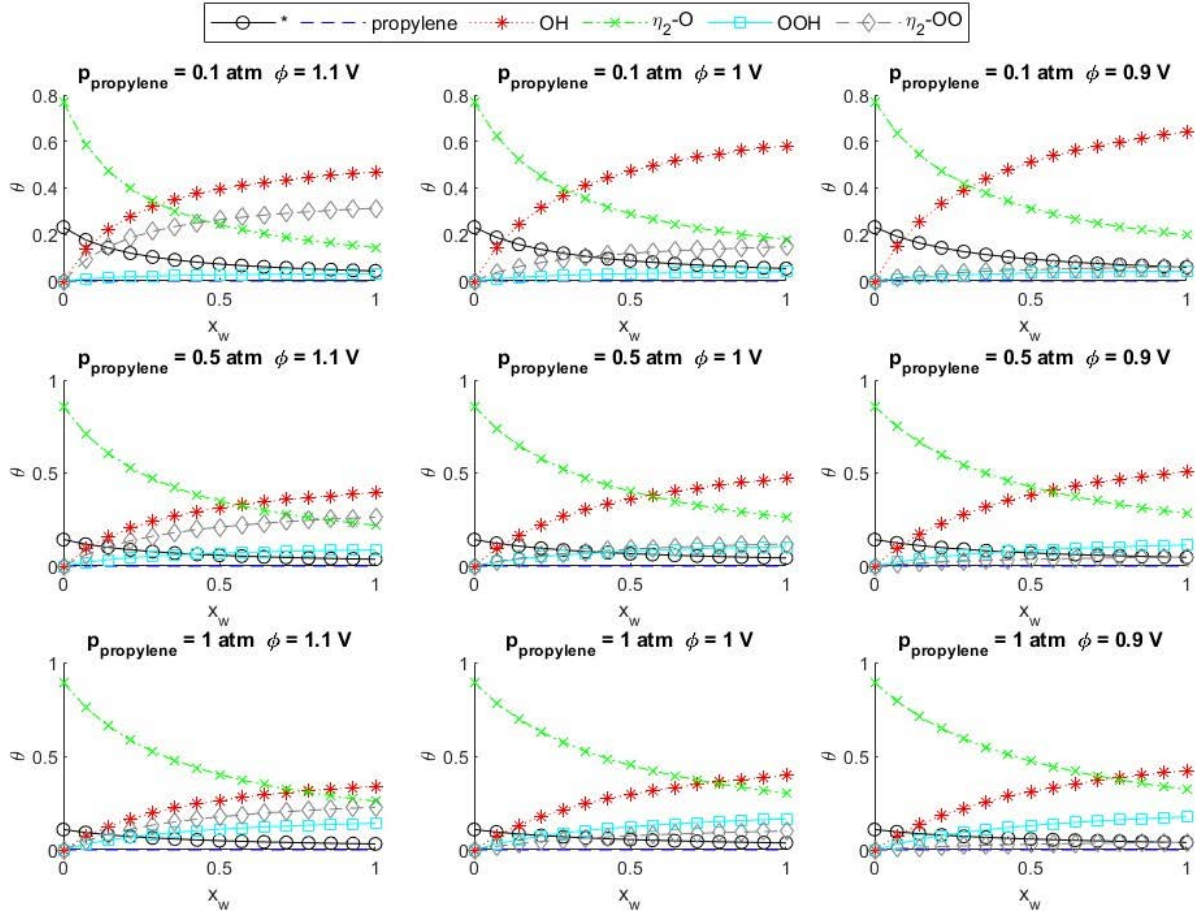
$$1 = \theta_* \left(1 + K_1' p_{olefin} + \frac{k_2''}{k_3''} x_w + \sqrt{\frac{k_2''(k_6' p_{olefin} + k_7')}{k_4'' k_7'}} + \frac{k_2''(k_6' p_{olefin} + k_7')}{k_5'' k_7'} x_w + \sqrt{\frac{k_2''}{k_7'}} x_w \right)$$

$$r_{epox} = r_6 = k_6' p_{olefin} \theta_{O-O}^2 = \frac{k_6' \frac{k_2''}{k_7'} p_{olefin} x_w^2}{\left(1 + K_1' p_{olefin} + \frac{k_2''}{k_3''} x_w + \sqrt{\frac{k_2''(k_6' p_{olefin} + k_7')}{k_4'' k_7'}} + \frac{k_2''(k_6' p_{olefin} + k_7')}{k_5'' k_7'} x_w + \sqrt{\frac{k_2''}{k_7'}} x_w \right)^2}$$

5

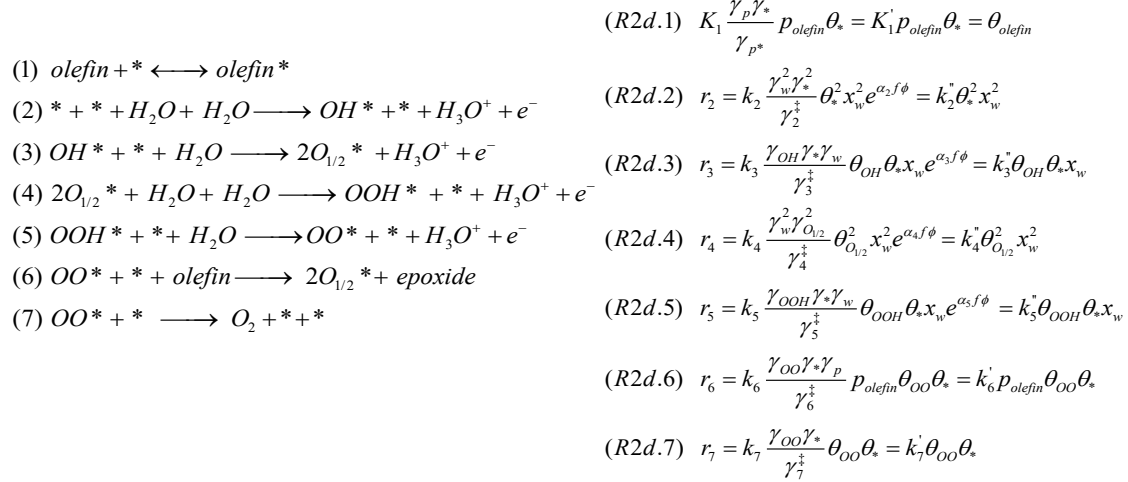
With the parameters estimated from data fitting, coverages of various surface sites at different water concentrations, propylene pressures, and applied potentials (vs. Fc/Fc⁺) are simulated as follows:

case 2c: η_2 -OO catalyzes epoxidation, η_2 -O and η_2 -OO intermediates



Case 2d: assuming OO^* as the reactive oxygen species for epoxidation, bridged O^*

A series of elementary steps and rate equations for each step are proposed as below:



5 η_2 - O^* is shown as $O_{1/2}^*$. Quasi-equilibrium between the vacant site and the propylene-adsorbed site was invoked, and PSSH was applied for each water oxidation species.

We apply the PSSH to the various water oxidation intermediates:

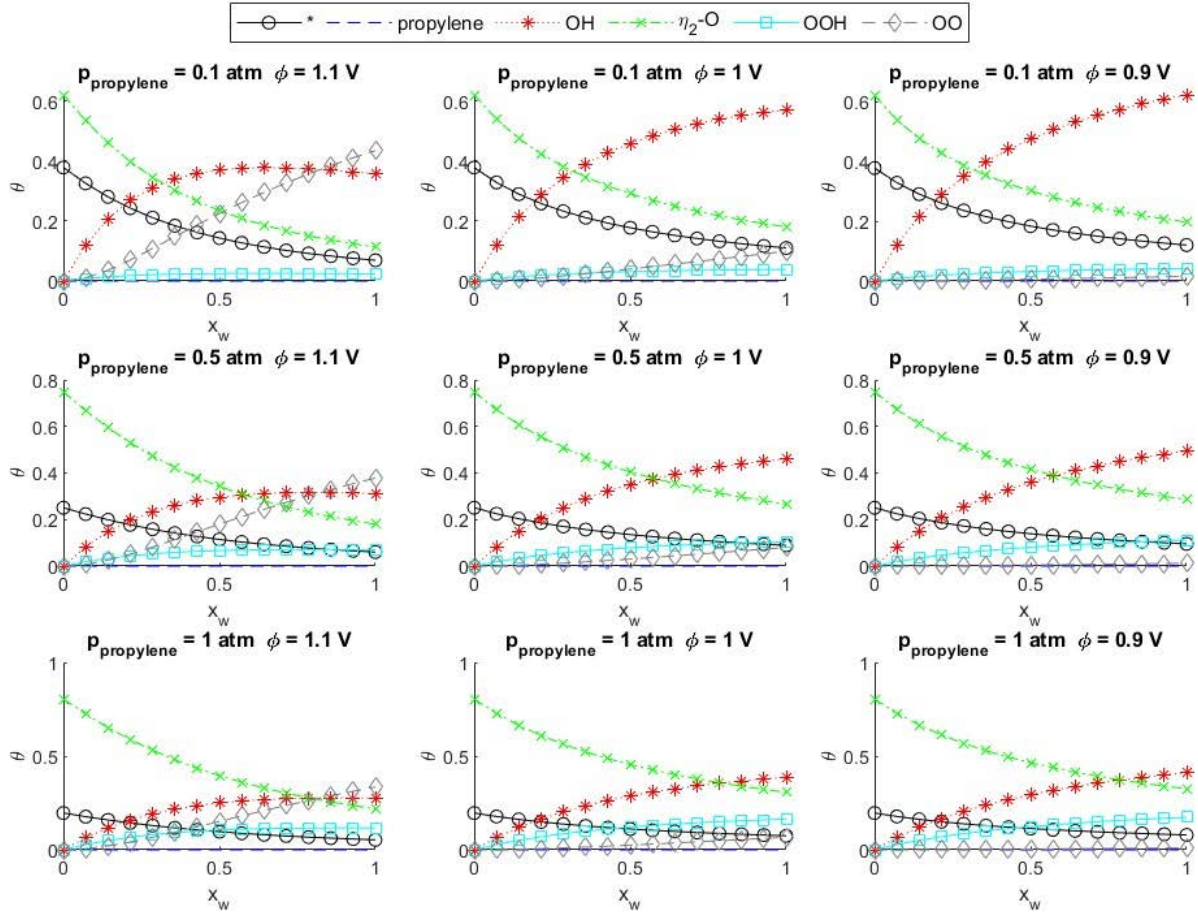
$$\frac{d\Gamma_{OH}}{dt} = r_2 - r_3 = 0, \quad \frac{d\Gamma_{\eta_2-O}}{dt} = 2r_3 + 2r_6 - 2r_4 = 0, \quad \frac{d\Gamma_{OOH}}{dt} = r_4 - r_5 = 0, \quad \frac{d\Gamma_{OO}}{dt} = r_5 - r_6 - r_7 = 0$$

10 Further rearrangement after substituting in equations R2d.2– R2d.7 yields surface coverages and rate equation as below:

$$\begin{aligned}
 \theta_{olefin} &= K_1' p_{olefin} \theta_* & \theta_{OH} &= \frac{k_2''}{k_3''} x_w \theta_* & \theta_{O_{1/2}} &= \sqrt{\frac{k_2'' (k_6' p_{olefin} + k_7')}{k_4'' k_7'}} \theta_* \\
 \theta_{OOH} &= \frac{k_2'' (k_6' p_{olefin} + k_7')}{k_7'' k_5''} x_w \theta_* & \theta_{OO} &= \frac{k_2''}{k_7''} x_w^2 \theta_* \\
 1 &= \theta_* \left(1 + K_1' p_{olefin} + \frac{k_2''}{k_3''} x_w + \sqrt{\frac{k_2'' (k_6' p_{olefin} + k_7')}{k_4'' k_7'}} + \frac{k_2'' (k_6' p_{olefin} + k_7')}{k_7'' k_5''} x_w + \frac{k_2''}{k_7''} x_w^2 \right) \\
 r_{epox} &= r_6 = k_6' p_{olefin} \theta_{OO} \theta_* = \frac{k_6' \frac{k_2''}{k_7''} p_{olefin} x_w^2}{\left(1 + K_1' p_{olefin} + \frac{k_2''}{k_3''} x_w + \sqrt{\frac{k_2'' (k_6' p_{olefin} + k_7')}{k_4'' k_7'}} + \frac{k_2'' (k_6' p_{olefin} + k_7')}{k_5'' k_7''} x_w + \frac{k_2''}{k_7''} x_w^2 \right)^2}
 \end{aligned}$$

With the parameters estimated from data fitting, coverages of various surface sites at different water concentrations, propylene pressures, and applied potentials (vs. Fc/Fc⁺) are simulated as follows:

case 2d: OO catalyzes epoxidation, η_2 -O intermediate



5

Case 2e: assuming OO* as the reactive oxygen species for epoxidation, bridged OO*

A series of elementary steps and rate equations for each step are proposed as below:

- (1) $\text{olefin} + * \rightleftharpoons \text{olefin}^*$
- (2) $* + * + \text{H}_2\text{O} + \text{H}_2\text{O} \longrightarrow \text{OH}^* + * + \text{H}_3\text{O}^+ + e^-$
- (3) $\text{OH}^* + * + \text{H}_2\text{O} \longrightarrow \text{O}^* + * + \text{H}_3\text{O}^+ + e^-$
- (4) $\text{O}^* + * + \text{H}_2\text{O} + \text{H}_2\text{O} \longrightarrow \text{OOH}^* + * + \text{H}_3\text{O}^+ + e^-$
- (5) $\text{OOH}^* + * + \text{H}_2\text{O} \longrightarrow 2\text{O}_{-O}^* + \text{H}_3\text{O}^+ + e^-$
- (6) $2\text{O}_{-O}^* + \text{olefin} \longrightarrow \text{O}^* + * + \text{epoxide}$
- (7) $2\text{O}_{-O}^* \longrightarrow \text{O}_2 + * + *$

10

$$(R2e.1) \quad K_1 \frac{\gamma_p \gamma_*}{\gamma_{p^*}} p_{olefin} \theta_* = K_1' p_{olefin} \theta_* = \theta_{olefin}$$

$$(R2e.2) \quad r_2 = k_2 \frac{\gamma_w^2 \gamma_*^2}{\gamma_2^*} \theta_*^2 x_w^2 e^{\alpha_2 f \phi} = k_2'' \theta_*^2 x_w^2$$

$$(R2e.3) \quad r_3 = k_3 \frac{\gamma_{OH} \gamma_* \gamma_w}{\gamma_3^*} \theta_{OH} \theta_* x_w e^{\alpha_3 f \phi} = k_3' \theta_{OH} \theta_* x_w$$

$$(R2e.4) \quad r_4 = k_4 \frac{\gamma_O \gamma_* \gamma_w^2}{\gamma_4^*} \theta_O \theta_* x_w^2 e^{\alpha_4 f \phi} = k_4'' \theta_O \theta_* x_w^2$$

$$(R2e.5) \quad r_5 = k_5 \frac{\gamma_{OOH} \gamma_* \gamma_w}{\gamma_5^*} \theta_{OOH} \theta_* x_w e^{\alpha_5 f \phi} = k_5'' \theta_{OOH} \theta_* x_w$$

$$(R2e.6) \quad r_6 = k_6 \frac{\gamma_{olefin} \gamma_{O-o}^2}{\gamma_6^*} p_{olefin} \theta_{O-o}^2 = k_6' p_{olefin} \theta_{O-o}^2$$

$$(R2e.7) \quad r_7 = k_7 \frac{\gamma_{O-o}^2}{\gamma_7^*} \theta_{O-o}^2 = k_7' \theta_{O-o}^2$$

η_2 -OO* is shown as O_{-o}^* . Quasi-equilibrium between the vacant site and the propylene-adsorbed site was invoked, and PSSH was applied for each water oxidation species.

5 We apply the PSSH to the various water oxidation intermediates:

$$\frac{d\Gamma_{OH}}{dt} = r_2 - r_3 = 0, \quad \frac{d\Gamma_O}{dt} = r_3 + r_6 - r_4 = 0, \quad \frac{d\Gamma_{OOH}}{dt} = r_4 - r_5 = 0, \quad \frac{d\Gamma_{\eta_2-OO}}{dt} = 2r_5 - 2r_6 - 2r_7 = 0$$

Further rearrangement after substituting in equations R2e.2– R2e.7 yields surface coverages and rate equation as below:

$$\theta_{olefin} = K_1' p_{olefin} \theta_* \quad \theta_{OH} = \frac{k_2''}{k_3^*} x_w \theta_* \quad \theta_O = \frac{k_2'' (k_6' p_{olefin} + k_7')}{k_4^* k_7'} \theta_*$$

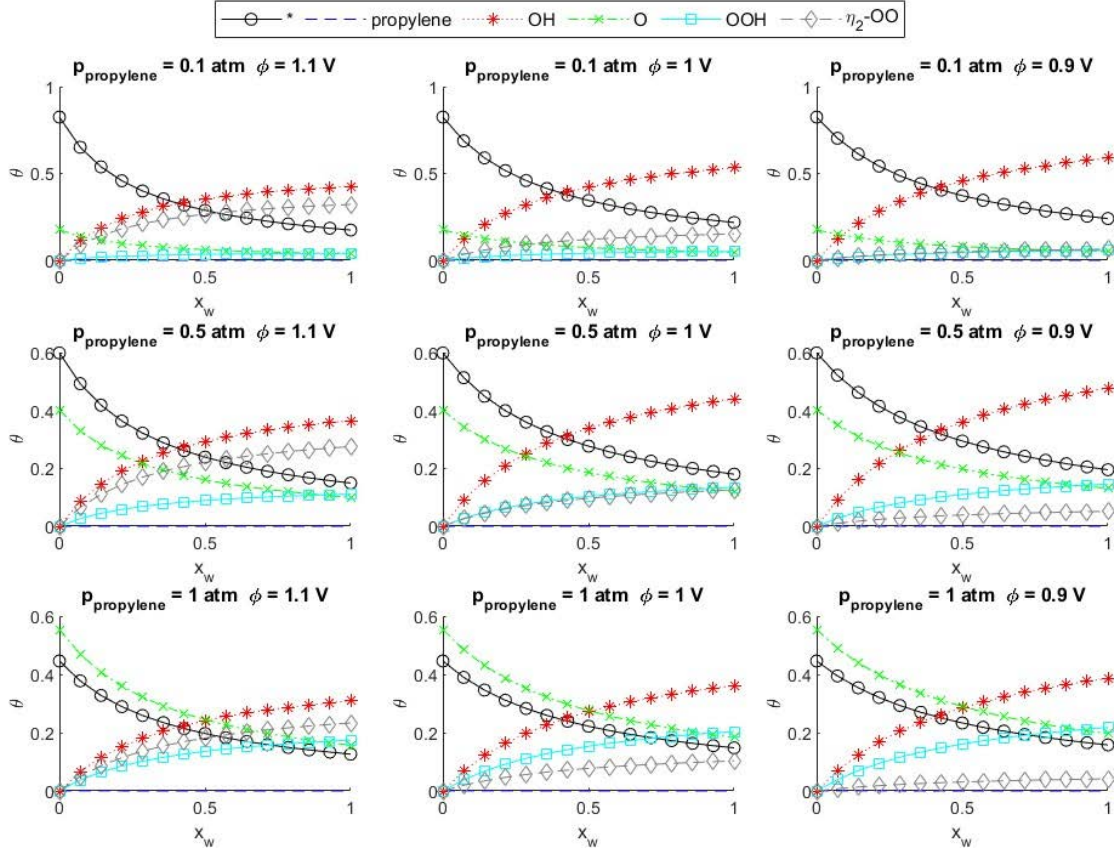
$$\theta_{OOH} = \frac{k_2'' (k_6' p_{olefin} + k_7')}{k_5^* k_7'} x_w \theta_* \quad \theta_{O-o} = \sqrt{\frac{k_2''}{k_7'}} x_w \theta_*$$

$$1 = \theta_* \left(1 + K_1' p_{olefin} + \frac{k_2''}{k_3^*} x_w + \frac{k_2'' (k_6' p_{olefin} + k_7')}{k_4^* k_7'} + \frac{k_2'' (k_6' p_{olefin} + k_7')}{k_5^* k_7'} x_w + \sqrt{\frac{k_2''}{k_7'}} x_w \right)$$

$$r_{epox} = r_6 = k_6' p_{olefin} \theta_{O-o}^2 = \frac{k_6' \frac{k_2''}{k_7'} p_{olefin} x_w^2}{\left(1 + K_1' p_{olefin} + \frac{k_2''}{k_3^*} x_w + \frac{k_2'' (k_6' p_{olefin} + k_7')}{k_4^* k_7'} + \frac{k_2'' (k_6' p_{olefin} + k_7')}{k_5^* k_7'} x_w + \sqrt{\frac{k_2''}{k_7'}} x_w \right)^2}$$

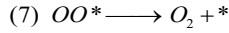
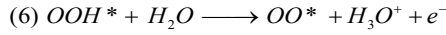
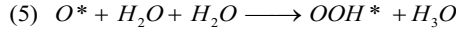
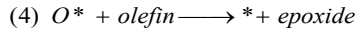
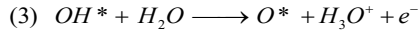
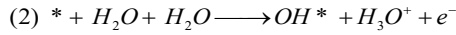
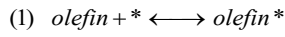
With the parameters estimated from data fitting, coverages of various surface sites at different water concentrations, propylene pressures, and applied potentials (vs. Fc/Fc⁺) are simulated as follows:

case 2e: η_2 -OO catalyzes epoxidation, η_1 -O intermediate



Case 3: assuming O^* as the reactive oxygen species for epoxidation

A series of elementary steps and rate equations for each step are proposed as below:



$$(R3.1) \quad K_1 \frac{\gamma_{olefin} \gamma^*}{\gamma_{olefin}^*} p_{olefin} \theta_* = K_1' p_{olefin} \theta_* = \theta_{olefin}$$

$$(R3.2) \quad r_2 = k_2 \frac{\gamma_w^2 \gamma^*}{\gamma_2^*} \theta_* x_w^2 e^{\alpha_2 f \phi} = k_2^* \theta_* x_w^2$$

$$(R3.3) \quad r_3 = k_3 \frac{\gamma_w \gamma_{OH}}{\gamma_3^*} \theta_{OH} x_w e^{\alpha_3 f \phi} = k_3^* \theta_{OH} x_w$$

$$(R3.4) \quad r_4 = k_4 \frac{\gamma_{olefin} \gamma_O}{\gamma_4^*} \theta_O p_{olefin} = k_4' p_{olefin} \theta_O$$

$$(R3.5) \quad r_5 = k_5 \frac{\gamma_w^2 \gamma_O}{\gamma_5^*} \theta_O x_w^2 e^{\alpha_5 f \phi} = k_5^* \theta_O x_w^2$$

$$(R3.6) \quad r_6 = k_6 \frac{\gamma_w \gamma_{OOH}}{\gamma_6^*} \theta_{OOH} x_w e^{\alpha_6 f \phi} = k_6^* \theta_{OOH} x_w$$

$$(R3.7) \quad r_7 = k_7 \frac{\gamma_{OO}}{\gamma_7^*} \theta_{OO} = k_7' \theta_{OO}$$

We apply the PSSH to the water oxidation intermediates:

$$\frac{d\Gamma_{OH}}{dt} = r_2 - r_3 = 0, \quad \frac{d\Gamma_O}{dt} = r_3 - r_4 - r_5 = 0, \quad \frac{d\Gamma_{OOH}}{dt} = r_5 - r_6 = 0, \quad \frac{d\Gamma_{OO}}{dt} = r_6 - r_7 = 0$$

Further rearrangement after substituting in equations R3.2 –7 yields:

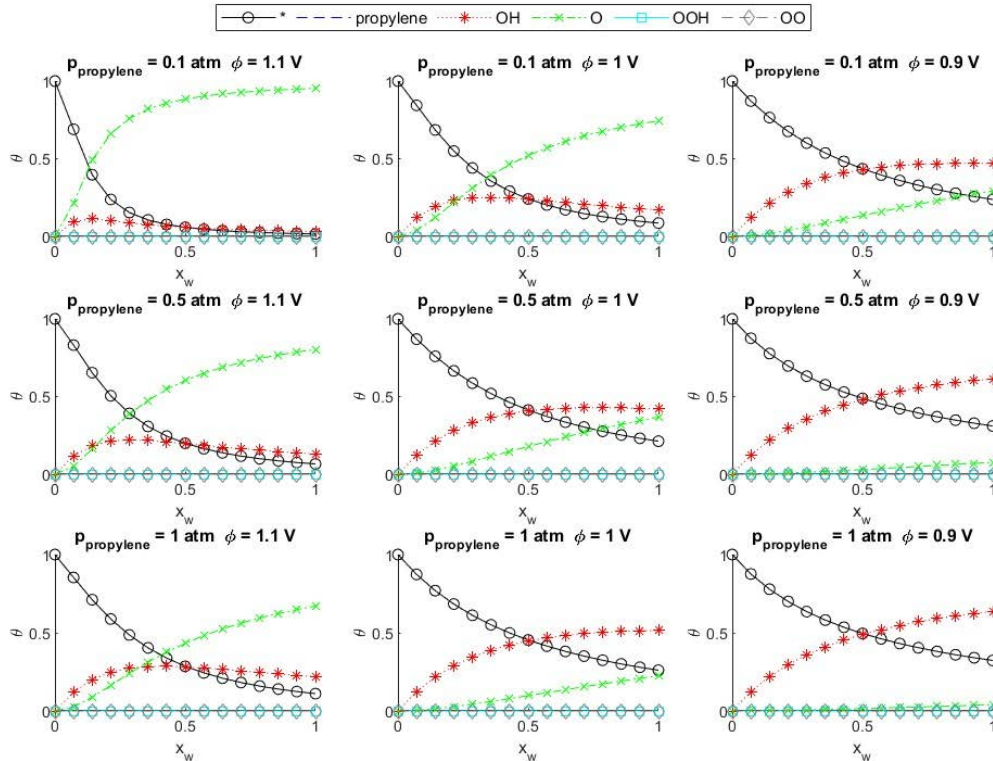
$$\begin{aligned}\theta_{olefin} &= K_1' p_{olefin} \theta_* & \theta_{OH} &= \frac{k_2' x_w}{k_3'} \theta_* \\ \theta_O &= \frac{k_3' x_w}{k_4' p_{olefin} + k_5' x_w^2} \theta_{OH} = \frac{k_2' x_w^2}{k_4' p_{olefin} + k_5' x_w^2} \theta_* \\ \theta_{OOH} &= \frac{k_5' x_w}{k_6'} \theta_O = \frac{(k_2' k_5' / k_6') x_w^3}{k_4' p_{olefin} + k_5' x_w^2} \theta_* \\ \theta_{OO} &= \frac{k_6'}{k_7'} a_{H_2O} \theta_{OOH} = \frac{(k_2' k_5' / k_7') x_w^4}{k_4' p_{olefin} + k_5' x_w^2} \theta_*\end{aligned}$$

With the assertion of quasi-equilibrium on step 1, all of the sites can be expressed in terms of the density of vacant sites. Then, the epoxidation rate equation can be formulated as below:

$$\begin{aligned}1 &= \theta_* + \theta_{olefin} + \theta_{OH} + \theta_O + \theta_{OOH} \\ &= \theta_* \left(1 + K_1' p_{olefin} + \frac{k_2' x_w}{k_3'} + \frac{k_2' x_w^2}{k_4' p_{olefin} + k_5' x_w^2} + \frac{(k_2' k_5' / k_6') x_w^3}{k_4' p_{olefin} + k_5' x_w^2} + \frac{(k_2' k_5' / k_7') x_w^4}{k_4' p_{olefin} + k_5' x_w^2} \right) \\ r &= r_4 = k_4' p_{olefin} \theta_O = \frac{k_2' k_4' p_{olefin} x_w^2}{k_4' p_{olefin} + k_5' x_w^2} \theta_* \\ &= \frac{\frac{k_2' k_4' p_{olefin} x_w^2}{k_4' p_{olefin} + k_5' x_w^2}}{\left(1 + K_1' p_{olefin} + \frac{k_2' x_w}{k_3'} + \frac{k_2' x_w^2}{k_4' p_{olefin} + k_5' x_w^2} + \frac{(k_2' k_5' / k_6') x_w^3}{k_4' p_{olefin} + k_5' x_w^2} + \frac{(k_2' k_5' / k_7') x_w^4}{k_4' p_{olefin} + k_5' x_w^2} \right)}\end{aligned}$$

With the parameters estimated from data fitting, coverages of various surface sites at different water concentrations, propylene pressures, and applied potentials (vs. Fc/Fc⁺) are simulated as follows:

case 3: O catalyzes epoxidation, single site mechanism



S2. Alkene electrophilicity as a descriptor for direct electrochemical epoxidation

Epoxidation using the chemical oxidant m-chloroperbenzoic acid and direct ethylene epoxidation using molecular oxygen on silver catalysts are categorized as electrophilic epoxidation, which is driven by the electrophilic attack of the oxidant on the alkene (27–29). However, the role of electrophile and nucleophile can be inverted in certain cases when the oxidant has highly nucleophilic oxygens, and when the metal withdraws electron density from the olefin (31). Olefins have been shown to be activated as an electrophile by coordinating with an electron-deficient metal ion. In the context of homogeneous catalysis, it has been reported that a bimolecular reaction of a PtOOH species with a Pt-coordinated olefin species can give the epoxide and PtOH (32). A heterogenous analog of this reaction could occur on an electrode surface, as applying an anodic potential to an electrode makes electron-deficient platinum species, which would facilitate the electrophilic activation of light olefins toward attack by a nucleophile.

As described above, previous studies suggest that either electrophilic or nucleophilic epoxidation can be a reasonable reaction pathway for electrochemical propylene epoxidation on late-transition metal catalysts. Identification of the nucleophile and electrophile in the reaction allows for a better understanding of the reaction and can guide catalyst engineering. To probe this question, we invoked the alkene electrophilicity index as a descriptor for direct anodic epoxidation.

The global electrophilicity index (ω) can be defined as:

$$\omega = \frac{\mu^2}{2\eta} = \frac{(\text{chemical potential})^2}{\text{chemical hardness}}$$

The chemical potential and the chemical hardness are defined as $\mu = \frac{\partial E}{\partial N}$, $\eta = \frac{\partial^2 E}{\partial N^2}$, respectively, where E is the energy of the system, and N is the number of electrons, meaning ω has a unit of energy (33, 34). A higher value of ω means the molecule is more electrophilic. The local reactivity at each atom of a molecule can be obtained using the Fukui function, which predicts the redistribution of electron density in a molecule upon electrophilic, nucleophilic, and radical attacks. Therefore, the global electrophilicity multiplied by the value of the Fukui function for the removal of an electron describes the local electrophilicity (ω^-) at each atom center. A linear correlation between the average ω^- of the two carbons in the alkene and the logarithm of epoxidation current density was expected as ω^- has a unit of energy (Fig. 5).

S3. Extended descriptions on deuterium kinetic isotope effects in water and acetonitrile isotopologues

To better understand the role of water and water oxidation intermediates in the epoxidation pathway, we compared reaction rates obtained with various isotopologues of water and acetonitrile (fig. S15 and table S10). We tested combinations of H₂O+CH₃CN, D₂O+CH₃CN, H₂O+CD₃CN, and D₂O+CD₃CN with fixed water (10 M) and propylene (1 atm) concentrations at 1.1 V vs. Fc/Fc⁺. The kinetic isotope effect (k_H/k_D) was determined by calculating the ratio of the epoxidation rate obtained from H₂O+CH₃CN to that obtained from deuterated conditions.

In both CH₃CN- and CD₃CN-based electrolytes, using D₂O instead of H₂O resulted in the primary kinetic isotope effect, suggesting that kinetically relevant steps involve O-H bond cleavage from water. In our proposed mechanism, water oxidation steps involving O-H bond cleavages are kinetically relevant steps. When the water identity was fixed as either H₂O or D₂O,

using CD₃CN instead of CH₃CN yielded epoxidation rates that were statistically indistinguishable, indicating that the cleavage of the C-H bond in acetonitrile did not occur or its cleavage product did not participate in the reaction.

S4. Investigation on water activity in water-acetonitrile electrolyte and the dependence of epoxidation rate on water activity

At lower water concentrations (1-5 M), the observed dependence of propylene epoxidation rates on water activity exhibited ~4th order behavior, which decreased to ~2nd order at higher water concentrations (5-12 M; fig. S12). The 4th order dependence on water activity at lower water concentrations may be attributed to disparate non-ideal behaviors of the transition state and reactant molecules. From a qualitative perspective, the stabilization of the transition state in kinetically relevant steps involving water may be achieved by augmenting the system's water content. In the water-acetonitrile electrolyte system, positive deviation from Raoult's law is observed, indicating the species engage in self-interactions that are more favorable than interactions with the other component of the mixture. 4th-order dependence in water activity with 2nd-order dependence in water concentration suggests a more prominent positive deviation from Raoult's law for the transition state ($\gamma_{\ddagger} > \gamma_{\text{water}} > 1$). This indicates that the transition state has less favorable interaction with its solvating environment compared to water.

To demonstrate this mathematically, we could assume that a $\eta_2\text{-O}^*$ is the dominant species, with the epoxidation rate greatly exceeding OER. This is an extreme version of our proposed mechanism (Fig. 4D; case 2d in the Supplementary Text S1). Under such extreme circumstances,

the rate expression is simplified to $r = k_4 \frac{\gamma_{\text{O}}^2}{\gamma_{\ddagger}^2} \gamma_w^2 x_w^2$. The observed rate dependence can be

logically explained if $\gamma_{\ddagger}$ is proportional to the square of γ_{water} . If we consider $\gamma_{\text{O}1/2}$ as a constant on the premise that $\eta_2\text{-O}^*$ is less influenced by solvation compared to water and the transition

state, the rate equation simplifies to $r = k_4 \frac{\gamma_{\text{O}}^2}{\gamma_{\ddagger}^2} \gamma_w^2 x_w^2 = k_4 x_w^2 = k_4 \frac{1}{\gamma_w^2} a_w^2$. Here, plotting the rate against activity will give higher apparent order dependence than actual since γ_{water} decreases (closer to 1) as water activity increases (gets closer to pure water).

In contrast, the 5-12 M region shows ~2nd order dependences on water activity as well as on water concentration. This can be rationalized if $\gamma_{\ddagger}$ remains relatively unchanged at higher water concentrations. This hypothesis aligns with our chemical intuition: once a sufficient quantity of water is present around the transition state, adding more water does not significantly alter the near-transition-state chemical environment, which makes the $\gamma_{\ddagger}$ relatively constant after reaching a certain water concentration. Therefore, it is reasonable to think of aqueous electrolyte as an extreme case of water-acetonitrile blended electrolyte where water mole ratio approaches one.

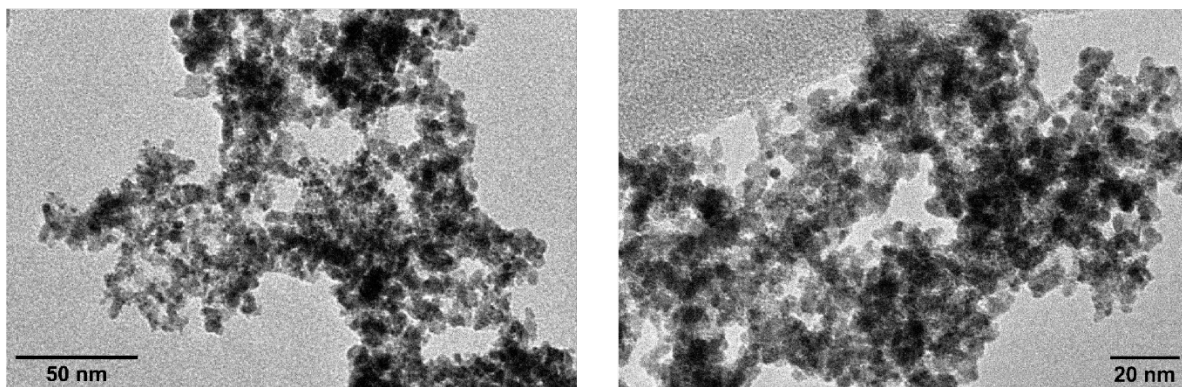
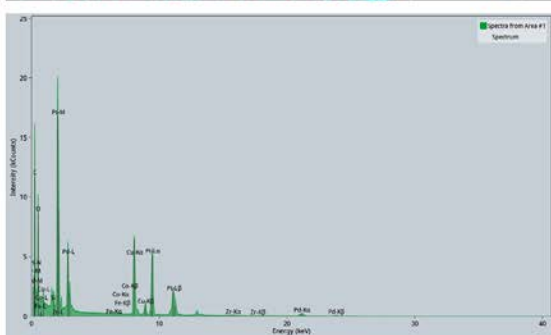
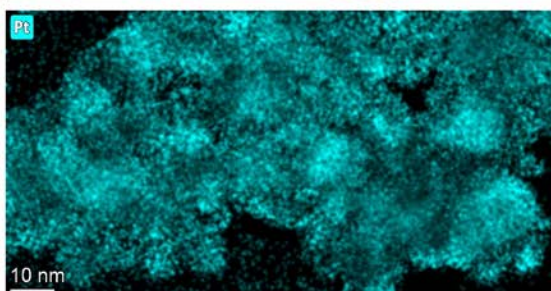
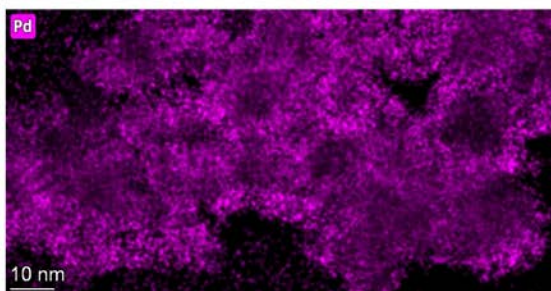
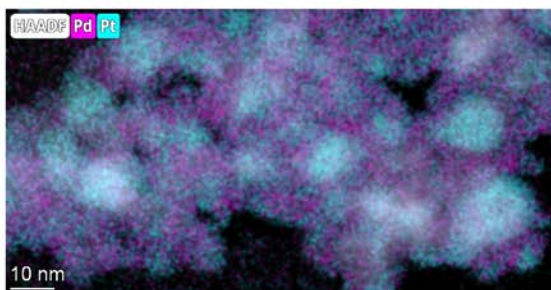


Fig. S1. Representative TEM image of images of PdPtO_x/C annealed at 400 °C.

Figure 1 consists of four panels. The top three panels are HAADF-EDS elemental maps for Pd (magenta), Pt (cyan), and Pt (cyan) respectively, with a 10 nm scale bar. The bottom panel shows the EDS spectrum with peaks labeled for various elements including Pd, Pt, Cu, and Zn.

43

40 wt% Pd₁Pt₃O_x/C, annealed at 500 °C for 3 h under static air



40 wt% Pd₃Pt₁O_x/C, annealed at 500 °C for 3 h under static air

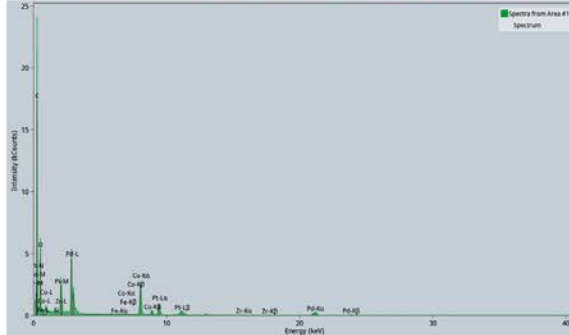
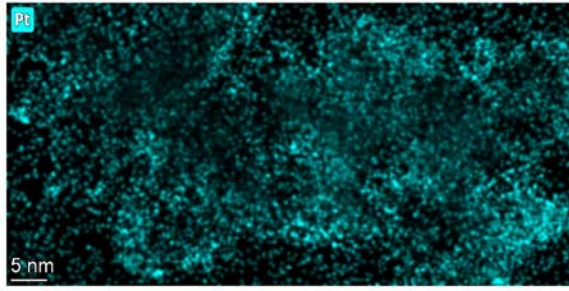
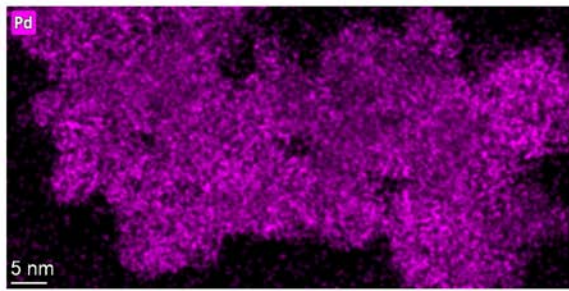
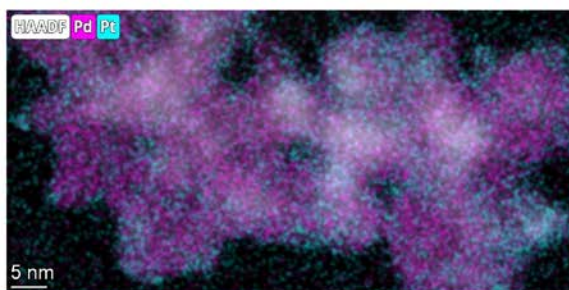


Fig. S3. High-resolution energy dispersive spectroscopy mapping of Pd₁Pt₃O_x/C and Pd₃Pt₁O_x/C annealed at 500 °C.

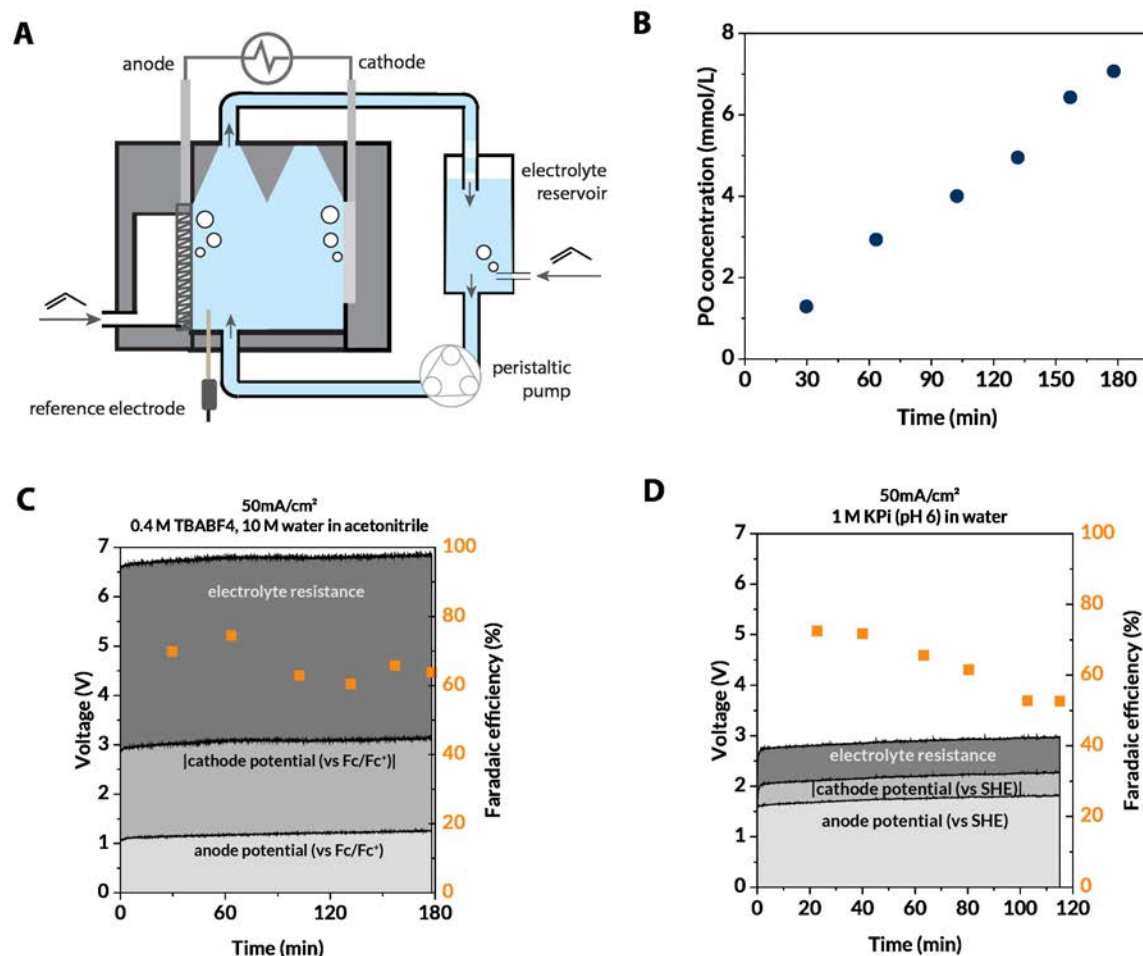


Fig. S4. Electrochemical cell setup with circulating liquid electrolytes. (A) A schematic of the flow cell for PO production. (B) Accumulation of PO in the electrolyte during the process in 0.4 M TBABF₄, 10 M water in acetonitrile electrolyte at a constant current density of 50 mA/cm². Cell performances with (C) water-acetonitrile blended electrolyte and (D) aqueous electrolyte. Uncompensated anode and cathode potentials were measured against Ag/AgCl reference electrode and later converted to SHE-referenced potentials. The resistive voltages between anode-to-reference and reference-to-cathode were determined by potentiostatic electrochemical impedance spectroscopy (PEIS) and combined as electrolyte resistance in the figure. Anode and cathode potentials in the figure are iR-compensated values after subtracting the resistive voltages. 1 atm of room-temperature propylene was flowed at 10 sccm during the measurements.

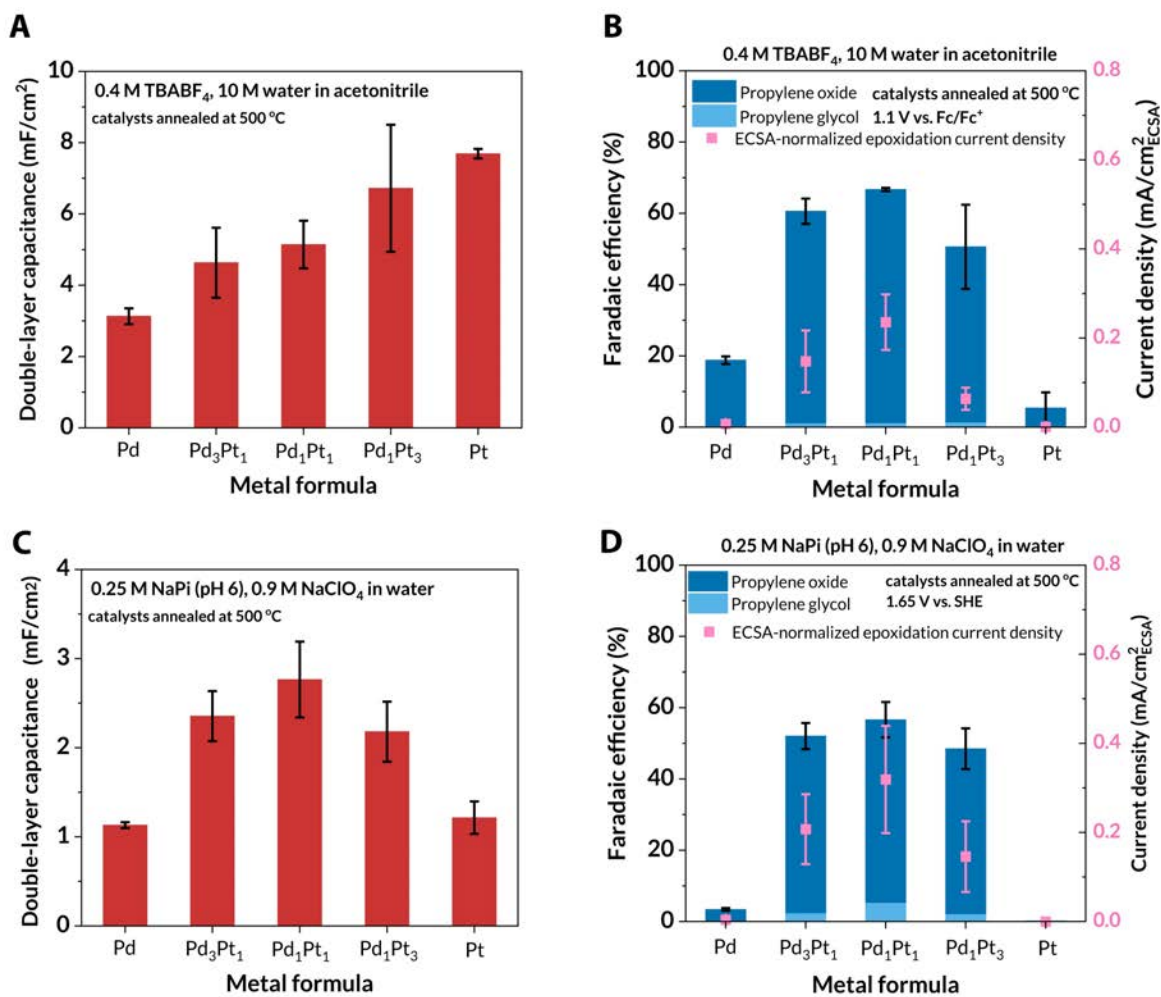


Fig. S5. Double-layer capacitance measurement and ECSA-normalized propylene epoxidation performance with different compositions of Pd-Pt-on-carbon catalysts. Potentials in this figure were 100% iR-compensated (i = current, R= resistance). Specific capacitance of 0.04 mF/cm² was assumed for calculating ECSA. The vertical error bars represent standard deviations from the mean of multiple replicates (n≥3) of the same experiment.

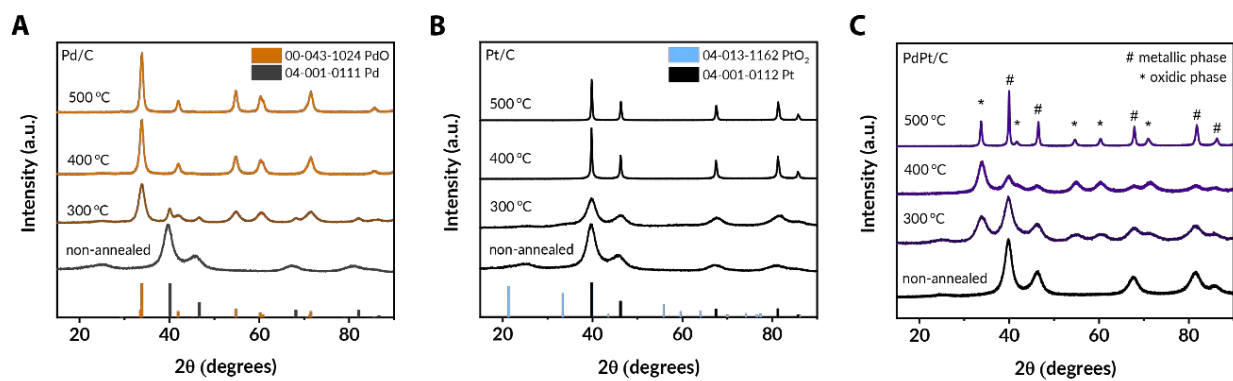


Fig. S6. XRD patterns of Pd/C, Pt/C, and PdPt/C catalysts

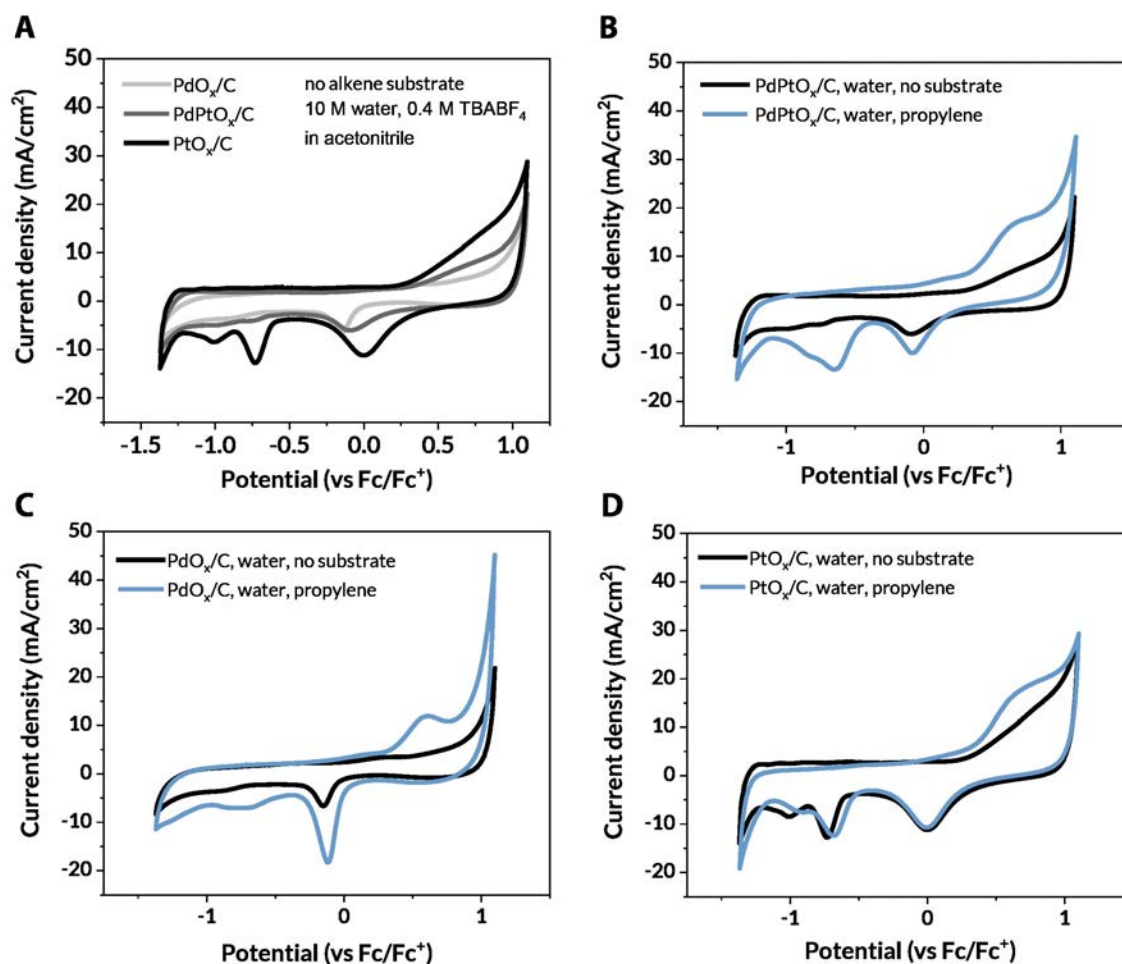


Fig. S7. Cyclic voltammetry traces of the catalysts. The scans were recorded at 50 mV/sec scan rate with 85% automatic iR-compensation (i = current, R= resistance), and the second scan from each CV is shown.

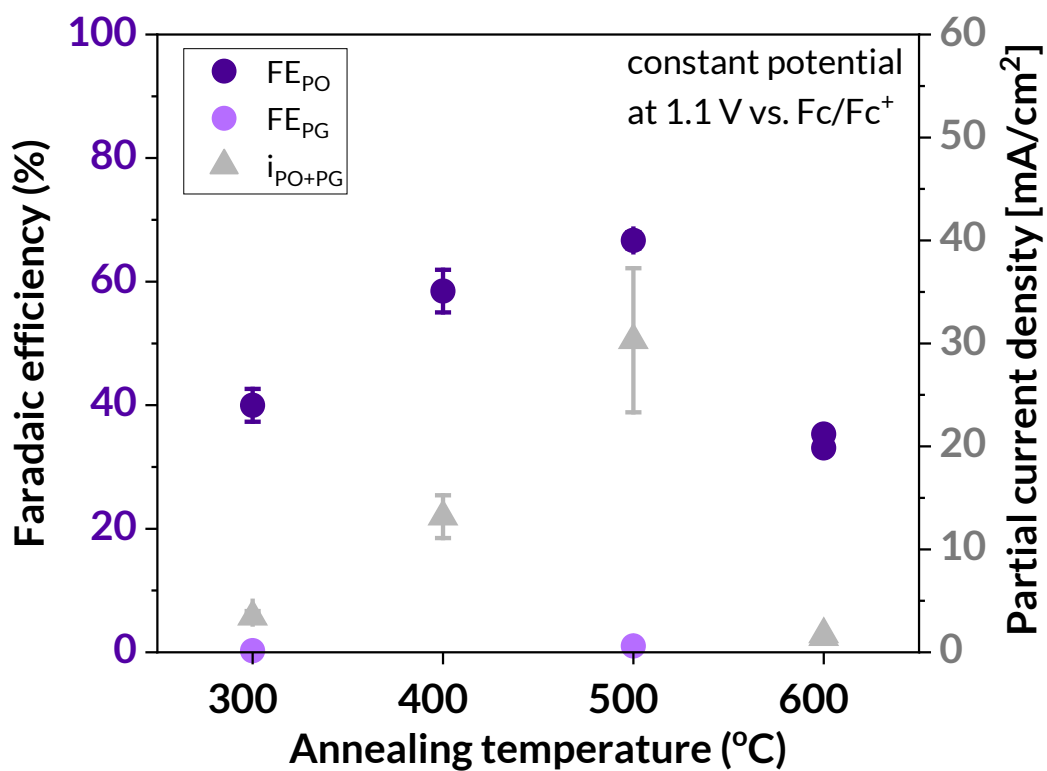


Fig. S8. Comparison of annealed PdPt/C catalyst performances at a constant potential of 1.1 V vs. Fc/Fc⁺ (100% iR-compensated). The vertical error bars represent standard deviations from the mean of multiple replicates (n=3) of the same experiment.

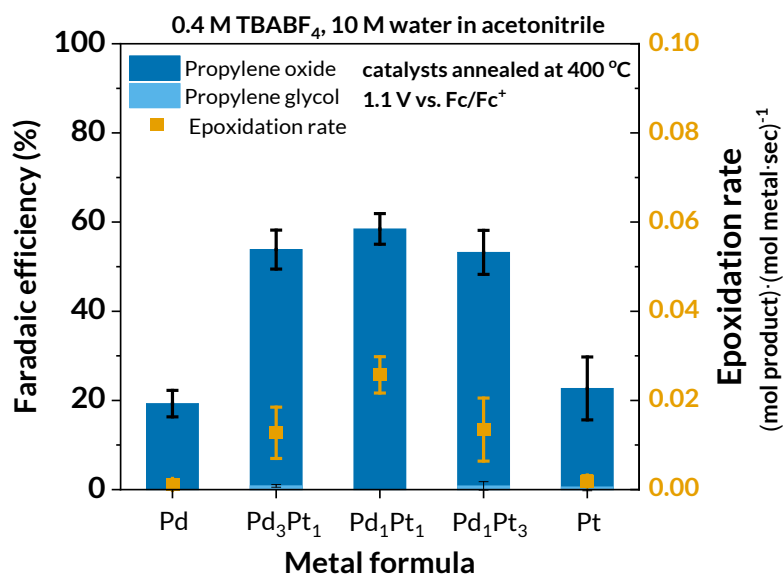


Fig. S9. Direct anodic propylene epoxidation performance with different compositions of Pd-Pt alloy catalysts annealed at 400 °C. This plot is analogous to Fig. 1D; note the different annealing temperatures and resulting differences in performances. 100% iR-compensated potential was applied. The vertical error bars represent standard deviations from the mean of multiple replicates ($n \geq 3$) of the same experiment.

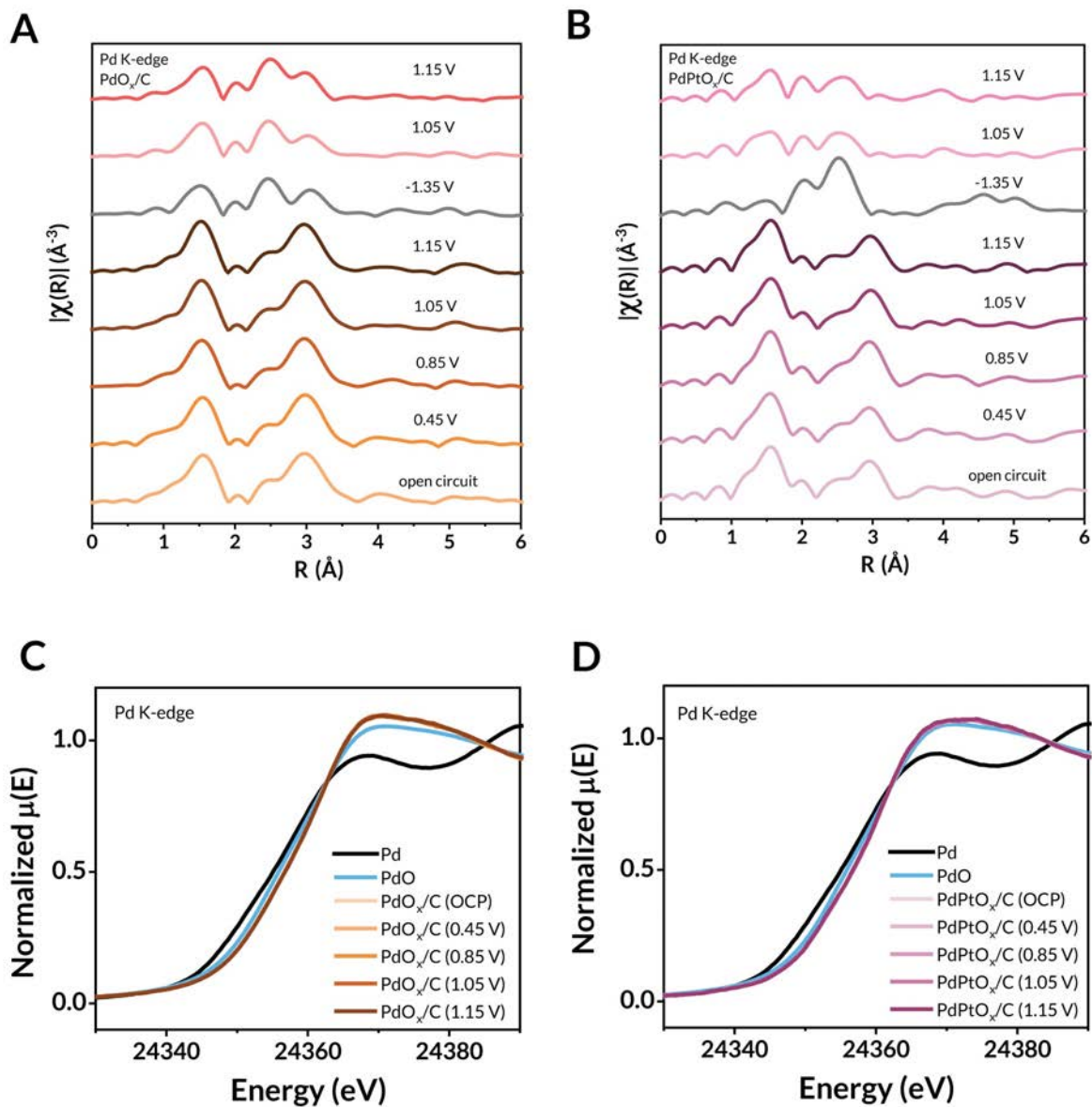


Fig. S10. Operando XAS at Pd K-edge. Potentials in this figure were 100% iR-compensated (i = current, R = resistance).

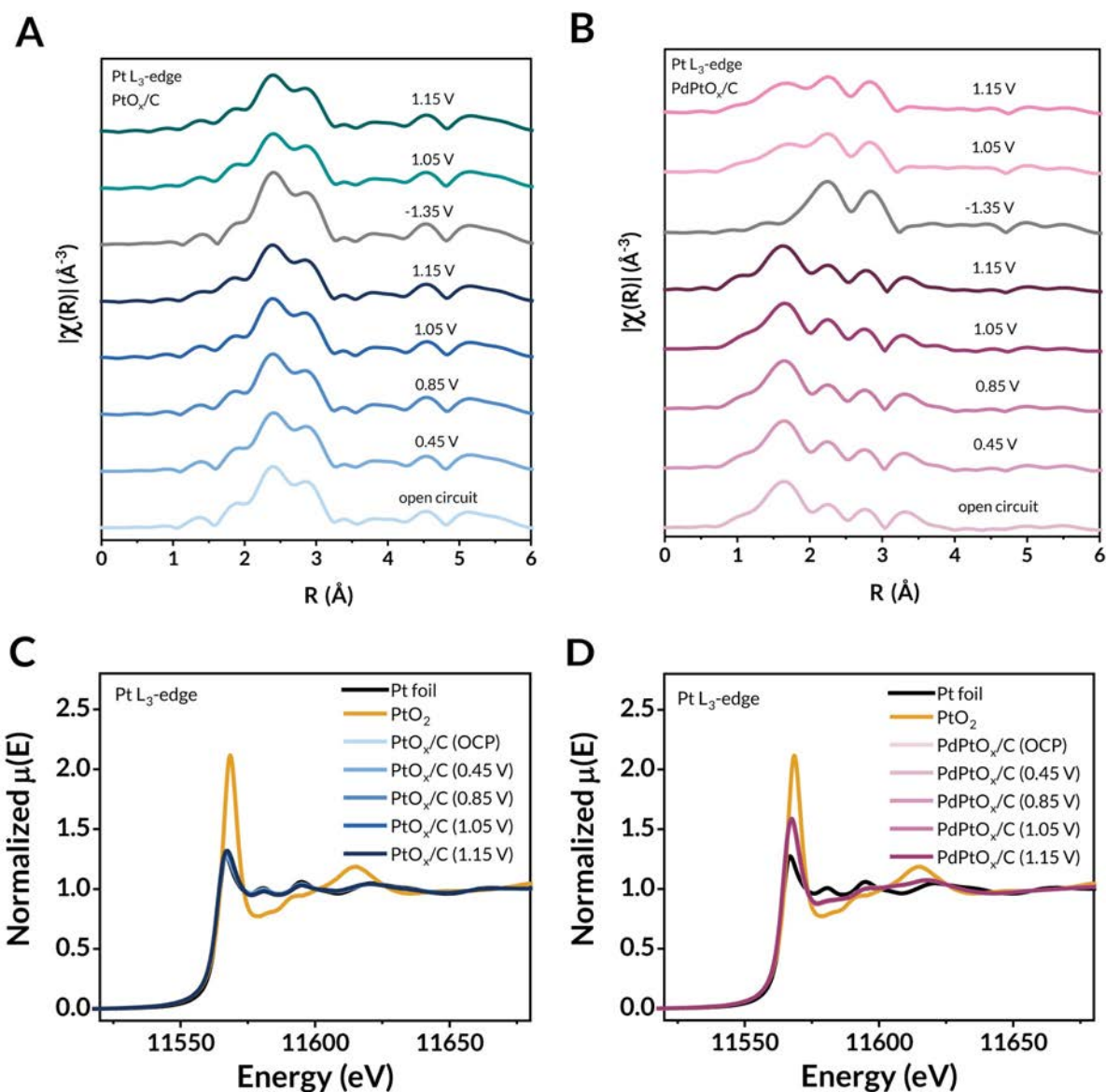


Fig. S11. Operando XAS at Pt L₃-edge. Potentials in this figure were 100% iR-compensated (i = current, R = resistance).

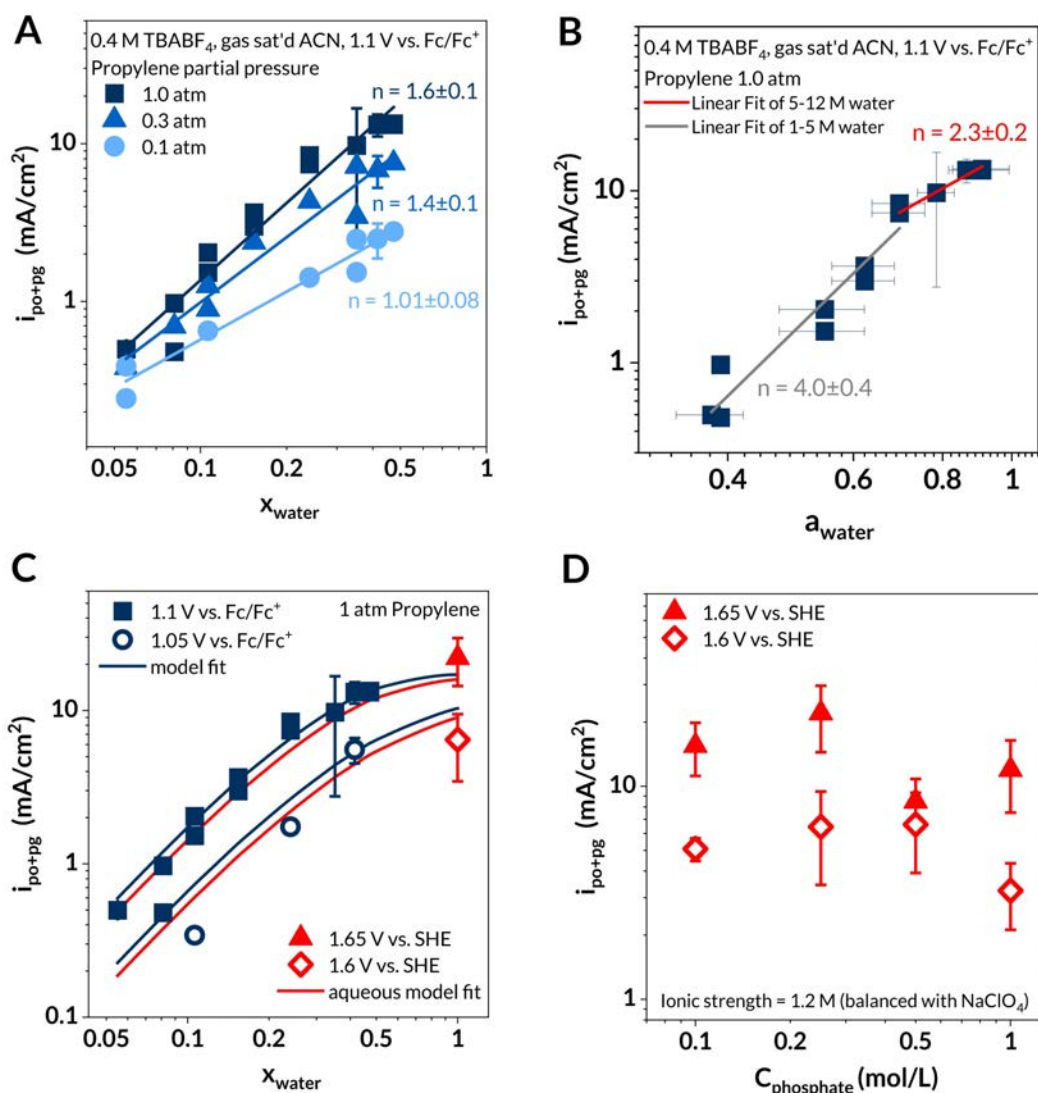


Fig. S12. Water dependences of propylene epoxidation current density. (A) Water concentration dependence of propylene epoxidation current density at different propylene partial pressures. The data set is same as that of Fig. 4B, but linear water concentration dependences were assumed in this plot. (B) Water activity dependence of propylene epoxidation current density. Different order dependences were assumed in high and low water contents. (C) Water molar concentration dependences of propylene epoxidation current density at different potentials. Water molar ratio of 1 was assumed for aqueous electrolytes (0.25 M sodium phosphate buffer (pH 6), 0.9 M NaClO₄ in water). Solid lines represent best-fit from kinetic model of proposed mechanism in Fig. 4D. Model fits for aqueous electrolytes assumed 1.65 V and 1.6 V vs. SHE as 1.09 V and 1.04 V vs. Fc/Fc⁺, respectively, reflecting the absolute potential values applied with respect to pseudo-reference Ag/AgCl electrode. (D) Phosphate concentration dependence of propylene epoxidation current density in aqueous electrolytes. The ratio of Na₂HPO₄ and NaH₂PO₄ was fixed to 1:9 for making pH 6 solution, and the ionic strength was fixed to 1.2 M using NaClO₄ as the balancing salt. The vertical error bars represent standard deviations from the mean of multiple replicates ($n \geq 3$) of the same experiment.

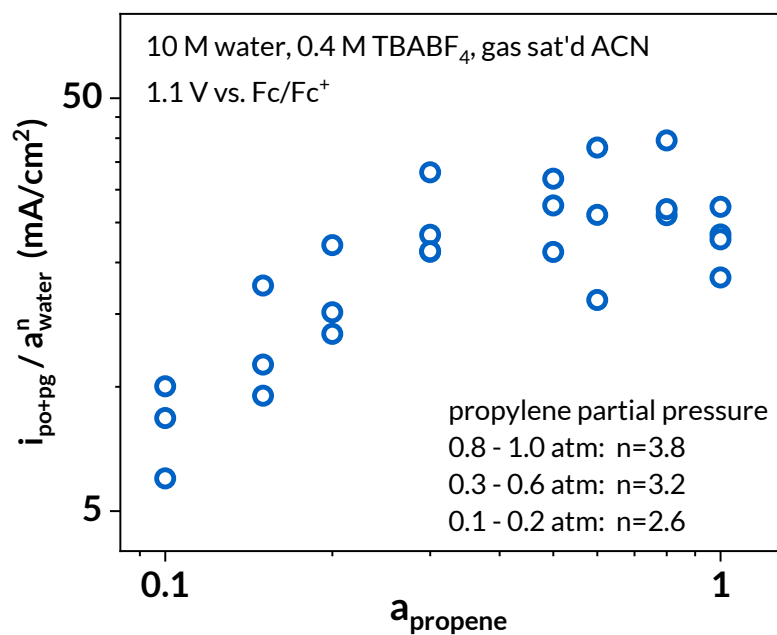


Fig. S13. Propylene dependence on propylene epoxidation current density. Adjustments were made to the current density to account for variations in water activity (fig. S17) and water activity dependences at different propylene pressures.

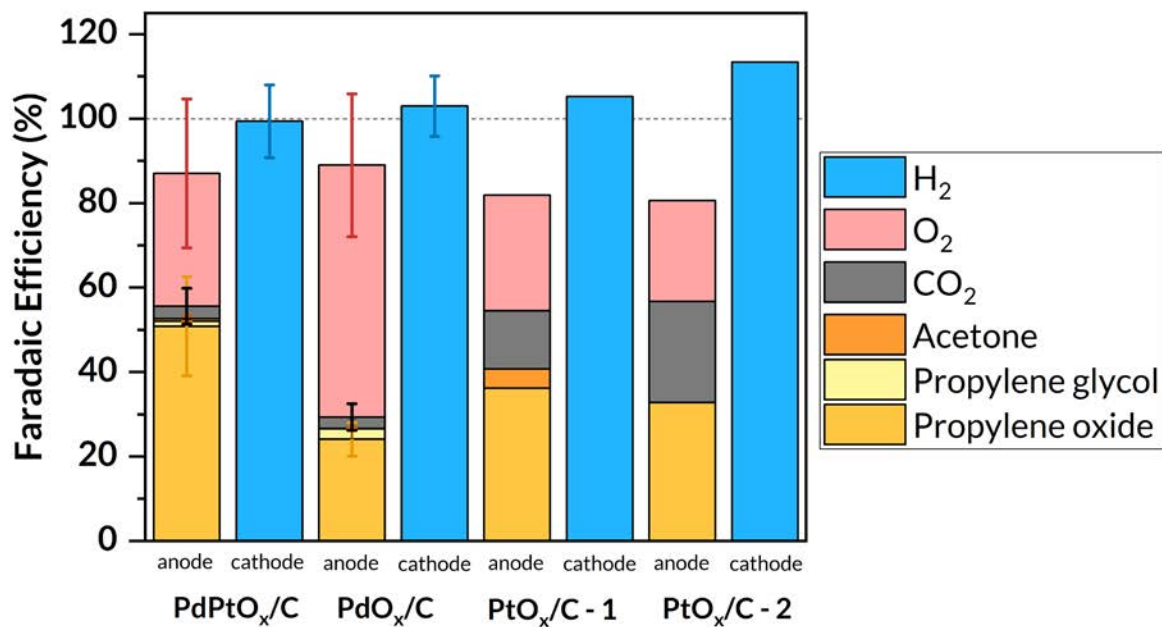


Fig. S14. Product distributions with different catalysts annealed at 400 °C, constant current at 40 mA/cm². 10 sccm of propylene was introduced to the electrolyte containing 0.4 M TBABF₄ and 10 M water in acetonitrile. The vertical error bars represent standard deviations from the mean of multiple replicates (n=3) of the same experiment. For PtO_x/C, both of duplicate data (PtO_x/C -1 and PtO_x/C -2) are presented.

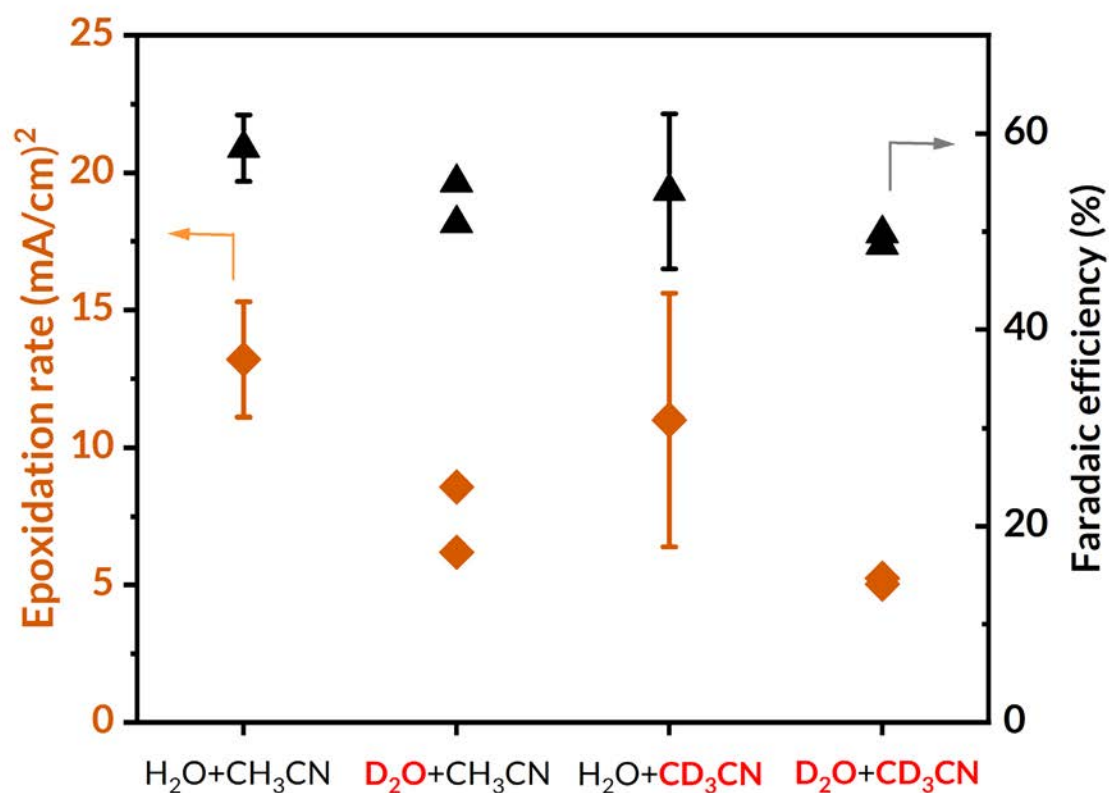


Fig. S15. Deuterium kinetic isotope effects on epoxidation rate catalyzed by PdPtO_x/C. 100% iR-compensated (*i* = current, *R*= resistance) potentials were applied. The vertical error bars represent standard deviations from the mean of multiple replicates (*n*=4) of the same experiment. See Supplementary Text S3 for extended descriptions.

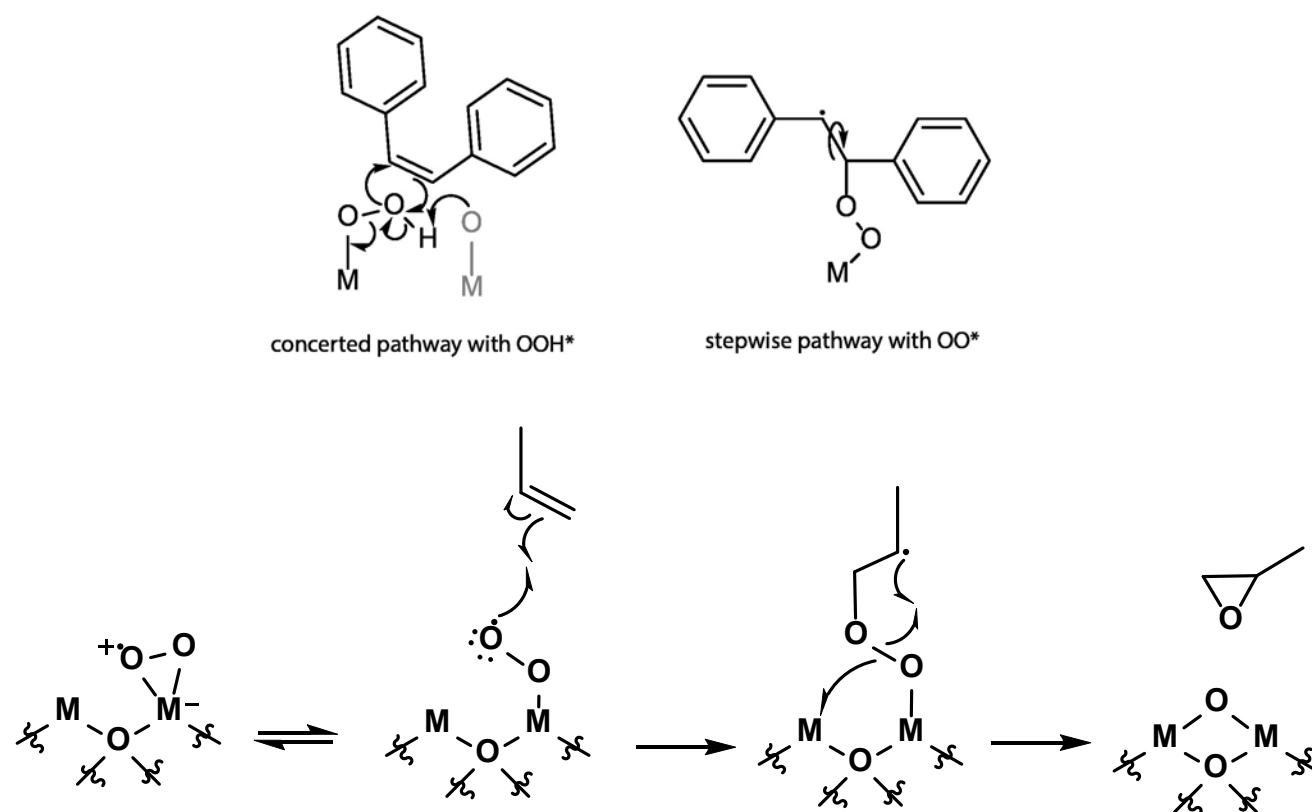


Fig. S16. Comparison of concerted and stepwise pathways for stilbene epoxidation and a proposed stepwise epoxidation pathway involving radical OO^* intermediate.

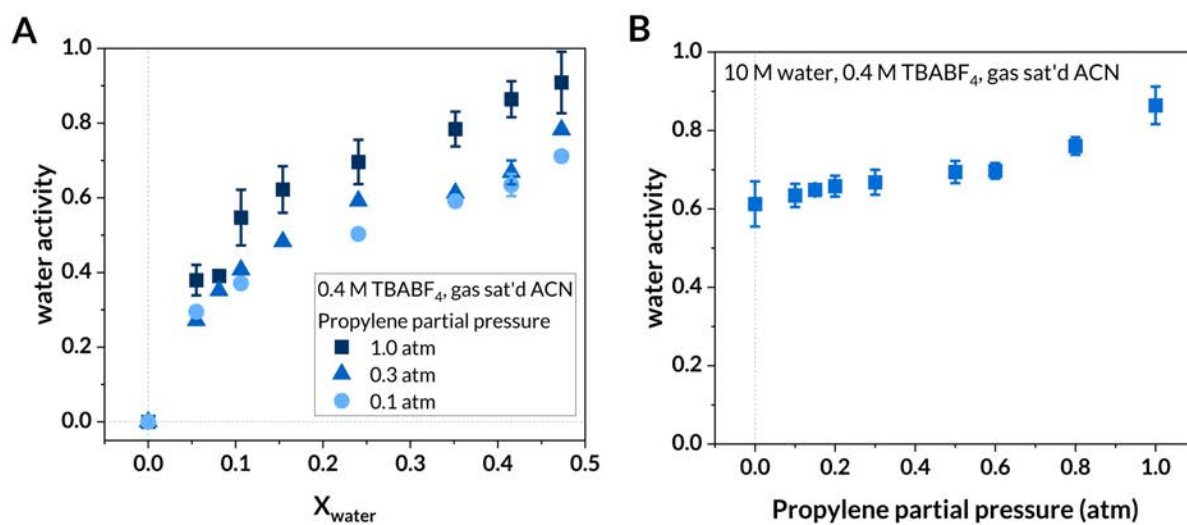


Fig. S17. Activity measurements using headspace-gas chromatography-TCD. The vertical error bars represent standard deviations from the mean of multiple replicates ($n \geq 3$) of the same experiment.

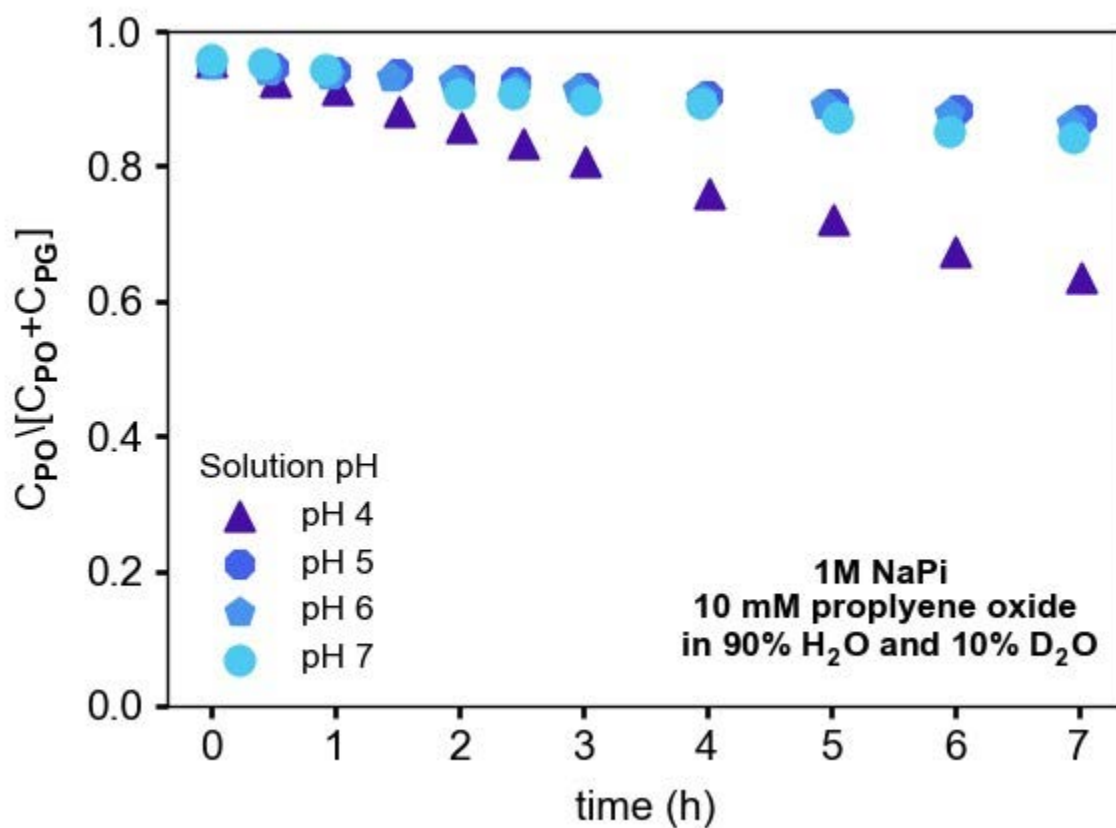


Fig. S18. Conversion of propylene oxide (PO) to propylene glycol (PG) in aqueous electrolytes with varying pHs. PO consumption and PG formation were monitored and quantified by ¹H-NMR.

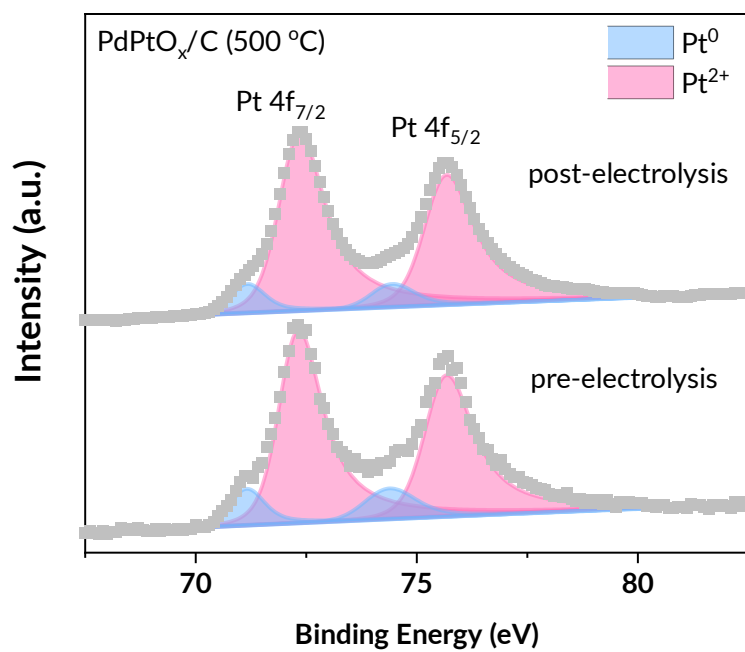


Fig. S19. High-resolution Pt 4f XPS spectra of PdPtO_x/C sample before and after electrolysis.

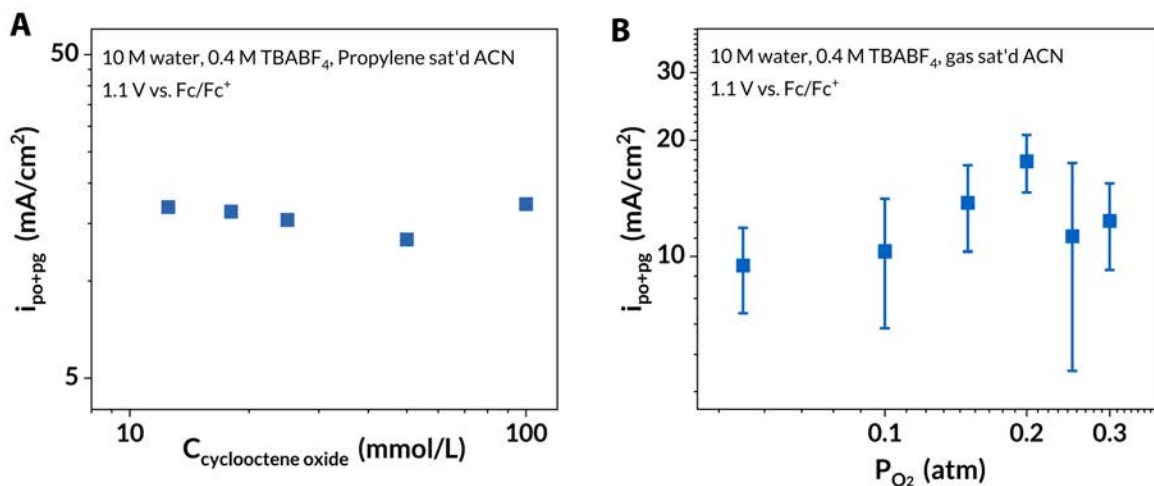


Fig. S20. Effects of epoxide and oxygen concentrations on propylene epoxidation rate. (A) Cyclooctene oxide dependence on propylene epoxidation rate. Cyclooctene oxide was utilized as a substitute for PO to differentiate between the epoxide produced from the reaction (PO) and the epoxide added at the start of the experiment (cyclooctene oxide). **(B)** Oxygen partial pressure dependence on propylene epoxidation rate. The oxygen partial pressure was adjusted by diluting it with propylene, ensuring the total flow rate remained at 10 sccm under a total gas pressure of 1 atm. This range for oxygen partial pressure was selected to prevent a reduction in rate caused by a decrease in propylene partial pressure. 100% iR-compensated potential was applied. The vertical error bars represent standard deviations from the mean of multiple replicates ($n=3$) of the same experiment.

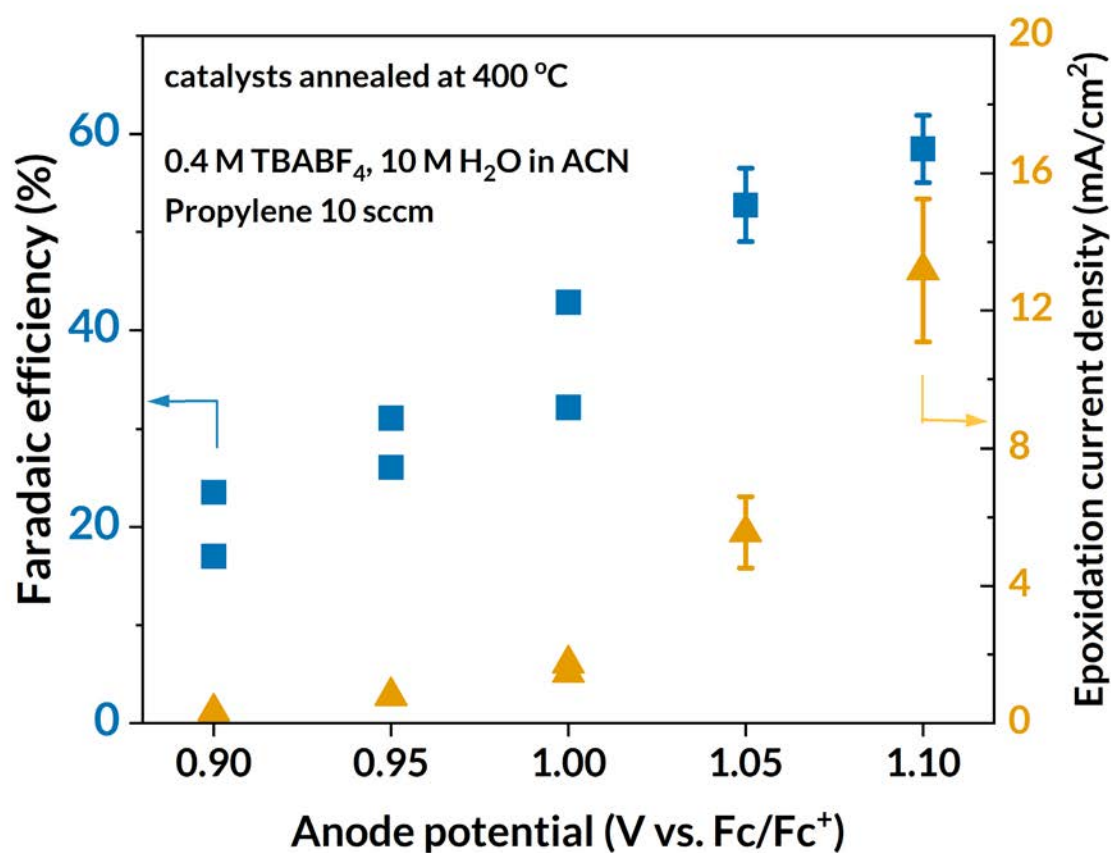


Fig. S21. Epoxidation Faradaic efficiencies and rates at different potentials (100% iR-compensated). This plot shares the data used for the Tafel plot (Fig. 4A). The vertical error bars represent standard deviations from the mean of multiple replicates ($n \geq 3$) of the same experiment

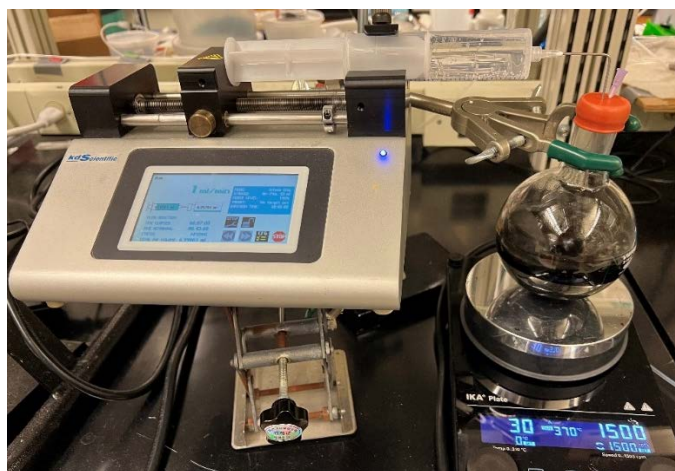


Fig. S22. A setup of a syringe pump adding NaBH₄ solution to the flask.

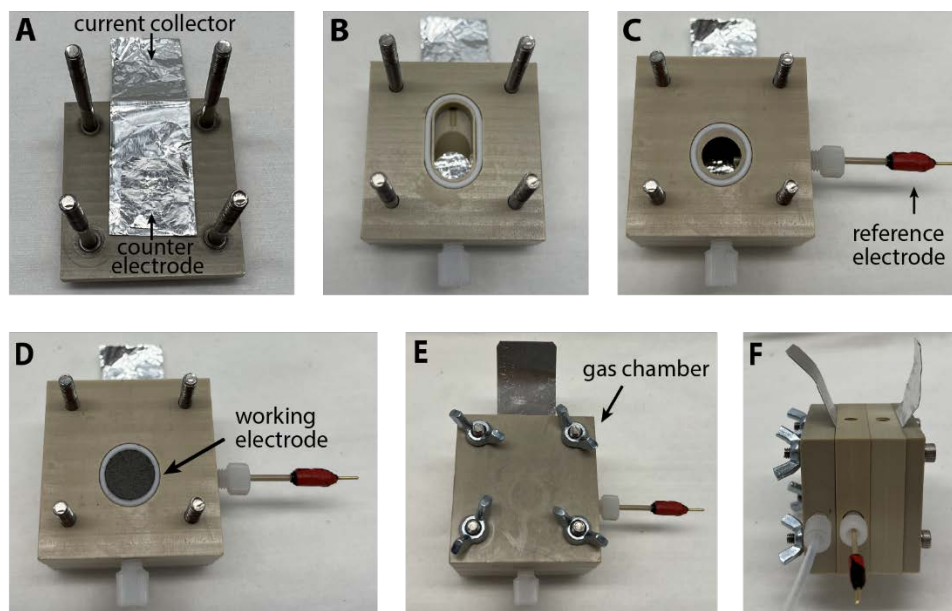


Fig. S23. Assembly and setup of the electrochemical cell used for the kinetic rate measurements. When liquid substrates were used, a backplate was used instead of a gas chamber. For operando XAS, a backplate with a hole covered with a Kapton polyimide film for the X-ray entrance was used instead.

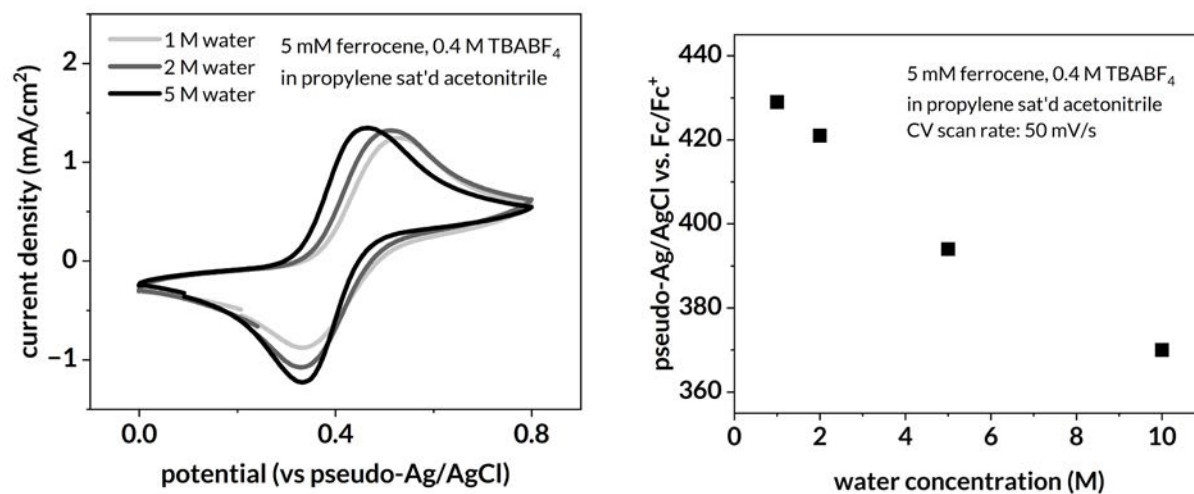


Fig. S24. Ferrocene redox calibration in water-acetonitrile electrolyte.

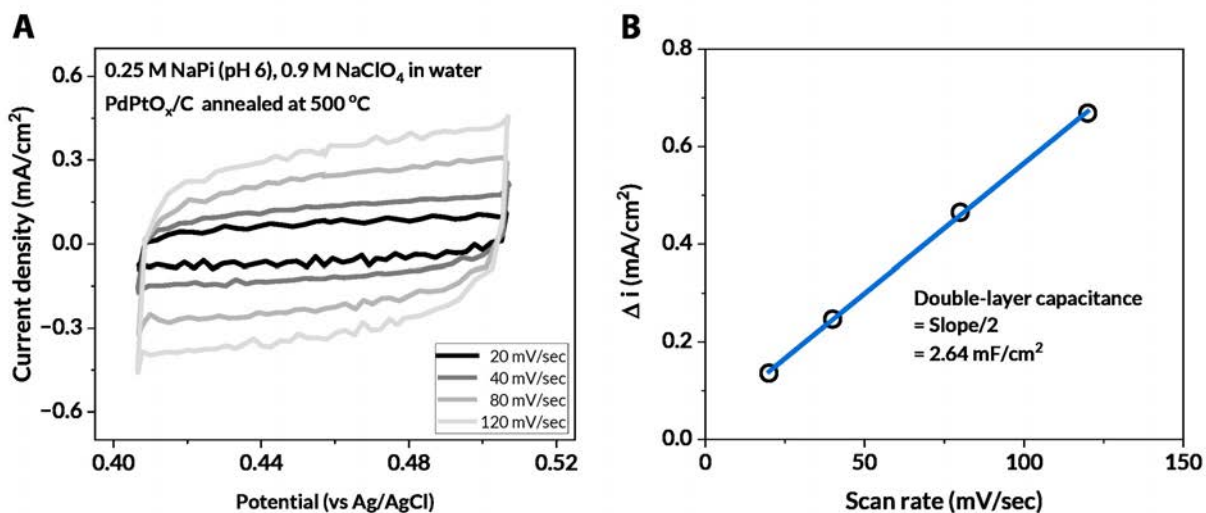


Fig. S25. A representative double-layer capacitance measurement. (A) CV curves at various scan rates over PdPtO_x/C catalyst in aqueous electrolyte. **(B)** Capacitive current densities at different scan rates. DLCs and errors reported from these measurements (fig. S5) are the average and standard deviation of multiple replicates (n=3-18).

Table S1. Compilation of previous reports on direct propylene epoxidation using water as the oxygen source at ambient temperature and pressure.

Entry	Anode material	electrolyte	Conditions*	Highest epoxide production rate ($\mu\text{mol}/\text{cm}^2/\text{hr}$) †	Epoxidation FE at the highest rate	Reference
1	Pt black (calcined at 400 °C air)	Aqueous 1 M H_3PO_4 between the electrodes, CH_2Cl_2 solvent at the anode	30 °C	84 (23%, 1.32V applied) 95 (13.7%, 1.38V vs. NHE applied)	~23%	(13)
2	Ag_3PO_4 cube	0.1 M phosphate (pH 7) buffer	2.4 V vs. RHE	9.2	18.7%	(17)
3	Ag	8.7 mM NaOH		<0.02	> 0	(16)
4	PtO_x	Nafion membrane-electrode assembly (0.5M H_2SO_4), CH_2Cl_2 solvent at the anode	1.6 V vs. SHE	37	7.4 %	(14)
5	PtO_x (N_2H_4)			4.7	40 %	
6	PtO_2 + graphite	1 M H_3PO_4	30 °C, ambient pressure	19	17 %	(15)
7	Electrodeposited Pd on glassy carbon	0.1 M HClO_4	1.2 V vs. RHE	<0.5 0.0037 $\mu\text{mol}/\text{cm}^2$ (ECS A)/hr	22 % (propylene glycol)	(10)
8	Pd functionalized titanium hollow fiber	0.1 M HClO_4	1.2 V vs. RHE	0.061 $\mu\text{mol}/\text{cm}^2$ (ECS A)/hr	19 % (propylene glycol)	(11)
9	Pd/C	0.1 M HClO_4	1.4 V vs. RHE	11.2	23.2 % (propylene glycol)	(9)
10	PdO/C			23.3	46.7 % (propylene glycol)	
11	Stainless steel screen	pH 13.8 KOH (0.63 M)	40 °C, Constant cell current at 24 mA/cm^2	353	79 %	(38)

12	PdPtO _x /C	0.4 M TBABF ₄ , 10 M water in acetonitrile	Constant cell current at 50 mA/cm ² , 1 h	695	74.5 %	<i>This work</i> (fig. S4C)
13			1.1V vs. Fc/Fc ⁺	473	66.0±0.8 %	<i>This work</i> (Fig. 1D)
14		0.25 M sodium phosphate buffer (pH 6), 0.9 M NaClO ₄	1.65 V vs. SHE	411	57±5 %	<i>This work</i> (Fig. 1E)
15		0.1 M HClO ₄	Constant cell current at 24 mA/cm ²	539	68.7% (propylene glycol)	<i>This work</i> (comparison to Entry 7-10)
16		0.1 M sodium phosphate (pH 7) buffer	2.4 V vs. RHE	308	38.4%	<i>This work</i> (comparison to Entry 2)

*Experiments were done at room temperatures and ambient pressures if not otherwise noted.

†Propylene epoxidation rate was compared in the unit of $\mu\text{mol}/\text{cm}^2/\text{hr}$, whereas the area indicates the geometric surface area of the electrode. Relevant information needed for the rate conversions was found in each paper's methods and supplementary material:

Entry 1 – FE(oxidation efficiency in the paper) and production rate are estimated from data points in Figure 2. Anodic potentials are estimated from Figure 10.

Entry 2 – Production rate is from Supplementary Table 2.

Entry 7 – Production rate is estimated from Figure 4d and Figure S3.

Entry 8- Production rate and FE are from Figure 8 and 9, respectively.

Entry 9,10 – Production rate is from Figure 1E, FE is calculated from the combination of Figure 1E and F.

Entry 11 – Although we only compared catalysts tested in ambient temperature and pressures, it is still worth mentioning this paper for reference, since the reported catalytic performance in elevated temperatures is notable. Previously studied catalysts can be clustered into Pd-, Pt-, and Ag- derived groups, with this single report using a stainless-steel electrode as an outlier. We attempted to reproduce this result, using the layers of stainless-steel cloth as the anode with the pretreatment and the electrolyte conditions described in the paper. However, no trace of propylene oxide or propylene glycol peak was shown in the ¹H-NMR spectrum of the post-electrolysis electrolyte.

Table S2. Comparison of catalyst performances for direct electrocatalytic cyclooctene epoxidation using water as the oxygen source.

	Anode material	electrolyte	conditions	Highest epoxide production rate (mA/cm²)	Epoxidation FE at highest rate	Reference
1	MnO _x	0.1 M TBABF ₄ , 5-10 M H ₂ O, and 200 mM cyclooctene in acetonitrile	1.45V vs. Fc/Fc ⁺	~2.5	~30 %	(8)
2	Ir _{single} -MnO _x	0.11 M TBABF ₄ , 10 M water in acetonitrile	1.45V vs. Fc/Fc ⁺	10.5±2.8	46±4 %	(7)
3	RuO ₂ /N _{0.12} C	0.025 M cyclooctene, 0.05 M LiClO ₄ , Ar-saturated ethanol/water (1:3)	1.2 V vs. Ag/AgCl	~3.2	~80 %	(39)
4	PdPtO _x /C (annealed at 400 °C)	0.11 M TBABF ₄ , 10 M water in acetonitrile	1.05 V vs. Fc/Fc ⁺	15.8±0.5	92±4 %	<i>This work</i>
5			Constant cell current at 40 mA/cm ²	34.1	85.2 %	

Table S3. XRD-determined crystallite sizes in Pd/C, Pt/C, and PdPt/C catalysts

(unit: nm)	Pd/C		Pt/C		PdPt/C	
	metal	oxide	metal	oxide	metal	oxide
non-annealed	3.4±1.1	-	3.2±1.3	-	3.2±0.6	-
300 °C	7.9±1.1	4.5±0.5	2.7±0.3	-	3.007±0.004	2.60±0.03
400 °C	-	8.7±0.4	23.9±2.9	-	5.7±0.7	4.0±1.1
500 °C	-	12.2±1.0	29.1±3.9	-	16.9±4.2	14.1±4.9

Table S4. EXAFS fitting of Pd and Pt foil standards.

	Bond type	N	S₀²	σ²	ΔE₀ (eV)	R_{eff} (Å)	ΔR (Å)	R-factor
Pd foil standard	Pd-Pd	12	0.74(3)	0.0053(3)	-5.6(4)	2.7505	-0.009(2)	0.004
Pt foil standard	Pt-Pt	12	0.79(6)	0.0041(5)	7.7(6)	2.7689	-0.007(4)	0.008

Table S5. EXAFS fitting of catalysts at Pd K-edge.

	Bond type	N	S_o^2	σ^2	ΔE_0 (eV)*	R (Å)	R-factor
PdO _x /C (open-circuit)	Pd-O	3.7(4)		0.0001(10)		2.014(9)	
	Pd-(di-μ-O-)Pd	4.5(18)	0.74	0.004(2)	0.7(16)	3.063(2)	0.01
	Pd-(mono-μ-O-)Pd	5.4(21)		0.003(2)		3.44(1)	
PdO _x /C (1.15 V vs. Fc/Fc ⁺)	Pd-O	3.7(4)		0.0004(10)		2.017(8)	
	Pd-(di-μ-O-)Pd	5.0(16)	0.74	0.004(2)	1.9(14)	3.06(1)	0.01
	Pd-(mono-μ-O-)Pd	5.4(21)		0.004(2)		3.45(1)	
PdPtO _x /C (open-circuit) Fit 1	Pd-O	3.1(5)				1.98(1)	
	Pd-Pd	0.62(55)				2.83(4)	
	Pd-Pt	1.2(10)	0.74	-0.002(2)	-2.4(25)	2.92(8)	0.02
	Pd-(di-μ-O-)Pd	1.2(11)				3.03(5)	
	Pd-(mono-μ-O-)Pd	1.7(6)				3.45(2)	
PdPtO _x /C (open-circuit) Fit2	Pd-O	3.2(5)				2.00(1)	
	Pd-(di-μ-O-)Pd	1.7(5)	0.74	-0.002(1)	0.0 (22)	3.06(1)	0.04
	Pd-(mono-μ-O-)Pd	1.4(5)				3.46(2)	
PdPtO _x /C (1.15 V vs. Fc/Fc ⁺) Fit 1	Pd-O	3.3(7)				1.99(2)	
	Pd-Pd	0.39(62)				2.85(8)	
	Pd-Pt	0.7(11)	0.74	-0.001(2)	-1.7(28)	2.9(2)	0.02
	Pd-(di-μ-O-)Pd	1.5(12)				3.04(5)	
	Pd-(mono-μ-O-)Pd	1.9(8)				3.45(3)	
PdPtO _x /C (1.15 V vs. Fc/Fc ⁺) Fit 2	Pd-O	3.5(4)				2.00(1)	
	Pd-(di-μ-O-)Pd	1.7(4)	0.74	-0.001(1)	-0.6(18)	3.05(1)	0.03
	Pd-(mono-μ-O-)Pd	1.8(5)				3.45(2)	

*Assigning incorrect atomic species for backscatter might have caused negative Debye-Waller factors for PdPtO_x/C samples.

Table S6. EXAFS fitting of catalysts at Pt L3-edge.

	Bond type	N	S_o^2	σ^2	ΔE_0 (eV)	R (Å)	R-factor
PtO _x /C (open-circuit)	Pt-Pt	9.5(8)	0.79	0.0049(7)	6.5(7)	2.752(5)	0.006
PtO _x /C (1.15 V vs. Fc/Fc ⁺)	Pt-Pt	9.8(16)	0.79	0.006(1)	6.0(13)	2.751(9)	0.02
PdPtO _x /C (open-circuit)	Pt-O	2.8(6)				2.01(2)	0.01
	Pt-Pt	2.2(12)	0.79	0.004(3)	11(3)	2.78(4)	
	Pt-Pd	0.94(78)				2.82(5)	
PdPtO _x /C (1.15 V vs. Fc/Fc ⁺)	Pt-O	2.8(5)				2.01(2)	0.007
	Pt-Pt	2.5(11)	0.79	0.005(3)	11(2)	2.79(3)	
	Pt-Pd	1.3(7)				2.83(4)	

Table S7. Near-surface compositions of PdPt/C catalysts derived from XPS survey spectra

	C	O	Pd	Pt
	weight %			
PdPt/C	52.47	7.42	12.84	27.28
PdPtO_x/C (400 °C)	10.63	19.61	25.94	42.37
PdPtO_x/C (500 °C)	7.54	20.79	25.5	46.16
	atomic %			
PdPt/C	85.78	9.1	2.37	2.75
PdPtO_x/C (400 °C)	32.55	47.50	9.48	8.65
PdPtO_x/C (500 °C)	26.11	54.07	9.97	9.84

Table S8. Quantification of catalyst compositions using ICP-OES

	metal weight percent	Pt molar composition in metal
Pt/C	44.0%	1.00
Pd ₁ Pt ₃ /C	48.6%	0.75
Pd ₁ Pt ₁ /C	34.8%	0.49
Pd ₃ Pt ₁ /C	37.8%	0.24
Pd/C	35.7%	0.00

Table S9. Surface concentrations of metal species evaluated by XPS analysis

	Pt(0)	Pt(II)	Pd(0)	Pd(II)
PdPtO _x /C (400 °C)	surface concentration (%)		surface concentration (%)	
	49.28	50.72	16.09	83.91
	4f _{7/2} peak position (eV)		3d _{5/2} peak position (eV)	
	70.9	72.1	335.5	337.2
PdPtO _x /C (500 °C)	surface concentration (%)		surface concentration (%)	
	13.5	86.5	0	100
	4f _{7/2} peak position (eV)		3d _{5/2} peak position (eV)	
	71.1	72.3	-	337.2
PdPtO _x /C (500 °C) Post-electrolysis (fig. S19)	surface concentration (%)		surface concentration (%)	
	10.27	89.73	0	100
	4f _{7/2} peak position (eV)		3d _{5/2} peak position (eV)	
	71.1	72.3	-	337.1
PdO _x /C (500 °C)			surface concentration (%)	
			0	100
			3d _{5/2} peak position (eV)	
			-	337.2
PtO _x /C* (500 °C)	surface concentration (%)			
	100	0		
	4f _{7/2} peak position (eV)			
	71.0	-		

*We tried to fit the PtO_x spectrum with both Pt(0) and Pt(II) species, but the best fit was achieved with the 4f_{7/2} peaks at 71.0 eV and 71.4 eV. However, since 71.4 eV does not align with the expected Pt(II) position, we opted to only fit the spectrum with the Pt(0) 4f_{7/2} and 4f_{5/2} peaks. When we attempted to fit the spectrum with a fixed Pt(II) 4f_{7/2} peak position between 71.1-71.3 eV, the 4f_{7/2} peak became smaller compared to the 4f_{5/2} peak, so we discarded that fit.

Table S10. Deuterium kinetic isotope effects on epoxidation rate catalyzed by PdPtO_x/C*

	FE _{PO}	i _{PO} (mA/cm ²)	k _H / k _D
10 M H ₂ O, CH ₃ CN	58.5±3.4 %	13.2±2.1	
10 M D₂O , CH ₃ CN	52.9±3.0 %	7.4±1.7	1.8±0.5
10 M H ₂ O, CD₃CN	54.1±7.9 %	11.0±4.6	1.2±0.5
10 M D₂O , CD₃CN	49.2±0.8 %	5.1±0.2	2.6±0.4

*This table shares data with fig. S15.

Table S11. Reaction conditions for studying correlation between alkene electrophilicity index and the direct anodic epoxidation rate

Substrate	Deviation from standard condition *	Faradaic efficiency (%)			ω^- average (kJ/mol)	Epoxidation current density (mA/cm ²)
		epoxide	diol	ketone		
2-methyl-2-butene	1.02 V vs. Fc/Fc ⁺ †	41.7	36.5	-	12.11	11.67
2-hexene	-	79.1	-	-	18.08	11.08
cyclooctene	-	92.4	-	-	19.15	15.80
cycloheptene	-	71.9	-	-	19.16	5.68
norbornene		15.9			19.93	0.64
1-hexene	-	53.0	-	-	25.70	1.73
1-octene	0.16 M substrate ‡	43.9	-	28.5	25.81	1.19
propylene	Saturated electrolyte with 1atm substrate flowing at 10 sccm	54.4	-	-	25.97	5.12
allyl chloride	-	19.7	4.63	-	35.52	0.97
ethylene	Saturated electrolyte with 1atm substrate flowing at 10 sccm	24.5	-	-	36.40	0.76
3,3,3-trifluoropropene	Saturated electrolyte with 1atm substrate flowing at 10 sccm	1.96	-	-	43.30	0.033

5 *Standard condition: 0.11 M TBABF₄, 0.2 M substrate, 10 M water in acetonitrile, annealed PdPt/C on carbon paper (at 400 °C) as an anode, Pt foil as a cathode, no separator, stirred using a magnetic bar, 1.05 V vs. Fc/Fc⁺ applied to the anode

†Slightly lowered the potential to due to instrumental current overload at 1.05 V vs. Fc/Fc⁺.

‡Lowered substrate concentration to avoid phase separation.

Table S12. Recipe for Pd-Pt alloys on carbon.

	PdCl ₂ (mmol)	0.1 M HCl(aq) (mL)	H ₂ PtCl ₆ ·6H ₂ O (mmol)	milliQ water (mL)	Vulcan XC-72R (mg)
Pd/C	0.5	10			79.8
Pd ₃ Pt ₁ /C	0.375	7.5	0.125	2.5	96.4
PdPt/C	0.25	5	0.25	5	113.1
Pd ₁ Pt ₃ /C	0.125	2.5	0.375	7.5	129.7
Pt/C			0.5	10	146.3